

Trigonometry on the Surcomplex Plane

Canonical Phases, Arbitrary-Scale Geometry,
and Infinitesimal Degeneration

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Abstract

The surcomplex plane is the algebraically closed class field $\mathbf{SC} = \mathbf{No}[i]$ with Euclidean norm $|x + iy| = \sqrt{x^2 + y^2}$. Its lengths may be infinite or infinitesimal, but every direction has a *finite* surreal angle. We make that observation into a trigonometric theory. Hahn summation gives canonical sine and cosine on finite surreal arguments, and identifies the full unit circle $\mathbb{T}(\mathbf{No})$ with the additive group of finite surreals modulo $2\pi\mathbb{Z}$; the circle splits canonically as an ordinary phase times an additive infinitesimal phase, and its angular and chordal distances have the same valuation.

We prove exact analogues of the angle-sum, cosine, sine, tangent, half-angle, Heron, Euler, angle-bisector, trigonometric Ceva, inscribed-angle and Ptolemy theorems at arbitrary surreal scale, together with the spherical sine and cosine laws and hyperbolic triangle laws in the full surreal Poincaré disk. Valuation formulas convert each of these into an exact scale dictionary; a relative flatness threshold and a quadratic triangle-inequality defect formula quantify degeneration; and near-tangent intersections exhibit exact square-root splitting with a quantified loss of angular precision. Three different stability certificates are proved and kept apart: a sharp $\sigma > 2\kappa$ root-stability bound with its sharpness proposition, a conditioned inverse-cosine expansion with second-order term, and a support-controlled exact cluster multiplicity. A canonical complex extension exists on the strip of arguments with finite real part and arbitrary surreal imaginary part; there the sine is surjective with completely determined fibres, trigonometric polynomials algebraize to ordinary-degree polynomials, and one obtains sharp root counts, finite Fourier inversion and Parseval identities, and a Fejér–Riesz factorization over the real closed surreal field, with a degree-two polynomial that is positive at every ordinary angle and negative at a finite nonordinary one.

Finally we classify the freedom in extending circular trigonometry to infinite real arguments. The Ehrlich–Kaplan integer-part normalization is one member of a character family; every member necessarily acquires infinite periods, and none admits a bounded common-domain Hahn-analytic description at infinite frequency. Two further sections treat the rotation group $\mathrm{SO}(2, \mathbf{No})$ and the arithmetic of the periods, where exponentials with the initial kernel-multiplier ring $\mathbb{Z}$ give a partial negative answer to a robustness question of Ehrlich and Kaplan, which stays open.

Status and provenance. This is one merged report assembled from six independently written manuscripts of September 21–22, 2026 ([section 1.1](#)). It develops three supplied unpublished manuscripts rather than presenting their results as published literature. Classical identities and established surreal analytic machinery are identified as such. The literature search behind it was targeted rather than exhaustive; no priority claim and no claim to have solved a named published open problem is made; nothing here is formally verified, and the accompanying symbolic checks validate displayed examples rather than machine-checking the general proofs. Per-theorem non-claims are stated where they arise and collected in [section 20.4](#).

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1 The organizing distinction: lengths versus angles

Let $z = x + iy$ with $x, y \in \mathbf{No}$ and $i^2 = -1$; this is the meaning of a surcomplex number throughout. The field structure and real closedness of $\mathbf{No}$, and the algebraic closedness of $\mathbf{SC}$, are established parts of surreal number theory [4, 5, 6, 7, 10]. They already give distances with nonnegative square roots, two-dimensional determinants measuring oriented area, and exact solutions of finite linear systems.

The analytic difficulty is different. The expression

$$\sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

cannot simply be declared to define $\sin x$ for every surreal x . At a nonzero ordinary real argument, infinitely many summands already share one Hahn exponent. At a positive infinite argument the monomial supports run in the wrong direction and their union is not well ordered. Ordinary real convergence repairs neither defect in the surreal fine topology.

There is nevertheless a canonical and sufficient theory of geometric angles. If $z \neq 0$, its normalized direction $z/|z|$ has coordinates in $[-1, 1]$, hence an ordinary standard part on the usual unit circle together with an infinitesimal angular correction. Consequently the trigonometry of *all* surcomplex triangles requires no choice of $\sin \omega$. The construction applies to $1 + i\omega$, to $\omega + i$, and to $\omega^{-\omega} + i\omega^\omega$ exactly as it applies to $1 + i$.

Three assertions must be kept apart, and this article keeps them apart everywhere.

1. An angle has a cosine and a sine defined as the coordinates of a unit vector. This is algebra over a real closed field; it needs no transcendental evaluation.
2. A *finite* surreal number has a canonical cosine and sine, obtained from the ordinary analytic germs by strongly summable infinitesimal Taylor evaluation.
3. Every surreal number, ω included, has a uniquely specified oscillatory phase compatible with those finite values.

We prove that (1) and (2) agree and suffice for all plane, spherical and hyperbolic geometry below. Assertion (3) requires additional data; [section 17](#) says exactly which data, and what is lost whichever choice is made, and [section 19](#) studies the arithmetic of the periods that every such choice creates.

Object	Domain used here	Governing construction
Lengths and coordinates	All of $\mathbf{No}$ and $\mathbf{SC}$	Ordered-field algebra and square roots
Circular angles	Finite $\theta \in \mathbf{No}$, modulo $2\pi\mathbb{Z}$	Ordinary value plus infinitesimal Taylor series
Inverse tangent	Every $x \in \mathbf{No}$	Direction of $1 + ix$
Complex sine and cosine	$\{x + iy : x \text{ finite}, y \in \mathbf{No}\}$	Finite phase and Gonshor exponential
Trigonometric polynomials	Ordinary finite degree, arbitrary surcomplex coefficients	Laurent-polynomial algebra on the unit circle
Sine at infinite real inputs	Only after an extension is specified	An additional additive-group character

1.1 What comes from the supplied manuscripts

The supplied analysis manuscript [1] provides the notation $t^\gamma = \omega^{-\gamma}$ for Conway normal-form monomials, the set-sized support convention, strong summability, standard parts, Taylor evaluation

on infinitesimal thickenings, coefficientwise contour and integral functionals, the fine-local logarithm, and examples of nonunique global fine-analytic exponentials. We retain those conventions and reprove the elementary parts needed here.

The two further supplied manuscripts [2, 3] concern finite *zero algebras* and analytic intersections rather than Euclidean triangle geometry. Their relevant themes are conservation of multiplicity, positive-support coefficient recursion, collision-stable residues, and loss of precision near a singular Jacobian; the threshold $\sigma > 2\kappa$ of [3, §9] is the direct antecedent of [theorem 9.7](#) and [theorem 9.9](#). We do not assume their several-variable Noetherianity, division, Nullstellensatz or residue theorems to prove the geometric results below; each trigonometric instance is proved directly.

Ehrlich and Kaplan construct canonical global trigonometric functions and a canonical surcomplex exponential from the initial integer part [8, §11]; that construction is recorded in [definition 17.1](#) and [theorem 17.2](#) and is *not* claimed as a new definition. Restricted analytic evaluation on surreal fields is likewise established [7, Theorem 2.1]. Classical trigonometric identities and inverse-function expansions are recorded in [11]; algebraic approaches to trigonometry over general fields are in [13, 14]; the classical scalar Fejér–Riesz context is in [12]. A proof of an analogue is not by itself evidence that its exact formulation is absent from all previous literature.

The six component manuscripts. The report itself is built from six manuscripts, numbered locally 16–21; the numbers prefix their code and data files. Sources 16, 17 and 18 — *Finite Radians, Arbitrary-Scale Geometry, and Infinitesimal Angular Phenomena; Canonical Phases, Arbitrary-Scale Triangles, and Infinitesimal Degeneration; Canonical Angles, Infinite Triangles, Infinitesimal Contact, and Hahn-Analytic Oscillation*, all of September 21, 2026 — are three drafts of one article on the trigonometry of **SC**; 18 is the base text of Sections 1–17, and 16 and 17 contribute the results the directory’s README lists. They were written from the three supplied manuscripts above and record no repository commit. Source 19, *Rotations of the Surreal Plane: The Norm-One Torus, Infinitesimal Angles, and Topology at Every Scale* (September 22, 2026), was written against repository commit `dcf86662b574`, at which this article stood unchanged; it contributes [section 18](#). Its structure theorem re-derives [proposition 4.1](#) and [theorem 4.2](#), and [section 18.1](#) lists every result it shares with sources 16–18, each printed once with both credited. Sources 20 and 21, *Exponential Kernels and Arithmetic Rigidity in the Surcomplex Field* and *Period Arithmetic and Noncanonical Surcomplex Exponentials* (both September 22, 2026), were written independently against repository commit `465a54b479a1`, at which this article stood unchanged; they build on [theorems 17.4](#) and [17.6](#) and together contribute [section 19](#), with source 21 as its base text. [Section 19.1](#) lists what they share, the credits they owe, and where the merge had to choose.

1.2 Conventions: one symbol per concept

Because sources 16–18 are three drafts with three notations, the following choices are fixed once and used without exception thereafter. Synonyms appearing in those manuscripts (“the sources” below) are recorded here and nowhere else; those of source 19 are recorded in [section 18.1](#), and those of sources 20 and 21 in [section 19.1](#).

- **Unit circle.** $\mathbb{T}(\mathbf{No}) = \{u \in \mathbf{SC} : u\bar{u} = 1\}$, with $\mathbb{T}(\mathbb{R})$ its ordinary complex counterpart. The sources also write $\mathbb{U}$ and $S^1(\mathbb{C})$ for these; we do not.
- **Finite phase.** $\text{cis } \theta = \cos \theta + i \sin \theta$ for finite real θ . The sources also write $E(\theta)$ and $\Phi(\theta)$ for this map; the letter E is reserved below for the strip exponential of [theorem 10.1](#).
- **Finite and infinitesimal elements.** $\mathcal{O}_{\mathbb{R}}, \mathfrak{m}_{\mathbb{R}} \subset \mathbf{No}$ and $\mathcal{O}_{\mathbb{C}}, \mathfrak{m}_{\mathbb{C}} \subset \mathbf{SC}$. The sources also

write $\mathcal{O}, \mathfrak{m}$ without subscripts.

- **Standard part.** The map st to $\mathbb{C}$ on finite elements, and the topology it induces, are called the *standard part* and the *standard-part topology*. The source manuscripts variously call this map or topology the reduction, the shadow and the residue; those three words are not used again. The genuinely different *fine* topology — generated by all balls of positive surreal radius — keeps its own name throughout.
- **Triangles.** Vertices A, B, C ; opposite sides $a = |B - C|$, $b = |C - A|$, $c = |A - B|$; interior angles $\alpha, \beta, \gamma \in (0, \pi)$; area Δ , circumradius R , inradius r , semiperimeter s . One source uses the letters A, B, C for both vertices and angles; we do not.
- **Angle defect and angular metric.** $d(\alpha) = \min\{\alpha, \pi - \alpha\}$ is the *defect* of an interior angle (a source writes e_A , another d_α). $\angle(u, v) \in [0, \pi]$ is the *angular metric* on $\mathbb{T}(\mathbf{No})$ (a source writes d_θ).
- **Small parameters.** h denotes a generic nonzero infinitesimal increment or angle; ε a generic positive infinitesimal; τ the ramification parameter of a cosine extremum or a near tangency, as in $\arccos(1 - \tau)$; δ the side gap $b + c - a$ of [theorem 8.4](#). The sources overload δ across all four roles.
- **Asymptotics.** For surreal or surcomplex a, b with $b \neq 0$, $a \prec b$ means $a/b \in \mathfrak{m}_{\mathbb{C}}$, and $a \sim b$ means $a/b - 1 \in \mathfrak{m}_{\mathbb{C}}$. These are exact algebraic relations, never limits along a sequence. For a fixed nonzero infinitesimal h , $O(h^m)$ denotes a remainder whose quotient by h^m is finite; it is an algebraic size statement, not a statement about an ordinary integer tending to infinity.
- **Rotation-group symbols.** Source 19 uses its own symbols for the objects above and for the new ones of [section 18](#); they are converted once, in [section 18.1](#), and not used elsewhere.
- **Period-arithmetic symbols.** Angles are radians everywhere, including [section 19](#). Sources 20 and 21 measure phases in turns; their objects are converted once, in [section 19.1](#), where periods divided by 2π form the normalized period group $\mathcal{I}$.
- **Ordinary finiteness.** Every integer n , polynomial degree N or n , Fourier grid size M , and polygon size in this article is an *ordinary* finite integer unless the contrary is said explicitly. “Finite coefficients” of a trigonometric polynomial means instead that each Laurent coefficient lies in $\mathcal{O}_{\mathbb{C}}$, i.e. is finite *in magnitude*; this second sense is used only in [theorem 9.7](#) and is flagged there.

1.3 Conventions: which coefficient ring each statement lives over

Several distinct rings of Hahn-analytic data circulate in this subject under nearly identical notation, and theorems about one of them are not theorems about another. Fix a set-sized divisible ordered additive subgroup $\Gamma \subset \mathbf{No}$ and write $t^\gamma = \omega^{-\gamma}$, so that $F_\Gamma = \mathbb{R}((t^\Gamma))$ is real closed and $K_\Gamma = \mathbb{C}((t^\Gamma))$ is algebraically closed [\[10, 7\]](#). In one variable z the four rings are:

- (R1) **the fixed-domain ring** $\mathcal{O}(J)((t^\Gamma))$. A datum is $F = \sum_{\gamma \in S} f_\gamma t^\gamma$ with $S \subset \Gamma$ one well-ordered support and *every* coefficient f_γ analytic on one fixed ordinary domain J . In the real case J is an ordinary real interval or neighbourhood; this is also the *uniform-support* ring used for integration along an ordinary parameter interval.
- (R2) **the common-domain germ ring.** The same, with J allowed to shrink: the coefficients are germs at a point admitting *some* common domain of definition.

(R3) the radius-free ring $\mathbb{C}\{z\}((t^\Gamma))$. Each f_γ is an individually convergent germ, with *no* common radius of convergence required.

(R4) the formal ring $\mathbb{C}[[z]]((t^\Gamma))$. Each f_γ is a formal power series.

The inclusions $(R1) \subset (R2) \subset (R3) \subset (R4)$ are all strict; the separating witnesses are not reproduced here, no statement below depending on them, and are proved in the companion report on infinitesimal analytic geometry. *Every coherence statement in this article names its ring.* Concretely: the obstruction theorems [theorem 17.7](#) and [theorem 17.8](#) are stated over $(R1)$ – $(R2)$ and over nothing else; the coefficientwise integral of [section 16](#) uses the uniform-support avatar of $(R1)$ along an ordinary real interval; and no theorem below is stated over the radius-free ring $(R3)$ or the formal ring $(R4)$, which appear here only to locate the two that are used. No statement is transferred from one of these rings to another anywhere in this article.

By contrast, the *scalar* Taylor lift of [equation \(2.5\)](#), which carries the bulk of the article, is not a statement about any of these rings: it evaluates one ordinary germ at one finite surreal point, and the only hypothesis is that the germ be defined near the standard part of that point.

1.4 The workspace, and what it does not authorize

Every normal-form support, coefficient family and recursion used here is a *set*; **No** is a proper class and a class theory such as NBG may be used. Any finite tuple of surcomplex numbers lies in some set-sized divisible workspace K_Γ , obtained as the divisible hull of the group generated by its normal-form supports, and any fixed calculation may be placed there. This enlargement rule never authorizes a proper-class sum or a proper-class polynomial ring, and no sum below has a proper-class index set. Workspace enlargement does not change any formula: the coefficients and strong sums have the same normal forms after embedding into a larger Hahn field, and the algebraic factorization and root statements are interpreted in the full field **SC**, so none of them is tied to a rank-one subfield.

2 Surreal scalars and support-controlled evaluation

2.1 Norm, valuation, and standard part

Conjugation, norm, inner product and oriented determinant are

$$\overline{x + iy} = x - iy, \quad |z| = \sqrt{z\bar{z}}, \quad (2.1)$$

$$\langle z, w \rangle = \operatorname{Re}(\bar{z}w), \quad [z, w] = \operatorname{Im}(\bar{z}w). \quad (2.2)$$

Expanding coordinates gives the Gram identity

$$\langle z, w \rangle^2 + [z, w]^2 = |z|^2 |w|^2, \quad (2.3)$$

whence $|zw| = |z||w|$, the Cauchy–Schwarz inequality $|\langle z, w \rangle| \leq |z||w|$, and the triangle inequality $|z+w| \leq |z|+|w|$. Equality in the last, for nonzero z, w , holds exactly when w/z is a positive element of **No**; squaring the equality gives $\langle z, w \rangle = |z||w|$, which selects the nonnegative proportionality factor. Here and below, “real” as a coordinate condition means belonging to **No**, while “ordinary real” means belonging to $\mathbb{R}$.

For a nonzero normal form $a = \sum_{\gamma \in S} a_\gamma t^\gamma$ with S well ordered, put

$$v(a) = \min S, \quad \operatorname{lc}(a) = a_{v(a)}, \quad v(0) = +\infty.$$

Then $v(ab) = v(a) + v(b)$, $v(a + b) \geq \min(v(a), v(b))$, and $v(|a|) = v(a)$. The value group is not assumed Archimedean; every comparison of valuations below uses only its ordered-group operations. Write

$$\mathcal{O}_{\mathbb{R}} = \{x : |x| < n \text{ for some } n \in \mathbb{N}_{>0}\}, \quad \mathfrak{m}_{\mathbb{R}} = \{x : |x| < 1/n \text{ for every } n \in \mathbb{N}_{>0}\},$$

and $\mathcal{O}_{\mathbb{C}} = \mathcal{O}_{\mathbb{R}} + i\mathcal{O}_{\mathbb{R}}$, $\mathfrak{m}_{\mathbb{C}} = \mathfrak{m}_{\mathbb{R}} + i\mathfrak{m}_{\mathbb{R}}$. Every finite element decomposes uniquely as

$$z = c + \eta, \quad c = \text{st}(z) \in \mathbb{C}, \quad \eta \in \mathfrak{m}_{\mathbb{C}}, \quad (2.4)$$

with $c \in \mathbb{R}$ for real elements. Standard part is a ring homomorphism on finite elements, and equals the coefficient at exponent zero. A finite element is *appreciable* when its standard part is nonzero; a finite nonzero element has valuation zero exactly when it is appreciable.

2.2 Strong summability and the support lemma

Definition 2.1 (Strong summability). A set-indexed Hahn family $(A_j)_{j \in J}$ is *strongly summable* when the union of its normal-form supports is well ordered *and* every exponent occurs in the supports of only finitely many members. Its sum is formed by finite addition at each exponent.

The two clauses are independent, and both are needed.

Lemma 2.2 (Hahn–Neumann support calculus). *If S, T are well-ordered subsets of an ordered abelian group, then $S + T$ is well ordered and each exponent has only finitely many representations $s + t$. If $T \subset \Gamma_{>0}$ is well ordered, the monoid*

$$T^* = \{\tau_1 + \cdots + \tau_k : k \geq 0, \tau_j \in T\}$$

is well ordered, and each exponent has only finitely many representations as a finite word in T , including only finitely many possible word lengths. In particular, for finitely many infinitesimals $\eta_1, \dots, \eta_d \in \mathfrak{m}_{\mathbb{C}}$ and arbitrary ordinary complex coefficients c_α , the family

$$\sum_{\alpha \in \mathbb{N}^d} c_\alpha \eta_1^{\alpha_1} \cdots \eta_d^{\alpha_d}$$

is strongly summable. Products and formal substitutions with zero constant term may be regrouped coefficientwise, and $S + T^$ is a support certificate for the result.*

Proof. The first assertion follows by extracting a nondecreasing subsequence in one coordinate: either a decreasing sequence of sums, or infinitely many representations of one sum, would force the other coordinate to decrease. The positive-monoid assertion is Neumann’s lemma [9]. One convenient proof uses the well-quasi-ordering of finite words in a well-ordered alphabet under subsequence embedding with coordinatewise increase: such an embedding cannot decrease a sum of positive letters, and equality forces identical letters and identical length. This excludes both descending sums and infinitely many distinct nondecreasing words of a fixed sum, and each word has only finitely many permutations. Applying this to the union of the positive supports of the η_j gives the last assertions. $\square$

Three negative companions to lemma 2.2 are load-bearing and are stated here once, because every proof below tacitly relies on them.

Remark 2.3 (What strong summability is not). (i) *A strong sum is not a valuation limit.* With $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$, the strong sum of t^n over $n \geq 0$ is $(1 - t)^{-1}$, but the tails never reach valuation ω . No argument of the form “the valuation improves by a fixed positive amount at each iteration” justifies any construction in this article.

(ii) *“Finitely many dependencies at a given exponent” is not “finitely many exponents below it”.* The latter already fails below ω in $\mathbb{Q} + \mathbb{Q}\omega$. The distinction is exactly what makes finite-parameter replacement in the proofs of [theorems 9.9](#) and [12.1](#) legitimate.

(iii) *Well-ordering cannot be weakened to positivity:* the set $\{1/n : n \geq 1\}$ is positive and not well ordered. The lemma applies even when the value group has arbitrary rank: with $0 < \lambda \ll \mu$ one may have $n\lambda < \mu$ for every ordinary n while $\sum_{n \geq 0} t^{n\lambda}$ remains a perfectly valid Hahn series whose partial sums converge in no relevant sense.

Proposition 2.4 (Degeneracy of fine convergence). *In the fine topology — generated by all balls $|z - a| < \rho$ with ρ a positive surreal — every set-sized subset of $\mathbf{SC}$ is closed and discrete. A net with a set-sized directed index set which converges in that topology is eventually equal to its limit. In particular a nonconstant ordinary real-parameter path cannot be fine-continuous on a connected real interval.*

Proof. For a set $A \subseteq \mathbf{SC}$ and a point p , the cut $\{0 \mid \{|p - a| : a \in A, a \neq p\}\}$ is a positive surreal smaller than every nonzero distance from p to A ; its ball meets A in at most $\{p\}$. This proves closedness and discreteness. For a convergent net, a positive lower bound for all its nonzero distances from the alleged limit shows that eventually entering the corresponding ball forces eventual equality. The image of an ordinary real interval is a set, hence discrete; a continuous image of a connected space is connected, hence a singleton. $\square$

Thus sequence limits are rejected as a definition device once and for all. For example the strong series $\sum_{n \geq 0} \omega^{-n}$ exists, while its finite partial sums do not converge to it. This proposition is used again in [proposition 16.2](#) and in [section 16](#), where it forces the arc-length functional to be *defined* coefficientwise rather than obtained as a limit of inscribed polygons.

2.3 Lifting ordinary analytic germs

For an ordinary real-analytic f near $r \in \mathbb{R}$ define

$$f^\sharp(r + \varepsilon) = \sum_{n \geq 0} \frac{f^{(n)}(r)}{n!} \varepsilon^n, \quad \varepsilon \in \mathfrak{m}_{\mathbb{R}}, \quad (2.5)$$

and identically for ordinary holomorphic germs and complex infinitesimals. The ordinary centre is fixed by the standard part, so the definition is unambiguous: *the ordinary value $f(r)$ is taken first in ordinary analysis, and only the infinitesimal correction is summed.* This is the restricted-analytic evaluation of [\[1, 7\]](#). Formal identities among germs lift to exact identities, because both sides have identical Taylor coefficients and [lemma 2.2](#) permits the substitutions.

Proposition 2.5 (Local calculus and one-variable sign lifting). *Taylor lifting preserves sums, products, composition where defined, and differentiation, and it is the unique extension specified by the Taylor rule [equation \(2.5\)](#). If an ordinary analytic g is nonnegative on an ordinary closed interval $[a, b]$ and analytic on a neighbourhood of it, then $g^\sharp(x) \geq 0$ for every surreal $x \in [a, b]$, including points infinitesimally close to an endpoint.*

Proof. The algebraic and compositional statements are formal Taylor identities justified by lemma 2.2. Re-expansion at x gives $f^\sharp(x+h) = f^\sharp(x) + (f')^\sharp(x)h + h^2 R_x(h)$ for infinitesimal h , with $R_x(h)$ finite on a sufficiently small neighbourhood, either from the convergent ordinary germ before substitution or directly from its Taylor coefficients and their standard parts. The difference quotient therefore tends to $(f')^\sharp(x)$ in the fine ε - δ sense: for any positive surreal tolerance, choose h smaller than both an infinitesimal radius and that tolerance divided by a fixed real bound for R_x .

For the sign assertion write $x = r + \varepsilon$ with $r \in [a, b]$. If $g(r) > 0$ the standard part decides. If $g(r) = 0$ and the germ is not zero, factor it as $(X - r)^m u(X)$ with $u(r) \neq 0$; at an interior zero nonnegativity forces even m and $u(r) > 0$, and at an endpoint the leading term has the correct sign on the side allowed by $x \in [a, b]$. Substitution preserves these signs; a zero germ causes no difficulty. $\square$

The uniqueness assertion is deliberately about the *Taylor rule*, not about arbitrary functions satisfying the same differential equation in the fine topology. Moreover, *no general transfer of arbitrary infinite families of inequalities is asserted*: in particular, coefficients that are themselves infinitesimal are not automatically covered by an ordinary real-coefficient sign argument. Example 13.6 makes that distinction concrete.

Lemma 2.6 (Leading-term test for a lifted germ). *Suppose f has an ordinary zero of order m at c , with $a = f^{(m)}(c)/m! \neq 0$. Then for every nonzero infinitesimal η ,*

$$f^\sharp(c + \eta) = a\eta^m(1 + \xi), \quad \xi \in \mathbf{m}_{\mathbf{SC}}.$$

Hence $v(f^\sharp(c + \eta)) = m v(\eta)$ and $\text{lc}(f^\sharp(c + \eta)) = \text{lc}(a\eta^m)$.

Proof. Factor the ordinary germ as $f(c + X) = aX^m(1 + Xg(X))$; the evaluated series $\eta g^\sharp(\eta)$ has strictly positive support by lemma 2.2. $\square$

The two mechanisms used for every inequality below are exactly these: a strict ordinary inequality with a nonzero ordinary margin survives by taking standard parts, and an ordinary equality case is resolved by the first nonzero Taylor coefficient.

Finally, a disambiguation that is needed in every later section: *a derivative in this article is the derivative of a function of a surreal or surcomplex variable*, with limits understood using all positive surreal error tolerances. It is not a derivation on **No** differentiating the monomials t^γ ; the Berarducci–Mantova differential-field structure [16] is important background but is not an input to any argument here.

3 Canonical sine and cosine on finite angles

3.1 Definition without a divergent global series

For $\theta = r + \varepsilon \in \mathcal{O}_{\mathbb{R}}$ with $r \in \mathbb{R}$ and $\varepsilon \in \mathbf{m}_{\mathbb{R}}$ define

$$\sin \theta = \sin r \sum_{n \geq 0} \frac{(-1)^n \varepsilon^{2n}}{(2n)!} + \cos r \sum_{n \geq 0} \frac{(-1)^n \varepsilon^{2n+1}}{(2n+1)!}, \quad (3.1)$$

$$\cos \theta = \cos r \sum_{n \geq 0} \frac{(-1)^n \varepsilon^{2n}}{(2n)!} - \sin r \sum_{n \geq 0} \frac{(-1)^n \varepsilon^{2n+1}}{(2n+1)!}. \quad (3.2)$$

The ordinary trigonometric functions on the right are evaluated *only* at the ordinary real number r . These are exactly the scalar lifts [equation \(2.5\)](#), and

$$\operatorname{cis} \theta = \cos \theta + i \sin \theta = e^{ir} \sum_{n \geq 0} \frac{(i\varepsilon)^n}{n!}. \quad (3.3)$$

Unless an extension is explicitly named, lowercase sine, cosine and tangent below have finite surreal real arguments.

Theorem 3.1 (Elementary trigonometry on finite surreal angles). *For $\alpha, \beta \in \mathcal{O}_{\mathbb{R}}$,*

$$\cos^2 \alpha + \sin^2 \alpha = 1, \quad (3.4)$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta, \quad (3.5)$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta, \quad (3.6)$$

$$\operatorname{cis}(\alpha + \beta) = \operatorname{cis} \alpha \operatorname{cis} \beta. \quad (3.7)$$

Sine is odd, cosine even, both have period 2π , and $(\sin)' = \cos$, $(\cos)' = -\sin$, $(\operatorname{cis})' = i \operatorname{cis}$. For every ordinary integer n ,

$$(\operatorname{cis} \alpha)^n = \operatorname{cis}(n\alpha). \quad (3.8)$$

In particular the usual double-angle, half-angle, sum-to-product and product-to-sum identities hold, with every quotient restricted to a nonzero denominator.

Proof. Write $\alpha = r + \varepsilon$, $\beta = s + \eta$. The ordinary addition identities at r, s and the formal identities for the power series in ε, η give [equation \(3.7\)](#); all reorganizations are justified by [lemma 2.2](#). Conjugating [equation \(3.3\)](#) gives $\operatorname{cis}(-\alpha) = \overline{\operatorname{cis} \alpha}$, so the modulus is one. The remaining identities follow by substitution, conjugation, differentiation and ordinary finite induction; negative integer powers use $\overline{\operatorname{cis} \alpha} = (\operatorname{cis} \alpha)^{-1}$. $\square$

For reference, several consequences are

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}, \quad \sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}, \quad (3.9)$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}, \quad 2 \sin \alpha \sin \beta = \cos(\alpha - \beta) - \cos(\alpha + \beta), \quad (3.10)$$

$$\cos(n\alpha) = T_n(\cos \alpha), \quad \sin(n\alpha) = \sin \alpha U_{n-1}(\cos \alpha) \quad (n \geq 1), \quad (3.11)$$

with $T_0 = 1$, $T_1 = X$, $T_{n+1} = 2XT_n - T_{n-1}$ and $U_0 = 1$, $U_1 = 2X$, $U_{n+1} = 2XU_n - U_{n-1}$. The last two follow from the same recurrence, so no new summability issue enters; the index n is an ordinary finite integer. The conventional forms are recorded in [\[11, §4.21\]](#); the point here is their well-defined evaluation and exact validity in a non-Archimedean field.

3.2 Order, monotonicity, and exact leading orders

Proposition 3.2 (Signs and monotonicity). *Sine is positive on $(0, \pi)$ and negative on $(\pi, 2\pi)$; cosine is strictly decreasing on $[0, \pi]$, and sine is strictly increasing on $[-\pi/2, \pi/2]$. Jordan's inequality*

$$\frac{2x}{\pi} \leq \sin x \leq x \quad (0 \leq x \leq \pi/2) \quad (3.12)$$

holds, and $|\sin x| \leq |x|$ for every finite x . Moreover $0 < \sin x < x < \tan x$ for finite $0 < x < \pi/2$.

Proof. An ordinary standard part strictly between 0 and π decides the positive sign; near 0, $\sin \varepsilon = \varepsilon(1 + \text{infinitesimal})$, and near π , $\sin(\pi - \varepsilon) = \sin \varepsilon$. These cover the remaining points of $(0, \pi)$, and translation by π gives the other sign. For $0 \leq a < b \leq \pi$ the identity $\cos a - \cos b = 2 \sin \frac{a+b}{2} \sin \frac{b-a}{2} > 0$ proves strict decrease, and the analogous sine difference uses a positive cosine of the midpoint on $[-\pi/2, \pi/2]$. The ordinary inequalities in [equation \(3.12\)](#) lift by [proposition 2.5](#); for $|x| \geq 1$ use $|\sin x| \leq 1$. For the last chain, standard parts in $(0, \pi/2)$ have ordinary margins; at positive infinitesimal x the leading terms $x - \sin x = x^3/6 + \dots$ and $\tan x - x = x^3/3 + \dots$ are positive by [lemma 2.6](#); and infinitesimally below $\pi/2$ the tangent is positive infinite. $\square$

This proof does not infer global monotonicity from the sign of a fine derivative. Such an inference would require additional hypotheses on a general surreal function.

Proposition 3.3 (Exact infinitesimal leading orders). *For $0 \neq h \in \mathfrak{m}_{\mathbb{R}}$,*

$$\sin h = h - \frac{h^3}{6} + \frac{h^5}{120} + O(h^7), \quad 1 - \cos h = \frac{h^2}{2} - \frac{h^4}{24} + O(h^6), \quad (3.13)$$

$$\tan h = h + \frac{h^3}{3} + \frac{2h^5}{15} + O(h^7), \quad (3.14)$$

and, exactly,

$$v(\sin h) = v(\tan h) = v(h), \quad v(1 - \cos h) = 2v(h), \quad (3.15)$$

with $\sin h \sim h$, $\tan h \sim h$ and $1 - \cos h \sim h^2/2$.

Proof. The truncations come from the strongly summable formal series. Dividing the three expressions by h, h^2, h gives finite elements with nonzero standard parts $1, 1/2, 1$, and such units have valuation zero. $\square$

These asymptotic equivalences are statements about *a single arbitrary infinitesimal*. They are strictly stronger than standard-part equalities and do not mean that a sequence of real arguments converges in the fine topology.

3.3 What normalizes the finite pair

Proposition 3.4 (Normalization of the finite trigonometric pair). *Suppose two functions S, C on $\mathcal{O}_{\mathbb{R}}$ agree with the ordinary sine and cosine at every ordinary real point, and satisfy, for every $r \in \mathbb{R}$ and every $\varepsilon \in \mathfrak{m}_{\mathbb{R}}$, the Taylor identities [equation \(3.1\)](#)–[equation \(3.2\)](#) with the ordinary derivatives at r . Then $S = \sin$ and $C = \cos$ on $\mathcal{O}_{\mathbb{R}}$. Equivalently, the prescribed ordinary germs together with their strong evaluations determine the pair uniquely.*

Proof. Every $x \in \mathcal{O}_{\mathbb{R}}$ has exactly one decomposition $r + \varepsilon$, so the hypothesis reproduces [equation \(3.1\)](#) and [equation \(3.2\)](#) verbatim. $\square$

This normalization is deliberately stronger than a differential equation with one initial value. Fine-local analyticity alone permits unrelated germs on disjoint monads [1], and [theorem 17.4](#) will show that even a global addition law does not determine the infinite-argument values.

4 The full unit circle and the group of angles

4.1 Infinitesimal exponential and logarithm

For $h, \eta \in \mathfrak{m}_{\mathbb{C}}$ put

$$\text{Exp}_{\mathfrak{m}}(h) = \sum_{n \geq 0} \frac{h^n}{n!}, \quad \text{Log}_{\mathfrak{m}}(1 + \eta) = \sum_{n \geq 1} \frac{(-1)^{n+1} \eta^n}{n}. \quad (4.1)$$

Formal power-series identities and [lemma 2.2](#) prove that these are mutually inverse bijections between $\mathfrak{m}_{\mathbb{C}}$ and $1 + \mathfrak{m}_{\mathbb{C}}$, that $\text{Exp}_{\mathfrak{m}}(h + k) = \text{Exp}_{\mathfrak{m}}(h) \text{Exp}_{\mathfrak{m}}(k)$, and that the logarithm turns products into sums; both commute with conjugation. In particular

$$v(\text{Log}_{\mathfrak{m}}(1 + \eta)) = v(\eta) \quad (\eta \neq 0), \quad \text{cis } h = \text{Exp}_{\mathfrak{m}}(ih) \quad (h \in \mathfrak{m}_{\mathbb{R}}). \quad (4.2)$$

Define

$$\mathbb{T}(\mathbf{No}) = \{u \in \mathbf{SC} : u\bar{u} = 1\},$$

whose elements all have finite coordinates even though $\mathbb{T}(\mathbf{No})$ is a proper class, and let $\mathbb{T}(\mathbb{R})$ be the ordinary complex unit circle. For nonzero z, w the *oriented angle* from z to w is, before any radian parameter, the group element

$$u(z, w) = \frac{\bar{z}w}{|z||w|} \in \mathbb{T}(\mathbf{No}), \quad (4.3)$$

whose real and imaginary coordinates are its cosine and sine: $\cos(z, w) = \langle z, w \rangle / (|z||w|)$ and $\sin(z, w) = [z, w] / (|z||w|)$. Changing either vector's length by a positive surreal factor does not change its angle. The *unoriented* angle is the unique $\alpha \in [0, \pi]$ with $\cos \alpha = \langle z, w \rangle / (|z||w|)$ and $\sin \alpha = |[z, w]| / (|z||w|)$; by [equation \(2.3\)](#) this is meaningful even when one vector has infinitesimal length and the other infinite length.

Proposition 4.1 (Algebraic angle addition and rotations). *Angle composition is multiplication: $u(z, w)u(w, q) = u(z, q)$, and if the angle coordinates are (c, s) and (d, t) the composite has coordinates $(cd - st, sd + ct)$. The map $u = c + is \mapsto \begin{pmatrix} c & -s \\ s & c \end{pmatrix}$ identifies $\mathbb{T}(\mathbf{No})$ with $\text{SO}(2, \mathbf{No})$. Multiplication by an element of $\mathbb{T}(\mathbf{No})$ preserves lengths, dot products, determinants and oriented angles; every orientation-preserving $\mathbf{No}$ -linear isometry of $\mathbf{No}^2$ is of this form; every orientation-preserving affine similarity is $z \mapsto a + \lambda uz$ with $a \in \mathbf{SC}$, $\lambda \in \mathbf{No}_{>0}$, $u \in \mathbb{T}(\mathbf{No})$, and the orientation-reversing ones are obtained by composing with conjugation. Finally every nonzero $z \in \mathbf{SC}$ is uniquely $z = \rho u$ with $\rho = |z| \in \mathbf{No}_{>0}$ and $u \in \mathbb{T}(\mathbf{No})$, and multiplication gives a group isomorphism $\mathbf{No}_{>0} \times \mathbb{T}(\mathbf{No}) \rightarrow \mathbf{SC}^{\times}$.*

Proof. The angle identity follows by cancellation, and multiplying $c + is$ by $d + it$ gives the displayed coordinates. The indicated matrix preserves the dot product and has determinant $c^2 + s^2 = 1$; conversely the first column of an orthogonal determinant-one matrix is some (c, s) of norm one, and orthogonality fixes the second column as $(-s, c)$. For a unit q one has $\bar{q}zqw = \bar{z}w$, which gives the preservation statements. Dividing a similarity's linear part by its positive scale factor reduces it to the isometry case. For the polar statement take $\rho = |z|$ and $u = z/|z|$; taking norms proves uniqueness and multiplicativity. $\square$

Thus addition, double-angle and finite multiple-angle identities already have an algebraic meaning over any real closed field. The linearity hypothesis on isometries can be dropped ([theorem 18.2](#), from source 19, which also proves the matrix identification). Surreal-specific analysis enters only when these unit directions are identified with numerical radians, which is the content of the next theorem.

4.2 The finite phase theorem

Theorem 4.2 (Polar decomposition and the angle group). *The map $\text{cis} : (\mathcal{O}_{\mathbb{R}}, +) \rightarrow (\mathbb{T}(\mathbf{No}), \cdot)$ is surjective with kernel exactly $2\pi\mathbb{Z}$, so that*

$$\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z} \cong \mathbb{T}(\mathbf{No}), \quad \text{and} \quad \mathbf{SC}^{\times} \cong \mathbf{No}_{>0} \times (\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z}). \quad (4.4)$$

More precisely, there is a canonical group isomorphism

$$\mathbb{T}(\mathbf{No}) \longrightarrow \mathbb{T}(\mathbb{R}) \times (\mathfrak{m}_{\mathbb{R}}, +), \quad u \longmapsto (u_0, -i \text{Log}_{\mathfrak{m}}(u/u_0)), \quad u_0 = \text{st}(u), \quad (4.5)$$

with inverse $(u_0, \delta) \mapsto u_0 \text{Exp}_{\mathfrak{m}}(i\delta)$. Consequently every $z \neq 0$ has a unique modulus $\rho = |z| > 0$ and a unique angle class $[\theta] \in \mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z}$ with $z = \rho \text{cis } \theta$; the modulus may be infinitesimal, finite or infinite, while the angle is always finite.

Proof. Taking standard parts in $u\bar{u} = 1$ gives $u_0 \in \mathbb{T}(\mathbb{R})$. Put $q = u/u_0 \in 1 + \mathfrak{m}_{\mathbb{C}}$; since $\bar{q} = q^{-1}$,

$$\overline{\text{Log}_{\mathfrak{m}} q} = \text{Log}_{\mathfrak{m}} \bar{q} = \text{Log}_{\mathfrak{m}}(q^{-1}) = -\text{Log}_{\mathfrak{m}} q,$$

so $\text{Log}_{\mathfrak{m}} q = i\delta$ for a unique $\delta \in \mathfrak{m}_{\mathbb{R}}$. Choosing any ordinary argument θ_0 of u_0 gives $u = u_0 \text{Exp}_{\mathfrak{m}}(i\delta) = \text{cis}(\theta_0 + \delta)$, proving surjectivity. The standard-part map and the logarithmic group law prove that equation (4.5) is a homomorphism with the stated inverse. If $\text{cis}(r + \varepsilon) = 1$, the standard part gives $r \in 2\pi\mathbb{Z}$, hence $\text{Exp}_{\mathfrak{m}}(i\varepsilon) = 1$, and logarithmic inversion gives $\varepsilon = 0$. Applying the circle result to $u = z/|z|$ and proposition 4.1 gives the polar statement. $\square$

This theorem is also the structure theorem of source 19 for $\text{SO}(2, \mathbf{No})$, proved there by the same argument; section 18 develops its group-theoretic consequences. The infinitesimal component in equation (4.5) is independent of an ordinary argument branch. A numerical representative of an angle class, however, requires a cut.

Corollary 4.3 (Representatives, roots, and torsion). *Every direction has exactly one representative in $[0, 2\pi)$ and exactly one in $(-\pi, \pi]$, obtained by adding or subtracting an ordinary multiple of 2π and then checking the actual surreal inequalities; near the endpoints the infinitesimal correction matters, so that the representative of $-\varepsilon$ with $\varepsilon > 0$ infinitesimal is $2\pi - \varepsilon$ and not an ordinary angle. The upper open semicircle corresponds exactly to $(0, \pi)$. For $u = \text{cis } \theta$ and an ordinary $n \geq 1$, the equation $v^n = u$ has exactly the n distinct solutions $\text{cis}((\theta + 2\pi k)/n)$, $0 \leq k < n$; in particular the n th roots of unity in $\mathbf{SC}$ are exactly the ordinary complex roots of unity, and $\mathbb{T}(\mathbf{No})$ has no nontrivial infinitesimal torsion. For $w = \rho \text{cis } \theta \neq 0$ the n th roots are $\rho^{1/n} \text{cis}((\theta + 2\pi k)/n)$, and for $n \geq 3$ a regular n -gon of radius $\rho \in \mathbf{No}_{>0}$ has side length $2\rho \sin(\pi/n)$.*

Proof. A finite number has an ordinary real standard part, so subtracting an ordinary multiple of 2π , with at most one endpoint adjustment, lands in either half-open interval; the kernel in theorem 4.2 gives uniqueness. On $(0, \pi)$ sine is positive by proposition 3.2, including the infinitesimal endpoint neighbourhoods, and parity treats the lower semicircle. The listed roots are distinct by the kernel and exhaust the roots of $X^n - u$ by degree. In the splitting equation (4.5), $n\delta = 0$ forces $\delta = 0$. The chord formula follows from equation (2.3) or the cosine double-angle identity. $\square$

Source 19 proves the root count by a second route through the splitting alone (remark 18.4), and adds the classification of finite subgroups (corollary 18.5).

Remark 4.4 (Two branch conventions, kept apart). A representative constrained to have *actual value* in $(-\pi, \pi]$ has a branch jump at the negative real axis, including across infinitesimal perturbations: an ordinary principal argument of $\text{st}(u)$ plus its infinitesimal correction need not already lie in that interval — it may be $\pi + \varepsilon$ with $\varepsilon > 0$ — and the representative then subtracts 2π . One may instead fix an argument β of the *standard* direction and retain its infinitesimal correction without reducing again,

$$\arg_{\beta} z = \beta - i \text{Log}_{\mathfrak{m}} \left(\frac{z/|z|}{\text{st}(z/|z|)} \right), \quad (4.6)$$

which is fine-locally analytic but may lie infinitesimally outside $(-\pi, \pi]$. The global logarithm of [1] uses the second, standard-part-pinned convention; the geometric representative convention of [corollary 4.3](#) is the first. They give the same direction and must not be identified as the same single-valued argument function. *Angle classes avoid the ambiguity entirely*, and are what all geometric theorems below use. When h/z is infinitesimal the unwrapped branch gives the local differential

$$\arg_{\beta}(z + h) - \arg_{\beta} z = \text{Im} \text{Log}_{\mathfrak{m}}(1 + h/z), \quad (4.7)$$

so that $d \arg z = \text{Im}(dz/z)$ and $d \log |z| = \text{Re}(dz/z)$.

4.3 Rational coordinates and angle division

Theorem 4.5 (Projective half-angle parametrization of every surreal direction). *The map*

$$C(t) = \frac{1+it}{1-it} = \frac{1-t^2}{1+t^2} + i \frac{2t}{1+t^2}, \quad t \in \mathbf{No}, \quad C(\infty) = -1, \quad (4.8)$$

is a bijection $\mathbb{P}^1(\mathbf{No}) \rightarrow \mathbb{T}(\mathbf{No})$; here ∞ is the projective point, not an infinitely large surreal. Restricted to $\mathbf{No}$ it is a bijection onto $\mathbb{T}(\mathbf{No}) \setminus \{-1\}$, with inverse $t = \text{Im } u / (1 + \text{Re } u)$. If $u = \text{cis } \theta \neq -1$ then $t = \tan(\theta/2)$ for the representative $-\pi < \theta < \pi$. In homogeneous coordinates $t = [p : q]$, multiplication of directions is

$$[p : q] \oplus [r : s] = [ps + qr : qs - pr], \quad (4.9)$$

which in affine coordinates is $(t + u)/(1 - tu)$ whenever the denominator is nonzero. Infinite t are allowed and describe directions infinitesimally close to -1 .

Proof. The displayed rational expression has norm one and cannot equal -1 , because $1 + \text{Re } C(t) = 2/(1 + t^2) > 0$. Conversely if $u = c + is$ has $c^2 + s^2 = 1$ and $c \neq -1$ then $1 + c > 0$ and substituting $t = s/(1 + c)$ recovers $c + is$; the only omitted direction is -1 , supplied by the projective point. The product formula follows by multiplying $(q + ip)/(q - ip)$ and $(s + ir)/(s - ir)$, and its two homogeneous outputs cannot both vanish because $(ps + qr)^2 + (qs - pr)^2 = (p^2 + q^2)(r^2 + s^2) > 0$. The half-angle identity is [theorem 3.1](#) on the chosen interval; the exceptional angle is handled projectively. $\square$

The homogeneous rule resolves the cases involving $t = \infty$: in particular $\infty \oplus \infty = 0$, corresponding to two half-turns, and one must not evaluate the affine formula by informal arithmetic with a symbol ∞ . This rational coordinate is valid for infinite t : $t = \omega$ gives a direction infinitesimally close to -1 , not an unspecified direction associated with ω radians. Algebraic half-slope coordinates have published predecessors in vector and rational trigonometry [13, 14]; their use here avoids unnecessary transcendental evaluation, and [theorem 13.4](#) and [theorem 9.7](#) both run through this chart.

5 Inverse functions and the angular metric

5.1 Inverse trigonometry on its full natural surreal domains

Theorem 5.1 (Inverse tangent, sine, and cosine). *There are unique inverse functions*

$$\arctan : \mathbf{No} \rightarrow (-\pi/2, \pi/2), \quad \arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2], \quad \arccos : [-1, 1] \rightarrow [0, \pi],$$

with the usual inverse identities; all values are finite surreal angles, and inverse tangent is an increasing bijection defined at every surreal slope, infinite ones included. They satisfy

$$\cos(\arctan t) = \frac{1}{\sqrt{1+t^2}}, \quad \sin(\arctan t) = \frac{t}{\sqrt{1+t^2}}, \quad (5.1)$$

$$\arcsin x = 2 \arctan \frac{x}{1 + \sqrt{1-x^2}}, \quad \arccos t = \frac{\pi}{2} - \arcsin t, \quad (5.2)$$

and at every point of their open domains — including infinite t for inverse tangent and points infinitesimally close to ± 1 for the other two —

$$(\arctan)'(t) = \frac{1}{1+t^2}, \quad (\arcsin)'(t) = \frac{1}{\sqrt{1-t^2}}, \quad (\arccos)'(t) = -\frac{1}{\sqrt{1-t^2}}. \quad (5.3)$$

Proof. The unit direction $(1+it)/\sqrt{1+t^2}$ has positive real part, so [theorem 4.2](#) and the sign rules of [proposition 3.2](#) give a unique angle in $(-\pi/2, \pi/2)$; this defines inverse tangent and proves its two coordinate formulas. If $x < y$ the determinant of $(1, x)$ and $(1, y)$ is $y - x > 0$, so the angle difference has positive sine and, both angles lying in an interval of length π , is positive: inverse tangent is increasing. Similarly the directions $\sqrt{1-t^2} + it$ and $t + i\sqrt{1-t^2}$ give inverse sine and inverse cosine on their stated ranges, whose uniqueness follows from strict monotonicity. The half-angle identity proves the inverse-sine formula, whose denominator is positive even at $x = \pm 1$.

For the derivatives: locally, square roots of nonzero positive elements have their binomial expansions after division by the base value, and a local argument difference is the imaginary part of the infinitesimal logarithm of a direction ratio ([equation \(4.7\)](#)). These give fine-local analytic expansions at every point under discussion, and differentiating the inverse identities gives [equation \(5.3\)](#); the relevant cosine or sine is nonzero. At an infinite inverse-tangent input the same computation is valid after taking increments small relative to $1+it$, and no finite bound on t is used. $\square$

For positive infinite t , hence infinitesimal $1/t$,

$$\arctan t = \frac{\pi}{2} - \arctan(1/t) = \frac{\pi}{2} - \frac{1}{t} + \frac{1}{3t^3} - \frac{1}{5t^5} + O(t^{-7}), \quad (5.4)$$

and for negative t one uses oddness; an infinite slope corresponds to a finite angle infinitesimally short of a right angle. For example

$$\text{Arg}(1+i\omega) = \frac{\pi}{2} - \omega^{-1} + \frac{\omega^{-3}}{3} - \frac{\omega^{-5}}{5} + \cdots,$$

a finite value describing an infinite slope without evaluating sine or cosine at an infinite angular argument. For infinitesimal h the strong identities

$$\arctan h = h - \frac{h^3}{3} + \frac{h^5}{5} - \cdots, \quad \arcsin h = h + \frac{h^3}{6} + \frac{3h^5}{40} + \cdots \quad (5.5)$$

hold, with $(\tan x)' = 1 + \tan^2 x$ wherever cosine is nonzero.

Remark 5.2 (An endpoint at which no surreal derivative exists). The endpoint derivative of inverse sine does not exist as a surreal-valued derivative: its difference quotient grows beyond every fixed surreal bound as the increment becomes sufficiently small. Allowing infinitely large elements as candidate derivative values does not repair this, because there is always a still smaller increment. This is a genuine failure, not an artefact of the choice of tolerance class, and it is the analytic shadow of the square-root ramification of [proposition 5.3](#).

Proposition 5.3 (Endpoint ramification of inverse cosine). *For $0 < \tau \in \mathfrak{m}_{\mathbb{R}}$,*

$$\begin{aligned} \arccos(1 - \tau) &= 2 \arcsin \sqrt{\tau/2} \\ &= \sqrt{2\tau} \sum_{n \geq 0} \frac{\binom{2n}{n}}{8^n (2n+1)} \tau^n \\ &= \sqrt{2\tau} \left(1 + \frac{\tau}{12} + \frac{3\tau^2}{160} + \frac{5\tau^3}{896} + O(\tau^4) \right), \end{aligned} \quad (5.6)$$

the series being strongly summable after adjoining $\sqrt{\tau}$, and

$$v(\arccos(1 - \tau)) = \frac{1}{2}v(\tau). \quad (5.7)$$

Proof. The double-angle identity gives $\cos(2 \arcsin \sqrt{\tau/2}) = 1 - \tau$; the angle is a positive infinitesimal, so inverse uniqueness gives the first equality. Substituting the ordinary formal inverse-sine series gives the rest, every remaining parenthesized factor having standard part one. $\square$

Inverse sine and inverse tangent are valuation-preserving at zero, whereas inverse cosine at 1 is a square-root branch. This single distinction governs every degeneration result in [section 8](#) and [section 9](#).

5.2 Chord distance and infinitesimal angular accuracy

For $u, v \in \mathbb{T}(\mathbf{No})$ let $\angle(u, v) \in [0, \pi]$ be the least absolute representative of their angle difference, equivalently $\angle(u, v) = \arccos \operatorname{Re}(\bar{u}v)$. This is a metric: choose shortest representatives for two successive differences, add them and reduce modulo 2π ; reduction cannot increase the least absolute value, which is the triangle inequality.

Theorem 5.4 (Angular and chordal comparison). *For $u, v \in \mathbb{T}(\mathbf{No})$, with $d = \angle(u, v)$,*

$$|u - v| = 2 \sin(d/2), \quad |u - v| \leq d \leq \frac{\pi}{2} |u - v|, \quad (5.8)$$

and for $u \neq v$,

$$v(\angle(u, v)) = v(u - v) \quad \text{exactly.} \quad (5.9)$$

If $\operatorname{st}(u) = \operatorname{st}(v)$, their unique infinitesimal oriented angle difference is

$$\delta = -i \operatorname{Log}_{\mathfrak{m}}(v/u), \quad v(\delta) = v(v - u), \quad \angle(u, v) = |\delta|. \quad (5.10)$$

Moreover, for infinitesimal d ,

$$|u - v| = d \left(1 - \frac{d^2}{24} + \frac{d^4}{1920} - \cdots \right), \quad 1 - \cos d = \frac{d^2}{2} \left(1 - \frac{d^2}{12} + \frac{d^4}{360} - \cdots \right). \quad (5.11)$$

Proof. Expanding $|u - v|^2 = 2 - 2\operatorname{Re}(\bar{u}v)$ and using the half-angle identity proves the first equality in [equation \(5.8\)](#); Jordan's inequality [equation \(3.12\)](#) proves both bounds. Bounding a positive quantity above and below by nonzero ordinary real multiples of another forces [equation \(5.9\)](#). Finally $v/u \in 1 + \mathfrak{m}_{\mathbb{C}}$, and [equation \(4.2\)](#) gives [equation \(5.10\)](#). The series come from substitution into the sine and cosine Taylor series, the parenthesized factors having standard part one. $\square$

Thus *angular precision is exactly the precision of normalized directions*, not merely comparable to it up to an unspecified non-Archimedean error. Two consequences deserve to be isolated.

Corollary 5.5 (Local valuation isometry of phase and rotation displacement). *If θ, ϕ are finite real angles with $0 \neq \theta - \phi \in \mathfrak{m}_{\mathbb{R}}$ then*

$$|\operatorname{cis} \theta - \operatorname{cis} \phi| \sim |\theta - \phi|, \quad v(\operatorname{cis} \theta - \operatorname{cis} \phi) = v(\theta - \phi), \quad \frac{\operatorname{cis} \theta - \operatorname{cis} \phi}{i \operatorname{cis} \phi (\theta - \phi)} = 1 + \text{an infinitesimal.} \quad (5.12)$$

For $z \neq 0$ and a nonzero infinitesimal rotation angle δ ,

$$|\operatorname{cis} \delta z - z| = 2|z| |\sin(\delta/2)| \sim |z| |\delta|. \quad (5.13)$$

Proof. Factor $\operatorname{cis} \theta - \operatorname{cis} \phi = \operatorname{cis} \phi (\operatorname{Exp}_{\mathfrak{m}}(i(\theta - \phi)) - 1)$; the first nonzero term of the last factor is $i(\theta - \phi)$, with infinitesimal relative remainder. [Equation \(5.13\)](#) is [equation \(5.8\)](#) followed by [proposition 3.3](#). $\square$

A rotation that is infinitesimal *as an angle* need not be infinitesimal *as a displacement*: rotating a vector of length ω by ω^{-1} moves its endpoint by a distance relatively equivalent to 1, and replacing ω by ω^2 makes that displacement infinite.

Corollary 5.6 (Stability of direction under a relative perturbation). *If $z \neq 0$ and $h/z = \eta \in \mathfrak{m}_{\mathbb{C}}$, then the local angle difference between $z + h$ and z is*

$$\Delta\theta = \operatorname{Im} \operatorname{Log}_{\mathfrak{m}}(1 + \eta) = \operatorname{Im} \eta - \frac{1}{2} \operatorname{Im}(\eta^2) + O(|\eta|^3).$$

In particular $v(\Delta\theta) \geq v(\eta)$ when the difference is nonzero, with equality when the leading coefficient of η has nonzero imaginary part.

Proof. The real part of the logarithm accounts for the change in modulus, its imaginary part for the change in direction; taking imaginary parts cannot lower a Hahn valuation, and a first coefficient with nonzero imaginary part cannot cancel against higher powers. $\square$

Although cosine loses first-order information near zero — $v(1 - \cos d) = 2v(d)$ — the full unit direction does not, because its sine coordinate retains the angular scale. At other angles one should use both coordinates, or a nonsingular tangent chart, rather than infer angular error solely from cosine error.

6 Triangles at arbitrary surreal scales

6.1 Angles and the exact angle sum

Let $A, B, C \in \mathbf{SC}$ be noncollinear, with side lengths $a = |B - C|$, $b = |C - A|$, $c = |A - B|$ and positive area $\Delta = \frac{1}{2}|[B - A, C - A]|$. All four quantities are positive surreals; *no finiteness*

assumption is made. Define the interior angle α at A by

$$\cos \alpha = \frac{\langle B - A, C - A \rangle}{bc}, \quad \sin \alpha = \frac{2\Delta}{bc}, \quad 0 < \alpha < \pi, \quad (6.1)$$

and β, γ cyclically. Equation (2.3) places the right-hand sides on the upper unit semicircle, so theorem 4.2 gives existence and uniqueness.

Theorem 6.1 (Euclidean angle sum over the surreal field). *Every noncollinear surcomplex triangle satisfies $\alpha + \beta + \gamma = \pi$.*

Proof. A translation, rotation and possibly a conjugation put $A = 0$, $B = c > 0$, $C = x + iy$ with $y > 0$. Let $\alpha = \text{Arg } C \in (0, \pi)$ and $\varphi = \text{Arg}(C - c) \in (0, \pi)$. Since $\text{Im}(\bar{C}(C - c)) = cy > 0$ we get $\sin(\varphi - \alpha) > 0$; the difference lies in $(-\pi, \pi)$, so $\varphi > \alpha$. The angle at B is $\pi - \varphi$ and the angle at C is $\varphi - \alpha$, both in $(0, \pi)$, and their sum with α is π . $\square$

The proof uses order, angle classes and determinants only. It applies equally to an infinitesimal triangle, an infinite triangle, or a triangle whose sides lie on three different scales, and the conclusion is an equality of finite surreals including their infinitesimal parts, not merely of standard parts. An equivalent argument multiplies the three interior-angle unit elements,

$$\frac{C - A}{B - A} \cdot \frac{A - B}{C - B} \cdot \frac{B - C}{A - C} = -1,$$

so that $\text{cis}(\alpha + \beta + \gamma) = -1$ with the sum in $(0, 3\pi)$; the kernel of theorem 4.2 then forces π . Neither argument uses compactness, a topological hypothesis, or an Archimedean parallel-postulate step.

6.2 Cosine, sine and tangent laws

Theorem 6.2 (Triangle laws). *For every noncollinear surcomplex triangle,*

$$a^2 = b^2 + c^2 - 2bc \cos \alpha, \quad (6.2)$$

$$\Delta = \frac{1}{2}bc \sin \alpha = \frac{1}{2}ca \sin \beta = \frac{1}{2}ab \sin \gamma, \quad (6.3)$$

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma} = \frac{abc}{2\Delta} = 2R, \quad (6.4)$$

$$\frac{a - b}{a + b} = \frac{\tan((\alpha - \beta)/2)}{\tan((\alpha + \beta)/2)}, \quad (6.5)$$

together with the cyclic versions of the cosine law. Here R is the radius of the unique circumcircle, and $R = abc/(4\Delta)$. All denominators are nonzero, and the sides may be infinite, infinitesimal, or of mutually different scales, as may the area and the circumradius. Those scales are not independent of one another: theorem 8.1 relates them, and in particular the least of $v(a), v(b), v(c)$ is always attained at least twice.

Proof. Expanding $|(C - A) - (B - A)|^2$ gives the cosine law, and equation (6.1) with its cyclic versions gives the area identity and the common sine-law value $abc/(2\Delta)$. For the circumcircle set $A = 0$, $B = c$, $C = x + iy$ with $y > 0$; a centre $O = X + iY$ equidistant from the three vertices satisfies $2cX = c^2$ and $2xX + 2yY = x^2 + y^2 = b^2$, two independent linear equations with the unique solution $X = c/2$, $Y = (b^2 - cx)/(2y)$. A direct expansion using $a^2 = (x - c)^2 + y^2$ gives $|O|^2 = a^2b^2/(4y^2)$, so $R = ab/(2y) = abc/(4\Delta)$; existence is thereby proved without a compactness

argument. Finally the sine law converts $(a - b)/(a + b)$ into $(\sin \alpha - \sin \beta)/(\sin \alpha + \sin \beta)$, and the sum-to-product identities give the tangent law; since $0 < \alpha + \beta < \pi$ and $|\alpha - \beta| < \pi$, every cancellation and tangent is legitimate. $\square$

Corollary 6.3 (Right triangles and the full range of slopes). *A triangle is right at C exactly when $c^2 = a^2 + b^2$. In a right triangle with positive legs p, q , the acute angle opposite q is $\arctan(q/p)$, and q/p may be any positive surreal, including an infinite one.*

Proof. The cosine law and the unique zero of cosine on $(0, \pi)$ give the first assertion; the second is the coordinate formula of [theorem 5.1](#). $\square$

6.3 Reconstruction from side or angle data

Theorem 6.4 (Existence, congruence, and similarity). *Positive surreals a, b, c are the side lengths of a noncollinear surcomplex triangle if and only if all three strict triangle inequalities $a < b + c$, $b < c + a$, $c < a + b$ hold; the triangle is then unique up to a Euclidean isometry, the two orientations being interchanged by reflection, and its angles are recovered by the cosine law and arccos. Conversely, positive finite α, β, γ with $\alpha + \beta + \gamma = \pi$ and an arbitrary positive scale $k \in \mathbf{No}$ determine a triangle with*

$$a = k \sin \alpha, \quad b = k \sin \beta, \quad c = k \sin \gamma,$$

unique up to similarity; specifying one positive side fixes k . Triangles with the same three angles have proportional corresponding sides.

Proof. Necessity is the norm triangle inequality with equality excluded by noncollinearity. For sufficiency place $A = 0$, $B = c$ and set

$$x = \frac{b^2 + c^2 - a^2}{2c}, \quad y = \sqrt{b^2 - x^2}.$$

The Heron factorization

$$4c^2(b^2 - x^2) = (a + b + c)(-a + b + c)(a - b + c)(a + b - c) \quad (6.6)$$

shows $y > 0$, so $C = x + iy$ has the required distances; the same equations force x and $|y|$, giving uniqueness. The cosine-law quotient lies strictly between -1 and 1 , so [theorem 5.1](#) recovers each interior angle.

For the converse, note that for $\alpha, \beta > 0$ with $\alpha + \beta < \pi$,

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} > 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha + \beta}{2} = \sin(\alpha + \beta),$$

the strict comparison following from cosine monotonicity on $[0, \pi]$ and $|\alpha - \beta| < \alpha + \beta$. Since $\sin \gamma = \sin(\alpha + \beta)$, the proposed sides satisfy the strict triangle inequalities cyclically, so a triangle exists. The identity $\sin^2 \beta + \sin^2 \gamma - \sin^2 \alpha = 2 \sin \beta \sin \gamma \cos \alpha$, which follows from $\alpha = \pi - \beta - \gamma$ and the addition formulas, shows via the cosine law that the constructed angle at A is exactly α , cosine being injective on $[0, \pi]$; similarly at the other vertices. Changing k is a similarity, and the sine law gives proportional sides for equal angles. $\square$

This is the usual side–side–side construction without any Archimedean assumption; side–angle–side is obtained from $C = b \operatorname{cis} \alpha$ with $0 < \alpha < \pi$, and the sine law gives angle–side–angle. *The usual side–side–angle ambiguity remains:* a ray–circle intersection is a quadratic problem and may have zero, one or two admissible solutions, and no additional infinite family of solutions appears merely because the coordinates are surreal. This last point is made quantitative by [theorem 9.1](#).

6.4 Area, half-angles, and the two radii

Put $s = (a + b + c)/2$; the strict triangle inequalities make $s - a$, $s - b$, $s - c$ positive.

Theorem 6.5 (Heron, half-angle, incircle and bisector formulas). *One has*

$$\Delta^2 = s(s - a)(s - b)(s - c), \quad r = \frac{\Delta}{s}, \quad (6.7)$$

$$\sin \frac{\alpha}{2} = \sqrt{\frac{(s - b)(s - c)}{bc}}, \quad \cos \frac{\alpha}{2} = \sqrt{\frac{s(s - a)}{bc}}, \quad (6.8)$$

$$\tan \frac{\alpha}{2} = \sqrt{\frac{(s - b)(s - c)}{s(s - a)}} = \frac{r}{s - a}, \quad (6.9)$$

all square roots being the positive ones. The incentre is

$$I = \frac{aA + bB + cC}{2s}, \quad (6.10)$$

and the internal bisector from A meets BC in the ratio $c : b$ and has length

$$\ell_A = \frac{2bc}{b + c} \cos \frac{\alpha}{2}. \quad (6.11)$$

Proof. With the coordinates of theorem 6.4, $\Delta = cy/2$, so equation (6.6) proves Heron's formula. Substituting the cosine law into $\sin^2(\alpha/2) = (1 - \cos \alpha)/2$ and $\cos^2(\alpha/2) = (1 + \cos \alpha)/2$ gives the half-angle squares; both half-angle functions are positive, which fixes the roots. Their quotient together with Heron's identity gives equation (6.9).

The point equation (6.10) has positive barycentric weights, hence lies strictly inside; its distance to BC , computed as an absolute determinant divided by a , is $\frac{a}{2s} \cdot \frac{2\Delta}{a} = \Delta/s$, and the other two distances agree cyclically. Thus it is the incentre with $r = \Delta/s$. For the bisector, put $A = 0$, $B = c$, $C = b \operatorname{cis} \alpha$; then

$$D = \frac{bB + cC}{b + c} = \frac{bc}{b + c}(1 + \operatorname{cis} \alpha) = \frac{2bc}{b + c} \cos(\alpha/2) \operatorname{cis}(\alpha/2)$$

lies in the interior of BC , has $|B - D|/|D - C| = c/b$, and its direction from A bisects the angle; its modulus is equation (6.11). Projection of an internal-bisector point onto either ray lies at positive distance along that ray, which places each contact point of the incircle on its own side segment. $\square$

Theorem 6.6 (Euler's incentre-circumcentre identity). *If O is the circumcentre and I the incentre then*

$$|O - I|^2 = R(R - 2r). \quad (6.12)$$

Consequently $R \geq 2r$, with equality exactly for an equilateral triangle.

Proof. Translate O to zero, so $|A| = |B| = |C| = R$. For arbitrary positive weights w_j and vectors Z_j , expansion of the inner products gives the weighted variance identity

$$\left| \frac{\sum_j w_j Z_j}{\sum_j w_j} \right|^2 = \frac{\sum_j w_j |Z_j|^2}{\sum_j w_j} - \frac{\sum_{j < k} w_j w_k |Z_j - Z_k|^2}{(\sum_j w_j)^2}.$$

Apply it with $(w_1, w_2, w_3) = (a, b, c)$; the numerator of the second term is $abc^2 + acb^2 + bca^2 = 2sabc$, so $|I|^2 = R^2 - abc/(2s) = R^2 - 2Rr$. Positivity gives $R \geq 2r$. If equality holds then $I = O$, so every side has the same positive distance r from the centre and therefore length $2\sqrt{R^2 - r^2}$; the triangle is equilateral, and the converse is immediate. $\square$

No limiting or Archimedean argument is hidden in any of these identities.

Corollary 6.7 (A scale-independent area inequality). *Every nondegenerate surcomplex triangle satisfies $\Delta \leq s^2/(3\sqrt{3})$, with equality exactly for an equilateral triangle.*

Proof. The positive quantities $s - a, s - b, s - c$ sum to s . The arithmetic–geometric mean inequality for three terms, proved algebraically over any real closed field, bounds their product by $(s/3)^3$; Heron’s formula converts this into the stated bound, and equality forces the three quantities to coincide. $\square$

7 Bisectors, concurrency, chords, and cyclic quadrilaterals

7.1 The cevian sine ratio and trigonometric Ceva

Theorem 7.1 (Sine ratios for a cevian). *Let D lie strictly inside the side BC of a noncollinear triangle. Then*

$$\frac{|B - D|}{|D - C|} = \frac{c \sin \angle BAD}{b \sin \angle DAC}. \quad (7.1)$$

In particular the internal bisector of α meets BC at the unique point with $|B - D|/|D - C| = c/b$.

Proof. The triangles ABD and ADC share an altitude to the line BC , so their areas are in the ratio $|B - D|/|D - C|$; computing the same areas from the common side AD and the included angles at A gives equation (7.1). For the bisector, equal sines of equal positive half-angles cancel. Uniqueness follows from the strict increase of $\sin x / \sin(\alpha - x)$ on $(0, \alpha)$, which follows in turn from the sine subtraction identity, whose numerator is $\sin \alpha \sin(y - x) > 0$ for $x < y$. $\square$

Theorem 7.2 (Trigonometric Ceva). *Let D, E, F lie strictly inside BC, CA, AB respectively. The lines AD, BE, CF are concurrent if and only if*

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} = 1. \quad (7.2)$$

Proof. All six split angles lie in $(0, \pi)$, so their sines are positive. For an interior point with positive barycentric coordinates (p_A, p_B, p_C) , its trace on BC has ratio $|B - D|/|D - C| = p_C/p_B$, the two cyclic ratios being p_A/p_C and p_B/p_A , so their product is one; conversely three prescribed positive ratios with product one determine such positive coordinates, hence a common point. This proves the ordinary ratio form of Ceva by field algebra alone, over **No**, with no topological hypothesis. Substituting equation (7.1) and its cyclic versions, the side factors $c/b, a/c, b/a$ cancel and give equation (7.2). $\square$

The restriction to interior traces avoids directed-length conventions. *A directed Menelaus or Ceva formula can be obtained in the same way after fixing signs, but is not needed for the present development.* Even when a cevian divides a side at an infinitesimal ratio, the sine-ratio criterion remains exact.

7.2 Chords, inscribed angles, and Ptolemy

A circle of centre $O \in \mathbf{SC}$ and radius $R \in \mathbf{No}_{>0}$ is *exactly* $\{O + R \operatorname{cis} \theta : \theta \in \mathcal{O}_{\mathbb{R}}\}$: by [theorem 4.2](#) the finite parameter θ already reaches all of its points, including directions differing by arbitrarily small infinitesimals. A circle of enormous radius therefore does *not* require the function $w \mapsto \operatorname{Sin}(Rw)$ to possess one Hahn support over an ordinary angular domain — a point made sharp in [theorem 17.7](#). If $z_j = O + R \operatorname{cis} \theta_j$, the addition identities give the exact factorization

$$z_2 - z_1 = 2iR \operatorname{cis} \frac{\theta_1 + \theta_2}{2} \sin \frac{\theta_2 - \theta_1}{2}, \quad |z_2 - z_1| = 2R \left| \sin \frac{\theta_2 - \theta_1}{2} \right|, \quad (7.3)$$

at every surreal radius.

Proposition 7.3 (Inscribed-angle theorem). *For three distinct points z_1, z_2, z_3 on a circle, the directed angle from $z_1 - z_3$ to $z_2 - z_3$, taken modulo π , is half the directed central angle from $z_1 - O$ to $z_2 - O$, halving sending a central angle modulo 2π to an angle modulo π . In particular points on a common arc subtend a fixed chord under equal interior angles, and an angle subtending a diameter is right.*

Proof. Apply [equation \(7.3\)](#) to both vectors based at z_3 : their quotient has phase $(\theta_2 - \theta_1)/2$ times a nonzero real scalar, and a real sign affects an argument only by π . Choosing the actual interior-angle ranges gives the stated consequences. The argument works equally for infinitesimally separated circle points. $\square$

Theorem 7.4 (Ptolemy inequality and its cyclic equality case). *For arbitrary four surcomplex points,*

$$|z_1 - z_3||z_2 - z_4| \leq |z_1 - z_2||z_3 - z_4| + |z_1 - z_4||z_2 - z_3|, \quad (7.4)$$

and if the four points are distinct and cyclically ordered on a circle, equality holds.

Proof. The algebraic identity

$$(z_1 - z_3)(z_2 - z_4) = (z_1 - z_2)(z_3 - z_4) + (z_1 - z_4)(z_2 - z_3)$$

and the norm triangle inequality prove [equation \(7.4\)](#). For cyclic points choose finite lifts $\theta_1 < \theta_2 < \theta_3 < \theta_4 < \theta_1 + 2\pi$. By [equation \(7.3\)](#) the two products on the right have the same unit phase and strictly positive real multiples of it, since all the half-difference sines involved have the prescribed sign; equality therefore holds in the norm triangle inequality. Equivalently, with $u = (\theta_2 - \theta_1)/2$, $v = (\theta_3 - \theta_2)/2$, $w = (\theta_4 - \theta_3)/2$, all positive with $u + v + w < \pi$, the equality reduces to the addition-formula identity $\sin(u + v) \sin(v + w) = \sin u \sin w + \sin v \sin(u + v + w)$, whose two sides differ by $\sin u \sin w (\cos^2 v + \sin^2 v - 1) = 0$. This includes the case in which some consecutive arcs are infinitesimal. $\square$

Here *cyclic order* is the order induced by finite representatives around the circle; it is *not* defined by fine-continuous motion of a real parameter, which [proposition 2.4](#) has already shown to be degenerate. This illustrates a general principle: algebraic identities among finitely many vectors often survive unchanged, while any analytic interpretation of the associated angles must still be supplied.

Corollary 7.5 (Finite regular polygons at surreal scale). *For an ordinary integer $n \geq 3$ the regular n -gon with vertices $O + R \operatorname{cis}(2\pi k/n)$ has side $2R \sin(\pi/n)$, perimeter $2nR \sin(\pi/n)$ and area $\frac{n}{2} R^2 \sin(2\pi/n)$, for every positive surreal radius R .*

Proof. The chord formula [equation \(7.3\)](#) gives the side; sum the finitely many equal sides, and the finitely many positively oriented centre triangles, each of determinant area $R^2 \sin(2\pi/n)/2$. $\square$

There is no literal polygon with “ ω vertices” defined by this finite construction, and [proposition 16.2](#) shows that the sequence of these perimeters must not be used to *define* the circumference.

7.3 Quadrance and spread

The *quadrance* of a side is its squared length and the *spread* of two lines is the square of the sine of either angle between them; these are the standard algebraic invariants of rational trigonometry [13, 14]. *They forget orientation and distinguish neither an angle from its supplement nor a directed angle from its negative. That loss is useful for rational formulas but must be remembered.* Writing $Q_a = a^2$ and $\sigma_\alpha = \sin^2 \alpha$ cyclically, squaring the sine and cosine laws gives

$$\frac{\sigma_\alpha}{Q_a} = \frac{\sigma_\beta}{Q_b} = \frac{\sigma_\gamma}{Q_c}, \quad (Q_b + Q_c - Q_a)^2 = 4Q_bQ_c(1 - \sigma_\alpha), \quad (7.5)$$

and the triple-spread relation is

$$(\sigma_\alpha + \sigma_\beta + \sigma_\gamma)^2 = 2(\sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\gamma^2) + 4\sigma_\alpha\sigma_\beta\sigma_\gamma. \quad (7.6)$$

To prove the latter, use $\sin \gamma = \sin(\alpha + \beta)$ to obtain $\sigma_\gamma - \sigma_\alpha - \sigma_\beta + 2\sigma_\alpha\sigma_\beta = 2\sin \alpha \sin \beta \cos \alpha \cos \beta$, square both sides, replace each squared cosine by one minus the corresponding spread, and expand. These formulas require no transcendental operation once the coordinates are known and remain meaningful at every surreal scale.

8 Valuation geometry and extremely thin triangles

8.1 An exact dictionary between sides, angles and area

For an interior angle $\alpha \in (0, \pi)$ its *defect* is $d(\alpha) = \min\{\alpha, \pi - \alpha\} \in (0, \pi/2]$, measuring proximity to either a zero angle or a straight angle. The defect rather than α itself is the right variable: an angle infinitesimally below π has infinitesimal sine but is not infinitesimal as a number.

Theorem 8.1 (Valuation form of the triangle laws). *For a noncollinear surcomplex triangle,*

$$v(\sin \alpha) = v(d(\alpha)), \quad v(a) - v(b) = v(d(\alpha)) - v(d(\beta)), \quad (8.1)$$

$$v(\Delta) = v(b) + v(c) + v(d(\alpha)), \quad 2v(\Delta) = v(s) + v(s-a) + v(s-b) + v(s-c), \quad (8.2)$$

$$v(R) = v(a) + v(b) + v(c) - v(\Delta), \quad v(r) = v(\Delta) - v(s), \quad (8.3)$$

together with the cyclic versions; equivalently $v(a) - v(d(\alpha)) = v(b) - v(d(\beta)) = v(c) - v(d(\gamma)) = v(2R)$. The minimum of $v(a), v(b), v(c)$ is attained at least twice. If α and β are both infinitesimal then $\alpha/\beta \sim a/b$.

Proof. Jordan’s inequality compares $\sin d(\alpha)$ with $d(\alpha)$ by nonzero ordinary constants, so their valuations agree; since $\sin \alpha = \sin d(\alpha)$, the first assertion follows. If $d(\alpha)$ is appreciable both quantities have positive ordinary standard part and valuation zero. Applying v to the sine law, the area formula, Heron’s identity and the radius formulas gives the rest, nonzero ordinary constants having valuation zero. If, say, $v(a) < v(b), v(c)$, then b/a and c/a are positive infinitesimals whose

sum is less than one, contradicting $a < b + c$; so no side can be the unique dominant valuation scale. Finally, when the two angles are infinitesimal, dividing the exact sine law by $\sin \alpha \sim \alpha$ and $\sin \beta \sim \beta$ gives the last relation. $\square$

The last statement is a non-Archimedean form of the triangle inequality. *It does not say that all three sides have equal valuation: a very short third side is possible.* Nor need one of $\sin \alpha, \sin \beta, \sin \gamma$ be appreciable: in a nearly flat triangle one angle is infinitesimally close to π and the other two are infinitesimal, so all three sines are infinitesimal.

Choosing a longest side and scaling it to length one, the remaining vertex is $x + iy$ with $y > 0$ and both coordinates finite, $0 < x < 1$; the normalized standard-part shadow is noncollinear exactly when y is appreciable. If y is infinitesimal the shadow loses information that remains visible in [equation \(8.3\)](#). Two further theorems quantify what it loses.

8.2 The triangle-inequality defect

Theorem 8.2 (A quadratic defect formula). *Let $u, v \neq 0$ with $a = |u|$, $b = |v|$, and let their unoriented angle be a nonzero infinitesimal δ . Set $D = a + b - |u + v|$. Then, exactly,*

$$D = \frac{4ab \sin^2(\delta/2)}{a + b + |u + v|}, \quad (8.4)$$

and

$$D \sim \frac{ab}{2(a + b)} \delta^2, \quad v(D) = v(a) + v(b) + 2v(\delta) - v(a + b). \quad (8.5)$$

No comparability assumption on a and b is required.

Proof. Squaring and subtracting, $(a + b)^2 - |u + v|^2 = 2ab(1 - \cos \delta) = 4ab \sin^2(\delta/2)$, and dividing by the positive sum of the square roots gives [equation \(8.4\)](#). Also

$$\frac{|u + v|^2}{(a + b)^2} = 1 - \frac{4ab}{(a + b)^2} \sin^2(\delta/2),$$

and since $0 < 4ab/(a + b)^2 \leq 1$ the subtracted term is infinitesimal, so the positive square root is 1 plus an infinitesimal. Hence the denominator in [equation \(8.4\)](#) is relatively equivalent to $2(a + b)$, while $4 \sin^2(\delta/2) \sim \delta^2$. $\square$

Unlike a bound that replaces $a + b$ by an ordinary real constant, this expression *detects the effective smaller scale* when a and b differ by many orders of magnitude: the factor $ab/(a + b)$ is comparable to the smaller of a and b .

8.3 Flat triangles: the normalized form and the relative threshold

The next two theorems are genuinely different statements and both are kept. The first normalizes coordinates and reads off every leading constant; the second imposes no normalization and isolates the correct *relative* flatness threshold.

Theorem 8.3 (Quadratic slack and reciprocal circumradius in normalized coordinates). *Let $A = 0$, $B = 1$, $C = x + iy$ where x is finite with $0 < \text{st}(x) < 1$ and $0 < y \in \mathfrak{m}_{\mathbb{R}}$, and put $a = \sqrt{(1 - x)^2 + y^2}$, $b = \sqrt{x^2 + y^2}$, $c = 1$. Then, exactly,*

$$\alpha = \arctan(y/x), \quad \beta = \arctan(y/(1 - x)), \quad \gamma = \pi - \alpha - \beta, \quad (8.6)$$

and

$$\alpha = \frac{y}{x} - \frac{y^3}{3x^3} + O(y^5), \quad \beta = \frac{y}{1-x} - \frac{y^3}{3(1-x)^3} + O(y^5), \quad (8.7)$$

$$a + b - 1 = \frac{y^2}{2x(1-x)} - \frac{y^4}{8} \left(\frac{1}{x^3} + \frac{1}{(1-x)^3} \right) + O(y^6), \quad (8.8)$$

$$R = \frac{x(1-x)}{2y} \left[1 + \frac{y^2}{2} \left(\frac{1}{x^2} + \frac{1}{(1-x)^2} \right) + O(y^4) \right], \quad r = \frac{y}{2} \left[1 - \frac{y^2}{4x(1-x)} + O(y^4) \right]. \quad (8.9)$$

Consequently

$$v(\alpha) = v(\beta) = v(\pi - \gamma) = v(y), \quad v(a + b - 1) = 2v(y), \quad v(R) = -v(y), \quad v(r) = v(y). \quad (8.10)$$

Proof. Both x and $1 - x$ are positive appreciable, so the base angles are acute and their slopes give [equation \(8.6\)](#); the inverse-tangent series [equation \(5.5\)](#) gives their expansions. Expanding $b = x\sqrt{1 + (y/x)^2}$ and $a = (1 - x)\sqrt{1 + (y/(1 - x))^2}$ by the binomial series — legitimate at these infinitesimal arguments by [lemma 2.2](#) — gives [equation \(8.8\)](#). Since $\Delta = y/2$, the exact radius formulas are $R = ab/(2y)$ and $r = y/(a + b + 1)$, and multiplication and geometric inversion yield the displayed expansions. Each indicated leading coefficient is positive appreciable, including the sum of the two leading angle coefficients, so taking valuations gives [equation \(8.10\)](#). $\square$

The theorem quantifies a phenomenon invisible in the ordinary standard-part shadow: the two small angles are first order in the height, the excess in the triangle inequality is second order, and the circumradius is of inverse order.

Theorem 8.4 (Exact gap identity and the relative flatness threshold). *Let a be a largest side of a nondegenerate triangle and put $B = b + c$, $\delta = B - a > 0$ and $h = \pi - \alpha$. Then $0 < \delta \leq \min(b, c)$ and*

$$4bc \sin^2(h/2) = \delta(2B - \delta), \quad \text{equivalently} \quad h = 2 \arcsin \sqrt{q}, \quad q = \frac{\delta(2B - \delta)}{4bc} \in (0, 1). \quad (8.11)$$

The defect h is infinitesimal if and only if

$$\delta \prec \frac{bc}{B}. \quad (8.12)$$

Under that criterion,

$$h = 2\sqrt{q} \left(1 + \frac{q}{6} + \frac{3q^2}{40} + \frac{5q^3}{112} + \cdots \right), \quad (8.13)$$

$$h \sim \sqrt{\frac{2B\delta}{bc}}, \quad \Delta \sim \sqrt{\frac{bcB\delta}{2}}, \quad R \sim \sqrt{\frac{Bbc}{8\delta}}, \quad (8.14)$$

and

$$\begin{aligned} v(h) &= \frac{1}{2}(v(\delta) + v(B) - v(b) - v(c)), \\ v(\Delta) &= \frac{1}{2}(v(\delta) + v(B) + v(b) + v(c)), \\ v(R) &= \frac{1}{2}(v(B) + v(b) + v(c) - v(\delta)). \end{aligned} \quad (8.15)$$

Proof. Since $a \geq \max(b, c)$, subtraction gives $\delta \leq \min(b, c)$. The cosine law with $\alpha = \pi - h$ gives $B^2 - a^2 = 2bc(1 - \cos h) = 4bc \sin^2(h/2)$, and $a = B - \delta$ turns this into [equation \(8.11\)](#); the half-angle formula $\sin^2(h/2) = \cos^2(\alpha/2) = s(s-a)/(bc)$ of [theorem 6.5](#) identifies the same quantity with q , and $h/2 \in (0, \pi/2)$ gives $h = 2 \arcsin \sqrt{q}$. The factor $2 - \delta/B$ lies between $3/2$ and 2 , because $\delta \leq \min(b, c) \leq B/2$; hence $\sin^2(h/2)$ is infinitesimal exactly when $\delta B/(bc)$ is, and since $0 < h/2 < \pi/2$ its sine is infinitesimal exactly when the angle is. This is the criterion.

Under the criterion δ/B is infinitesimal, because $\delta/B = (\delta B/(bc)) \cdot (bc/B^2)$ with $0 < bc/B^2 \leq 1/4$. The inverse-sine series gives [equation \(8.13\)](#). From [equation \(8.11\)](#) and $\sin(h/2) \sim h/2$ we get $bc h^2 \sim 2B\delta$, and positive square roots give the first equivalence in [equation \(8.14\)](#). The area law gives $\Delta = \frac{bc}{2} \sin h \sim \frac{bc}{2} h$, proving the second; and $a \sim B$ with $R = a/(2 \sin h) \sim B/(2h)$ gives the third. Taking valuations, and noting that $2B - \delta$ and B have the same valuation, proves [equation \(8.15\)](#). $\square$

The denominator bc/B is essential, and the hypothesis is deliberately *relative* rather than an absolute smallness assumption: since $bc/(b+c)$ is comparable to the smaller of b and c , the gap must be small relative to the *shorter adjacent side*, not merely relative to the total length. A defect that is small compared with the perimeter need not make the opposite angle close to π when one adjacent side is much shorter than the other. The square root of the side gap controls the angular defect while its reciprocal controls the circumradius; this is the same ramification as in [proposition 5.3](#).

Example 8.5 (A symmetric infinitesimal-height triangle). Take $x = 1/2$, $y = \varepsilon > 0$ infinitesimal in [theorem 8.3](#). Then

$$a = b = \frac{1}{2} \sqrt{1 + 4\varepsilon^2}, \quad \alpha = \beta = \arctan(2\varepsilon), \quad R = \frac{1}{8\varepsilon} + \frac{\varepsilon}{2},$$

$$r = \frac{\varepsilon}{1 + \sqrt{1 + 4\varepsilon^2}} = \frac{\varepsilon}{2} - \frac{\varepsilon^3}{2} + \varepsilon^5 + O(\varepsilon^7).$$

The circumradius is infinite although all vertices lie within a bounded ordinary region. Its existence is not contradicted by the fact that the standard-part points are collinear.

Example 8.6 (A symmetric flattening, seen through the gap). Let $0 < \tau \in \mathfrak{m}_{\mathbb{R}}$, $b = c = 1$, $a = 2 - \tau$, so $\delta = \tau$ and $B = 2$ in [theorem 8.4](#). Exactly,

$$H = \sqrt{\tau - \tau^2/4}, \quad \Delta = (1 - \tau/2)H, \quad R = \frac{1}{2H}, \quad h = 2 \arccos(1 - \tau/2), \quad \beta = \gamma = h/2,$$

with the strongly summable expansions

$$h = 2\sqrt{\tau} \left(1 + \frac{\tau}{24} + \frac{3\tau^2}{640} + \frac{5\tau^3}{7168} + \cdots \right),$$

$$H = \sqrt{\tau} \left(1 - \frac{\tau}{8} - \frac{\tau^2}{128} - \frac{\tau^3}{1024} + \cdots \right),$$

$$R = \frac{1}{2\sqrt{\tau}} \left(1 + \frac{\tau}{8} + \frac{3\tau^2}{128} + \frac{5\tau^3}{1024} + \cdots \right).$$

Here the side lengths have ordinary scale, the altitude and the area have square-root infinitesimal scale, and the circumradius is infinite.

Example 8.7 (An appreciable side gap can still be a flat degeneration). Take $b = c = \omega^2$ and $\delta = 1$, so $a = 2\omega^2 - 1$. The criterion [equation \(8.12\)](#) holds, since $bc/B = \omega^2/2$. Then $h \sim 2/\omega$, $\Delta \sim \omega^3$ and $R \sim \omega^3/2$: the triangle is angularly nearly flat although its area is infinite and its gap is the ordinary number 1. This is precisely why degeneration hypotheses must be stated relatively.

Example 8.8 (An infinite triangle of ordinary finite area). Let $L > 0$ be infinite and take $A = 0$, $B = L$, $C = L + i/L$. The legs at B are L and L^{-1} , and

$$\Delta = \frac{1}{2}, \quad \alpha = \arctan(L^{-2}), \quad \beta = \frac{\pi}{2}, \quad \gamma = \frac{\pi}{2} - \alpha,$$

with circumradius $\frac{1}{2}\sqrt{L^2 + L^{-2}}$ and inradius

$$r = \frac{L + L^{-1} - \sqrt{L^2 + L^{-2}}}{2} = \frac{1}{2L} - \frac{1}{4L^3} + O(L^{-7}).$$

The special case $L = \omega$ is the right triangle with vertices $0, \omega, i/\omega$, of legs ω and ω^{-1} , area exactly $1/2$, hypotenuse $\omega\sqrt{1 + \omega^{-4}}$, nonright angles $\arctan(\omega^{-2})$ and $\pi/2 - \arctan(\omega^{-2})$, and circumradius $\sim \omega/2$. Thus an infinite side, an infinitesimal side, an ordinary area and an infinitesimal angle coexist without disturbing a single triangle identity; standard parts of the normalized small angle alone would erase information essential to both the height and the area.

Example 8.9 (Two thin triangles on different scale hierarchies). Take $A = 0$, $B = L$, $C = x + ih$ with $0 < x < L$ and $0 < h < \min\{x, L - x\}$. Then, exactly, $\alpha = \arctan(h/x)$, $\beta = \arctan(h/(L - x))$, $\gamma = \pi - \alpha - \beta$, $\Delta = Lh/2$, and

$$R = \frac{\sqrt{x^2 + h^2}\sqrt{(L - x)^2 + h^2}}{2h} \sim \frac{x(L - x)}{2h}, \quad r \sim h/2.$$

For $L = \omega$, $x = \omega/2$, $h = 1$, both base angles equal $\arctan(2/\omega) = 2/\omega - 8/(3\omega^3) + \dots$, while $R \sim \omega^2/8$, $r \sim 1/2$ and $\Delta = \omega/2$: infinite area, infinite circumradius, finite inradius, infinitesimal base angles. For $L = 1$, $x = \varepsilon$, $h = \varepsilon^2$ with $\varepsilon > 0$ infinitesimal, the two base angles sit on *different* scales, $\alpha \sim \varepsilon$ and $\beta \sim \varepsilon^2$, with $R \sim 1/(2\varepsilon)$ and $r \sim \varepsilon^2/2$. Both families obey exactly the same identities as ordinary triangles, but their scale information cannot be represented by ordinary positive real lengths and angles alone.

9 Trigonometric intersections, contact, and conditioning

This section specializes the finite-intersection viewpoint of [2, 3] to angular equations. Rather than invoke their general division and residue theorems, every instance is computed directly; the geometric theory above is therefore independent of them.

9.1 A complete first-harmonic intersection theorem

Theorem 9.1 (Amplitude–phase reduction and intersection count). *Let $A, B, D \in \mathbf{No}$ with $(A, B) \neq (0, 0)$ and $\rho = \sqrt{A^2 + B^2} > 0$, and choose the finite angle class ϕ with $\cos \phi = A/\rho$ and $\sin \phi = B/\rho$. Then*

$$A \cos \theta + B \sin \theta = \rho \cos(\theta - \phi), \tag{9.1}$$

and the equation $A \cos \theta + B \sin \theta = D$ has, modulo 2π in finite angles, no solution if $|D| > \rho$, exactly one if $|D| = \rho$, and exactly two if $|D| < \rho$, namely $\theta = \phi \pm \arccos(D/\rho)$. The single solution

at $|D| = \rho$ has multiplicity two: the first angular derivative vanishes there and the second equals $\mp\rho \neq 0$. Writing $\mathcal{D} = \rho^2 - D^2$, the two unit points at $\mathcal{D} > 0$ are

$$(X, Y) = \frac{D}{\rho^2}(A, B) \pm \frac{\sqrt{\mathcal{D}}}{\rho^2}(-B, A), \quad (9.2)$$

the absolute angular derivative of the left-hand side at either simple solution is $\sqrt{\mathcal{D}}$, and the Jacobian determinant of $X^2 + Y^2 - 1$ and $AX + BY - D$ has absolute value $2\sqrt{\mathcal{D}}$.

Proof. Polar decomposition (theorem 4.2) gives ϕ , and the cosine subtraction identity proves equation (9.1). The range, strict monotonicity and inverse cosine on $[0, \pi]$ (proposition 3.2, theorem 5.1) give the count, including both endpoints, where the two signs coincide modulo 2π . For the explicit points use the orthonormal coordinates $U = (AX + BY)/\rho$ and $W = (-BX + AY)/\rho$: the equations become $U = D/\rho$ and $W^2 = 1 - D^2/\rho^2 = \mathcal{D}/\rho^2$. The angular derivative is $-A \sin \theta + B \cos \theta = BX - AY = -\rho W$, of absolute value $\sqrt{\mathcal{D}}$, and the stated Jacobian expresses the same transversality. $\square$

Geometrically this is the complete intersection classification of a line $AX + BY = D$ with the unit circle. Neither an infinite normal vector nor an infinitesimal distance from tangency creates an exception. If $A = B = 0$ the equation instead holds for every angle when $D = 0$ and for none otherwise. This also settles the side–side–angle ambiguity left open after theorem 6.4: the ray–circle problem is exactly this quadratic.

9.2 Square-root splitting and its conditioning

Theorem 9.2 (Tangency, angular splitting, and loss of valuation). *In theorem 9.1 suppose $D = \rho(1 - \tau)$ with $0 < \tau \in \mathfrak{m}_{\mathbb{R}}$. The two solutions near ϕ are*

$$\theta_{\pm} = \phi \pm \sqrt{2\tau} \left(1 + \frac{\tau}{12} + \frac{3\tau^2}{160} + O(\tau^3) \right), \quad (9.3)$$

their separation has valuation $\frac{1}{2}v(\tau)$, and the derivatives of the left-hand side at these roots have absolute value $\rho\sqrt{\tau(2-\tau)}$. If $e \neq 0$ and $e \prec \tau$, then with $q(\tau) = \arccos(1 - \tau)$ one has $\tau + e > 0$ and

$$q(\tau + e) - q(\tau) = \frac{e}{\sqrt{\tau(2-\tau)}}(1 + O(e/\tau)), \quad (9.4)$$

the correction factor having standard part one; in particular

$$v(q(\tau + e) - q(\tau)) = v(e) - \frac{1}{2}v(\tau). \quad (9.5)$$

Proof. Equation (9.3) is proposition 5.3, and the separation is $2q(\tau)$, whose valuation is unchanged by the ordinary factor two. Differentiating equation (9.1) gives magnitude $\rho \sin q(\tau)$ and $\sin q(\tau) = \sqrt{1 - (1 - \tau)^2} = \sqrt{\tau(2 - \tau)}$. For the perturbation formula write $e = \tau u$ with u infinitesimal; the exact expression $q(\tau) = \sqrt{2\tau}H(\tau)$ from equation (5.6), with $H(0) = 1$, may be expanded in the two infinitesimals τ, u . Its first derivative in e is $1/\sqrt{\tau(2-\tau)}$, and every later term contains, relative to that first term, a factor $u = e/\tau$ times a finite factor; the substitution is support-controlled by lemma 2.2. The first term being nonzero and the parenthesized multiplier a unit of standard part one gives equation (9.5). $\square$

The loss of one half of $v(\tau)$ is the inverse-Jacobian loss at a nearly double root. *The condition $e \prec \tau$ cannot be omitted from a uniform branch-matching theorem: $e = -\tau$ merges the roots, and $e < -\tau$ destroys the two real intersections.*

Example 9.3 (An infinite tangent coordinate). The line $X = -1 + \tau$ meets the unit circle near -1 in the two points $Y = \pm\sqrt{\tau(2-\tau)}$, whose half-angle coordinates from [equation \(4.8\)](#) are $t = \pm\sqrt{(2-\tau)/\tau}$, and these are *infinite* even though the points and their angles are finite. Rational circle coordinates and finite angles therefore complement each other rather than having identical size behaviour.

Theorem 9.4 (The cosine fold). *Let $\tau \in \mathbf{ms}_{\mathbf{C}}$. If $\tau \neq 0$, the solutions in the infinitesimal monad of zero of $\cos \theta = 1 - \tau$ are exactly*

$$\theta_{\pm} = \pm 2 \arcsin_{\inf} \sqrt{\tau/2} = \pm \sqrt{2\tau} \left(1 + \frac{\tau}{12} + \frac{3\tau^2}{160} + \frac{5\tau^3}{896} + \cdots \right), \quad (9.6)$$

where $\arcsin_{\inf}$ is the zero-centred formal inverse-sine germ and either square root may be selected. Both are simple, with $v(\theta_{\pm}) = \frac{1}{2}v(\tau)$ exactly. If $\tau = 0$, zero is the unique root there, of multiplicity two. For real $\tau > 0$ both roots are real; for real $\tau < 0$ there is no real solution near zero, but there are exactly two conjugate purely imaginary surcomplex solutions, each of multiplicity one.

Proof. On the infinitesimal monad, $1 - \cos \theta = 2 \sin^2(\theta/2)$, and $\theta \mapsto \sin(\theta/2)$ is a germ with nonzero linear coefficient, hence a bijection of that monad onto itself with inverse $w \mapsto 2 \arcsin_{\inf} w$; all its formal inverse identities evaluate by [lemma 2.2](#). The equation becomes $w^2 = \tau/2$, with exactly the two stated roots. At a nonzero root $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2) \neq 0$, so the derivative is nonzero; at $\tau = 0$, $\cos \theta - 1 = -\theta^2/2 + \cdots$ gives the double root. The expansion is [equation \(5.6\)](#). Its coefficients are real and its germ odd, so a real positive square root produces real roots and a purely imaginary square root produces purely imaginary ones, while a real square cannot equal $\tau/2 < 0$. $\square$

The complex multiplicity is constant across this real bifurcation, which is the one-variable version of the finite-algebra viewpoint of [2]. The explicit inverse germ supplies the proof; no sequence of Newton approximations is used as a convergence argument. The same phenomenon in the strip is visible in [example 12.2](#).

9.3 The rank-two algebra that survives the collision

Put $d = \mathcal{D}/\rho^2$ in [theorem 9.1](#). Over $\mathbf{SC}$ the coordinate algebra of the line-circle intersection is

$$\mathcal{A}_d = \mathbf{SC}[W]/(W^2 - d), \quad (9.7)$$

whose elements are the unique remainders $a + bW$ with $W^2 = d$ defining the multiplication; this interprets the notation as a class algebra without forming a collection of proper-class cosets. It has basis $1, W$ and dimension two *for every d* , including $d = 0$ and negative real d . When $d < 0$ there are no real points but the two complexified points still exist; when $d = 0$ there is a single point with the nonreduced algebra of dual numbers. This is the precise sense in which multiplicity survives the change in the number of real solutions.

Theorem 9.5 (A collision-stable residue pairing). *Define $\Lambda_d : \mathcal{A}_d \rightarrow \mathbf{SC}$ by $\Lambda_d(a + bW) = b$, the coefficient of W in the unique remainder. The pairing $(f, g) \mapsto \Lambda_d(fg)$ is nondegenerate for every d , with matrix*

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (9.8)$$

in the basis $1, W$, of determinant -1 . If $d \neq 0$ and $s^2 = d$ then for every polynomial g ,

$$\Lambda_d(g) = \frac{g(s)}{2s} + \frac{g(-s)}{-2s} = \frac{g(s) - g(-s)}{2s}, \quad (9.9)$$

which is exactly the sum of the two simple residues of $g(W)/(W^2 - d)$; at $d = 0$ the corresponding formula is $\Lambda_0(g) = g'(0)$. The trace of multiplication by g is $\Lambda_d(2Wg)$.

Proof. The relation $W^2 = d$ gives $\Lambda_d(1) = 0$, $\Lambda_d(W) = 1$ and $\Lambda_d(W^2) = \Lambda_d(d) = 0$, which is equation (9.8). Reducing g modulo $W^2 - d$ gives $g = a + bW$, whose values at $\pm s$ are $a \pm bs$, so the right-hand side of equation (9.9) is b . At $d = 0$ the reduction is $g(0) + g'(0)W$, and the coefficient of W^{-1} in $g(W)/W^2$ is $g'(0)$. Finally multiplication by $a + bW$ has matrix $\begin{pmatrix} a & bd \\ b & a \end{pmatrix}$, of trace $2a = \Lambda_d(2W(a + bW))$. $\square$

For infinitesimal positive d the two residues in equation (9.9) may individually be infinite while their sum remains well behaved: for $g = 1$ they are $\pm(2\sqrt{d})^{-1}$ and cancel exactly, while for $g = W$ they are both $1/2$ and sum to 1. This elementary model exhibits the structural phenomenon emphasized in [2, 3]: *the quotient and its pairing are more stable than the list of separated simple points.*

9.4 A coupled four-point angular collision

Consider two infinitesimal angles $\alpha, \beta \in \mathfrak{m}_{\mathbb{C}}$ and parameters $s, u \in \mathfrak{m}_{\mathbb{C}}$ with

$$\sin^2 \alpha + \sin^2 \beta = s, \quad \sin \alpha \sin \beta = u. \quad (9.10)$$

(Here, and only here, the letters s, u denote these parameters rather than a semiperimeter or a unit direction.) Set $x = \sin \alpha$, $y = \sin \beta$ and then $X = x + y$, $Y = x - y$; both coordinate changes are invertible analytic germs, and the system becomes

$$X^2 = s + 2u, \quad Y^2 = s - 2u. \quad (9.11)$$

The associated finite intersection algebra is

$$\mathcal{B}_{s,u} = \mathbf{SC}[X, Y]/(X^2 - s - 2u, Y^2 - s + 2u).$$

Division by the two monic quadratics gives the basis $1, X, Y, XY$, so $\dim_{\mathbf{SC}} \mathcal{B}_{s,u} = 4$ for every infinitesimal parameter value. Every support point lies in the infinitesimal monad, since its coordinate squares are infinitesimal; the total complex multiplicity there is therefore always four. *This is the algebra of the entire finite zero set, not the local algebra at one selected root:* it decomposes into the local algebras of the distinct roots, and the invertible analytic sine-coordinate changes preserve each local multiplicity. The angle solutions are

$$\alpha = \arcsin \frac{\epsilon_1 \sqrt{s+2u} + \epsilon_2 \sqrt{s-2u}}{2}, \quad \beta = \arcsin \frac{\epsilon_1 \sqrt{s+2u} - \epsilon_2 \sqrt{s-2u}}{2}, \quad (9.12)$$

with $\epsilon_1, \epsilon_2 \in \{1, -1\}$ and repetitions counted by local multiplicity; the square-root choices may be fixed arbitrarily and the sign choices exhaust them.

For real infinitesimal parameters there are four distinct real solutions exactly when $s + 2u > 0$ and $s - 2u > 0$; if precisely one of those quantities vanishes and the other is positive there are two distinct real solutions of multiplicity two; at $s = u = 0$ there is one solution of multiplicity four; if

either quantity is negative no solution has both angles real, because real X, Y have nonnegative squares; and if neither vanishes there are always four distinct complex solutions. The angular Jacobian is

$$J = 2 \cos \alpha \cos \beta (x^2 - y^2) = 2 \cos \alpha \cos \beta XY,$$

whose vanishing locus in parameter space is $s^2 - 4u^2 = 0$. The cosine factors are units in the infinitesimal monad, so they change neither the collision locus nor any multiplicity. This is the only genuinely multivariable instance in this report, and the dimension four is a *total* multiplicity, not a claim about any individual root.

9.5 Three stability certificates, and what each is for

The remainder of this section proves two sharp perturbation theorems; a third, of a different shape, is [theorem 12.1](#) below. They are genuinely different statements and none subsumes another, so all three are kept.

- [Theorem 9.7](#) bounds the *displacement of one simple root* of a trigonometric polynomial with finite coefficients under a perturbation of coefficient valuation σ , under the threshold $\sigma > 2\kappa$ with $\kappa = v(f'(a))$, and [proposition 9.8](#) proves both that the threshold cannot be weakened to $\sigma \geq 2\kappa$ and that both displayed exponents are attained. It is the sharpest statement available when one tracks a single named root of a polynomial with finite coefficients.
- [Theorem 9.9](#) instead inverts *one scalar function*, the inverse cosine, at a prescribed interior angle, and supplies the second-order term $-c\varepsilon^2/(2s^3)$ together with a remainder bound $v(\mathcal{R}) \geq 3\sigma - 5\kappa$. It is sharper than [theorem 9.7](#) whenever the second-order behaviour of the inverse is what one needs, and it applies at an angle rather than to a polynomial; its own sharpness construction is different and explicit.
- [Theorem 12.1](#) certifies instead an *exact multiplicity* for a whole cluster of roots, together with a coefficient-support bound, and makes no smallness demand beyond infinitesimality. It is the right tool when roots collide and individual displacements are not the object of interest, and it is the only one of the three that survives a multiple unperturbed root.

Lemma 9.6 (A simple residue root of a polynomial). *Let $H(Y)$ be a real polynomial with finite surreal coefficients whose coefficientwise standard part is $Y + b_0$, $b_0 \in \mathbb{R}$. Then H has exactly one root with standard part $-b_0$, and that root is simple.*

Proof. Put $y_0 = -b_0$ and $\eta = H(y_0) \in \mathfrak{m}_{\mathbb{R}}$. If $\eta = 0$ then y_0 is a root. Otherwise choose a positive infinitesimal ϱ with $|\eta| \prec \varrho$, for instance $\varrho = \sqrt{|\eta|}$. Polynomial expansion gives $H(y_0 \pm \varrho) = \eta \pm \varrho(1 + \vartheta_{\pm})$ with $\vartheta_{\pm} \in \mathfrak{m}_{\mathbb{R}}$, so the two values have opposite signs, and the polynomial intermediate-value property of the real closed field **No** supplies a root between them. For any finite y, z with the required standard part the divided difference is 1 plus an infinitesimal, because the polynomial is finite and its reduced polynomial is linear; hence it is nonzero, which proves uniqueness, and the corresponding derivative calculation proves simplicity. $\square$

This replaces a Newton-iteration limit by a single intermediate-value step with an infinitesimal test radius — the only available substitute, since by [proposition 2.4](#) a Newton sequence has no fine-topological limit to converge to.

Theorem 9.7 (Sharp angular root-stability bound). *Let f be a real-valued trigonometric polynomial with finite coefficients — i.e. each Laurent coefficient lies in $\mathcal{O}_{\mathbb{C}}$, finite in magnitude, not finite in*

number — and let $a \in \mathcal{O}_{\mathbb{R}}$ be a simple zero. Put $A = f'(a)$ and $\kappa = v(A) \geq 0$. Let g be a real-valued trigonometric polynomial all of whose Laurent coefficients have valuation at least σ , where

$$\sigma > 2\kappa. \quad (9.13)$$

Then $f + g$ has exactly one zero $a + h$ in the angular neighbourhood $v(h) > \kappa$; it is simple, and

$$v(h) \geq \sigma - \kappa, \quad (9.14)$$

$$v\left(h + \frac{g(a)}{f'(a)}\right) \geq 2\sigma - 3\kappa. \quad (9.15)$$

The assertion holds for value groups of arbitrary rank; σ and κ need not be real numbers.

Proof. All Taylor coefficients of $f(a + H)$ are finite, and those of $g(a + H)$ have valuation at least σ , since only finitely many frequencies occur and $\text{cis } a$ has valuation zero. Write

$$f(a + H) = AH + Q(H), \quad g(a + H) = g(a) + BH + Q_g(H),$$

where Q, Q_g have order at least two with coefficient valuations at least 0 and σ respectively, and $v(B) \geq \sigma$. Set $r = \sigma - \kappa > \kappa$ and $\lambda = t^r$. Formally substituting $H = \lambda Y$ and dividing by $A\lambda$ gives $Y + b + P(Y)$ with $b = g(a)/(A\lambda) \in \mathcal{O}_{\mathbb{R}}$, every nonconstant error coefficient of P having strictly positive valuation: the normalized degree- j term with $j \geq 2$ has valuation at least $(j - 1)r - \kappa > 0$.

To turn this into an existence proof without an iteration limit, use the exact rational chart of [theorem 4.5](#),

$$H = 2 \arctan(\lambda Y/2), \quad \text{cis}(a + H) = \text{cis } a \cdot C(\lambda Y/2).$$

Since f and g have finite trigonometric degree, their pullbacks are rational functions of Y ; clearing the common denominator, whose standard part is one for finite Y , and dividing by $A\lambda$, yields a finite polynomial whose coefficientwise reduction is $Y + \text{st}(b)$. The chart's difference from $H = \lambda Y$ begins in degree three and also has positive normalized valuation. [Lemma 9.6](#) now gives a finite root, whence h with $v(h) \geq r$: this is existence and [equation \(9.14\)](#).

For uniqueness on the whole stated neighbourhood take $v(h_1), v(h_2) > \kappa$. Strong Taylor divided differences give $(f + g)(a + h_1) - (f + g)(a + h_2) = (A + D)(h_1 - h_2)$ with $v(D) \geq \min\{v(h_1), v(h_2), \sigma\} > \kappa$, so $A + D \neq 0$; the same estimate for the derivative at the new root proves simplicity. These divided-difference series are legitimate strong sums: each nonlinear term introduces an infinitesimal factor, and the finitely many original coefficient supports give a common well-ordered support by [lemma 2.2](#).

Finally the root equation gives $h + A^{-1}g(a) = -A^{-1}(Q(h) + Bh + Q_g(h))$, and $v(h) \geq r$ bounds its valuation below by $\min\{2r - \kappa, \sigma + r - \kappa, \sigma + 2r - \kappa\} = 2\sigma - 3\kappa$, which is [equation \(9.15\)](#). $\square$

This is the scalar angular counterpart of the inverse-Jacobian theorem of [3, §9], with a direct algebraic existence argument available because a trigonometric polynomial becomes rational in a tangent chart.

Proposition 9.8 (The threshold and both exponents are sharp). *For uniform simple-root persistence, $\sigma > 2\kappa$ cannot be weakened to $\sigma \geq 2\kappa$, and the exponents in [equation \(9.14\)](#) and [equation \(9.15\)](#) are attained.*

Proof. Let $0 < \tau \in \mathfrak{m}_{\mathbb{R}}$ and $f(\theta) = \sin^2 \theta - \tau^2$. At $a = \arcsin \tau$, $f'(a) = 2\tau\sqrt{1 - \tau^2}$, so $\kappa = v(\tau)$. Adding the constant $g = \tau^2$, of valuation 2κ , merges the two roots near zero into the double root of $\sin^2 \theta$: at $\sigma = 2\kappa$ there is no uniform simple-root persistence.

For $g = d$ with $v(d) = \sigma > 2\kappa$, the root near a is $a + h = \arcsin \sqrt{\tau^2 - d}$, and expansion in the infinitesimal ratio d/τ^2 gives

$$h = -\frac{d}{2\tau\sqrt{1-\tau^2}} + \frac{(2\tau^2-1)d^2}{8\tau^3(1-\tau^2)^{3/2}} + \text{terms of higher valuation.}$$

The first coefficient has valuation $-\kappa$ and the second -3κ , while successive terms gain at least $\sigma - 2\kappa > 0$ in valuation. Thus $v(h) = \sigma - \kappa$ and $v(h + g(a)/f'(a)) = 2\sigma - 3\kappa$ for this family: both estimates are equalities. The expansion is justified by strong binomial and inverse-sine composition, not by any rank-one convergence assertion. $\square$

Theorem 9.9 (Conditioned inversion of cosine). *Let $\theta \in (0, \pi)$, $c = \cos \theta$, $s = \sin \theta > 0$, and let $0 \neq \varepsilon \in \mathbf{No}$ satisfy*

$$\sigma := v(\varepsilon) > 2\kappa, \quad \kappa := v(s) \geq 0. \quad (9.16)$$

Then $c + \varepsilon \in (-1, 1)$, and with $h = \arccos(c + \varepsilon) - \theta$,

$$h = -\frac{\varepsilon}{s} - \frac{c\varepsilon^2}{2s^3} + \mathcal{R}, \quad v(\mathcal{R}) \geq 3\sigma - 5\kappa, \quad v(h) = \sigma - \kappa, \quad v\left(h + \frac{\varepsilon}{s}\right) \geq 2\sigma - 3\kappa. \quad (9.17)$$

The strict threshold in [equation \(9.16\)](#) cannot be weakened to equality uniformly near an extremum.

Proof. The distance from c to the nearer endpoint of $[-1, 1]$ is $1 - |c| = s^2/(1 + |c|)$. Condition [equation \(9.16\)](#) says that ε/s^2 is infinitesimal, so $|\varepsilon|$ is smaller than that distance; hence $c + \varepsilon \in (-1, 1)$.

Set $u = \varepsilon/s^2$ and write $h = sq$. The addition formula gives

$$\frac{\cos(\theta + sq) - c}{s^2} = c \frac{\cos(sq) - 1}{s^2} - \frac{\sin(sq)}{s} = -q - \frac{c}{2}q^2 + \frac{s^2}{6}q^3 + \cdots, \quad (9.18)$$

every coefficient on the right being a polynomial with rational coefficients in the finite parameters c and s^2 , and the linear coefficient being -1 . Formal inversion therefore yields $q = -u - \frac{c}{2}u^2 + u^3H(u)$ with every coefficient of H again such a polynomial. Evaluation at the infinitesimal u is legitimate: write the finitely many parameters as ordinary standard parts plus infinitesimals; all expanded terms are supported in the positive monoid generated by those infinitesimals and u , and positive powers of u ensure a finite contribution at each exponent ([lemma 2.2](#), and [remark 2.3\(ii\)](#) for why finitely many parameters suffice). Thus $H(u)$ is finite, $v(H(u)) \geq 0$.

The constructed $h = sq$ has $h/s \in \mathbf{m}_{\mathbb{R}}$ and therefore stays inside the same interior-angle branch, the distance to either endpoint being at least s because $s = \sin \theta \leq \min(\theta, \pi - \theta)$ by [proposition 3.2](#). It solves the desired equation, so strict monotonicity of cosine identifies it with the stated difference of inverse values. Multiplying the inverse series by s gives $\mathcal{R} = su^3H(u)$ with $v(\mathcal{R}) \geq \kappa + 3(\sigma - 2\kappa) = 3\sigma - 5\kappa$, while the linear term has valuation $\sigma - \kappa$ and every later term strictly larger valuation.

For sharpness let $\eta > 0$ be infinitesimal and put $c = 1 - \eta$, so $s^2 = 2\eta - \eta^2$ has valuation $v(\eta)$. Taking $\varepsilon = \eta$ gives $\sigma = 2\kappa$ and moves the target to 1, merging the two real roots, so the perturbed root is no longer simple; taking $\varepsilon = 2\eta$, of the same valuation, moves the target above 1 and the real solutions disappear altogether. A uniform simple-root stability theorem therefore cannot replace $>$ by $\geq$. $\square$

The roles of the two powers of the small derivative are now explicit: one factor of s^{-1} amplifies the displacement, the other measures how close the target is to a critical value. The leading displacement has order $\sigma - \kappa$ and the nonlinear correction order at least $2\sigma - 3\kappa$, matching the inverse-Jacobian pattern of [3]. Near a noncritical ordinary angle, $\kappa = 0$ and the condition is simply that the target perturbation be infinitesimal.

10 Canonical complex trigonometry on a class-sized strip

10.1 Real exponential and hyperbolic functions

We use the established Gonshor exponential $\exp_{\mathbf{No}} : (\mathbf{No}, +) \rightarrow (\mathbf{No}_{>0}, \cdot)$ and its inverse $\log_{\mathbf{No}}$ [5, 7]. They are order-preserving group isomorphisms, agree with the ordinary real functions, and satisfy

$$\exp_{\mathbf{No}}(x + h) = \exp_{\mathbf{No}}(x) \operatorname{Exp}_{\mathbf{m}}(h) \quad (h \in \mathbf{m}_{\mathbb{R}}). \quad (10.1)$$

For every $y \in \mathbf{No}$ put $\sinh y = \frac{1}{2}(\exp_{\mathbf{No}} y - \exp_{\mathbf{No}}(-y))$ and $\cosh y = \frac{1}{2}(\exp_{\mathbf{No}} y + \exp_{\mathbf{No}}(-y))$; the hyperbolic addition identities, $\cosh^2 y - \sinh^2 y = 1$ and the usual derivatives follow from the group law and equation (10.1), and $\sinh y = 0$ exactly at $y = 0$. For $x \geq 1$ set $\operatorname{arcosh} x = \log_{\mathbf{No}}(x + \sqrt{x^2 - 1})$ and, for $|x| < 1$, $\operatorname{artanh} x = \frac{1}{2} \log_{\mathbf{No}} \frac{1+x}{1-x}$; the real exponential identities show these are the inverses of $\cosh$ on $\mathbf{No}_{\geq 0}$ and of $\tanh$.

10.2 Two complementary strips and an exponential quotient

Define the additive subclasses

$$\mathcal{H} = \{x + iy : x \in \mathbf{No}, y \in \mathcal{O}_{\mathbb{R}}\}, \quad \mathcal{V} = \{x + iy : x \in \mathcal{O}_{\mathbb{R}}, y \in \mathbf{No}\},$$

so that $i\mathcal{V} = \mathcal{H}$. These are *not* fixed-width ordinary strips. On the first define

$$E(x + iy) = \exp_{\mathbf{No}}(x) \operatorname{cis} y. \quad (10.2)$$

Theorem 10.1 (Canonical strip exponential). *The map $E : \mathcal{H} \rightarrow \mathbf{SC}^{\times}$ is a surjective group homomorphism with kernel $2\pi i\mathbb{Z}$. It is fine-locally analytic and satisfies $E(z + h) = E(z) \operatorname{Exp}_{\mathbf{m}}(h)$ for infinitesimal $h \in \mathbf{SC}$ and $E'(z) = E(z)$. Consequently*

$$\mathcal{V}/(2\pi\mathbb{Z}) \longrightarrow \mathbf{SC}^{\times}, \quad [z] \longmapsto E(iz), \quad (10.3)$$

is a group isomorphism.

Proof. The group law follows from the real exponential and finite-angle addition. If $E(x + iy) = 1$ its norm forces $x = 0$ and theorem 4.2 forces $y \in 2\pi\mathbb{Z}$. For $w \neq 0$ choose a finite angle θ of w ; then $E(\log_{\mathbf{No}} |w| + i\theta) = w$. The local formula follows by splitting an infinitesimal increment into real and imaginary parts and using equation (10.1); this is an exact strongly summable local expansion, valid even when $E(z)$ is infinite, since multiplication by a fixed surcomplex coefficient preserves strong summability and one may choose a sufficiently small fine neighbourhood. Multiplication by i proves equation (10.3). $\square$

The quotient is an algebraic statement, not a claim that this class strip is a classical topological universal covering space.

Definition 10.2 (Canonical strip sine and cosine). For $z \in \mathcal{V}$ set

$$\operatorname{Sin} z = \frac{E(iz) - E(-iz)}{2i}, \quad \operatorname{Cos} z = \frac{E(iz) + E(-iz)}{2}. \quad (10.4)$$

When z is a finite real surreal these are the lowercase functions already defined; when z is finite surcomplex they are the Taylor lifts of the ordinary entire complex sine and cosine.

Theorem 10.3 (Complex-variable identities and zero sets). *For $z = x + iy \in \mathcal{V}$,*

$$\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y, \quad \cos(x + iy) = \cos x \cosh y - i \sin x \sinh y, \quad (10.5)$$

$$|\sin(x + iy)|^2 = \sin^2 x + \sinh^2 y, \quad |\cos(x + iy)|^2 = \cos^2 x + \sinh^2 y. \quad (10.6)$$

On the entire strip the addition identities, $\sin^2 z + \cos^2 z = 1$, $\sin' = \cos$ and $\cos' = -\sin$ hold exactly, and $\sin, \cos$ are fine-locally analytic. The zeros of $\sin$ are precisely $\pi\mathbb{Z}$ and those of $\cos$ precisely $\pi/2 + \pi\mathbb{Z}$; all are simple, and $\tan = \sin / \cos$ has simple poles at the latter points.

Proof. Insert equation (10.2) into equation (10.4) and use the real hyperbolic definitions for the first two formulas; squaring real and imaginary parts and using $\cosh^2 y = 1 + \sinh^2 y$ gives the norm identities. All other identities follow algebraically from the group law for E and its local derivative. A zero in either norm identity forces $\sinh y = 0$, hence $y = 0$ since the real exponential is injective; a sum of nonnegative squares in $\mathbf{No}$ vanishes only if each term does. The kernel of cis then gives the listed zeros, whose derivatives are ± 1 , giving simplicity and the pole statement. $\square$

In particular $\sin(i\omega) = i \sinh \omega$ is canonically defined, even though this construction prescribes no value for $\sin \omega$: *infinite growth along an imaginary direction and infinite oscillation along a real direction are different problems.*

Theorem 10.4 (Surjectivity and complete fibres of strip sine). *Every $w \in \mathbf{SC}$ is $\sin z$ for some $z \in \mathcal{V}$. Modulo 2π it has exactly two preimages counted with multiplicity, and the two coincide exactly when $w = 1$ or $w = -1$. For $z_1, z_2 \in \mathcal{V}$,*

$$\sin z_1 = \sin z_2 \iff z_1 - z_2 \in 2\pi\mathbb{Z} \text{ or } z_1 + z_2 \in \pi + 2\pi\mathbb{Z}. \quad (10.7)$$

Cosine is likewise surjective, since $\cos z = \sin(z + \pi/2)$.

Proof. Put $u = E(iz)$; the equation $\sin z = w$ becomes the quadratic

$$u^2 - 2iwu - 1 = 0. \quad (10.8)$$

The algebraically closed field $\mathbf{SC}$ supplies two roots counted with multiplicity, both nonzero since their product is -1 , and each has exactly one preimage modulo 2π by theorem 10.1. The discriminant is $4(1 - w^2)$, which gives the multiplicity assertion. Alternatively, subtracting the two sine values and using $\sin z_1 - \sin z_2 = 2 \cos \frac{z_1 + z_2}{2} \sin \frac{z_1 - z_2}{2}$ together with the zero sets of theorem 10.3 gives equation (10.7). $\square$

Corollary 10.5 (Explicit inverse branches). *For $w \in \mathbf{SC}$ solve equation (10.8), $u = iw \pm \sqrt{1 - w^2}$; neither root is zero. For either root take a finite argument $x = \text{Arg}(u)$ and put*

$$z = x - i \log_{\mathbf{No}} |u|. \quad (10.9)$$

Then $\sin z = w$, and all solutions in $\mathcal{V}$ are $z + 2\pi k$ and $\pi - z + 2\pi k$, $k \in \mathbb{Z}$. For example, for real $w > 1$, including infinite w ,

$$z = \frac{\pi}{2} \pm i \log_{\mathbf{No}}(w + \sqrt{w^2 - 1})$$

are the two solution classes.

Proof. Equation (10.9) gives $E(iz) = u$ by theorem 10.1, and the second root of the quadratic is $-u^{-1}$, while $E(i(\pi - z)) = -E(iz)^{-1}$; theorem 10.4 does the rest. $\square$

Thus canonical complex trigonometry is confined neither to finite surcomplex values nor to finite imaginary arguments, and no phase at an infinite *real* argument is needed even to solve $\text{Sin } z = w$ for an arbitrary surcomplex target. This is a genuine extension of inverse complex trigonometry to infinitely large values, with the choice of real exponential stated explicitly. The unresolved phase choice is specifically at infinite real arguments, and [section 17](#) is devoted to it.

11 Trigonometric polynomials and sharp algebraic root counts

Let $n \in \mathbb{N}$ be ordinary and $c_{-n}, \dots, c_n \in \mathbf{SC}$ arbitrary. Define on $\mathcal{V}$

$$T(z) = \sum_{k=-n}^n c_k E(ikz). \quad (11.1)$$

This is a *finite* sum, so arbitrary surreal sizes of the coefficients cause no summability problem; on finite real angles it is the trigonometric polynomial $T(\theta) = \sum_k c_k \text{cis}(k\theta)$, equivalently the Laurent polynomial $T(u) = \sum_k c_k u^k$ on $\mathbb{T}(\mathbf{No})$. It is real-valued on finite real angles exactly when $c_{-k} = \bar{c}_k$ for all k .

Theorem 11.1 (Algebraization and the $2n$ root bound). *Suppose $T \neq 0$ and set*

$$P(U) = \sum_{k=-n}^n c_k U^{k+n}. \quad (11.2)$$

Under $U = E(iz)$, the zeros of T modulo 2π correspond bijectively, and with multiplicity, to the nonzero roots of P . Hence T has at most $2n$ zeros modulo 2π , counted with multiplicity, on all of $\mathcal{V}$, and exactly $2n$ if $c_{-n}c_n \neq 0$; in particular the number of finite real angular zeros is at most $2n$, a bound attained by $\sin(n\theta)$, which has the $2n$ distinct simple classes $\theta = k\pi/n$.

Proof. The identity $T(z) = U^{-n}P(U)$ and the bijection [equation \(10.3\)](#) give the set-theoretic correspondence. Near a point with value $U_0 \neq 0$,

$$E(i(z+h)) = U_0 \text{Exp}_{\mathbf{m}}(ih) = U_0 + iU_0h + O(h^2)$$

has nonzero linear coefficient, and substitution by such a formal local coordinate preserves the order of a zero; multiplication by the unit U^{-n} also preserves it. Polynomial factorization in $\mathbf{SC}$ gives the total number of nonzero roots as $\deg P - \text{ord}_0 P \leq 2n$, with equality when both endpoint coefficients are nonzero. $\square$

The degree bound is an ordinary finite integer even when the coefficients have arbitrary infinite normal forms. This is the definition of angular multiplicity used throughout: in a local angular coordinate the derivative is a unit, so the angular multiplicity of a zero of T is, and equals, the polynomial multiplicity of P . There cannot be a nonzero finite trigonometric polynomial vanishing at every ordinary real angle, since its associated polynomial would have infinitely many distinct ordinary unit-circle roots; this also proves uniqueness of the finite Fourier coefficients over $\mathbf{SC}$. Infinitesimal coefficients can split a multiple zero into a cluster, as in [theorem 9.4](#) and [theorem 12.1](#), but cannot violate the total-degree bound.

Corollary 11.2 (Chebyshev and extremum equations). *For an ordinary $n \geq 1$ and $w \in \mathbf{SC}$, the equation $\text{Cos}(nz) = w$ has exactly $2n$ solutions modulo 2π counted with multiplicity. For a nonconstant real trigonometric polynomial of degree at most n , its derivative has at most $2n$ finite real zeros modulo 2π .*

Proof. The associated equation for the first assertion is $U^{2n} - 2wU^n + 1 = 0$, with nonzero constant and leading coefficients. For the second, differentiation multiplies c_k by ik , preserves the frequency band, and does not annihilate a nonconstant polynomial in characteristic zero; apply [theorem 11.1](#). $\square$

The derivative statement bounds stationary angles; it does not invoke a general compactness or extreme-value theorem on a surreal interval. On all of $\mathbf{No}$, after a global extension has been chosen in [section 17](#), trigonometric polynomials repeat their zeros with the larger period class of [corollary 17.3](#), so a global count without a phase quotient would be a different and generally infinite assertion.

12 Support-controlled lifting of trigonometric zero clusters

This section specializes the positive-support philosophy of [\[2, 3\]](#). The proof is polynomial, so that the geometric theory does not come to depend on their more substantial analytic division results. Let

$$T_0(z) = \sum_{k=-n}^n a_k E(ikz), \quad a_k \in \mathbb{C}, \quad a_{-n}a_n \neq 0,$$

be an ordinary trigonometric polynomial, perturbed to $T(z) = \sum_k (a_k + e_k) E(ikz)$ with $e_k \in \mathbf{m}_{\mathbb{C}}$, possibly zero, and let $S \subset \Gamma_{>0}$ be the union of the supports of the nonzero e_k .

Theorem 12.1 (Exact cluster multiplicity and coefficient support). *Suppose $\theta_0 \in \mathbb{C}$ is an ordinary zero of T_0 of multiplicity m , and put $u_0 = e^{i\theta_0}$. Then T has exactly m zeros, counted with multiplicity, in the complex infinitesimal monad $\theta_0 + \mathbf{m}_{\mathbb{C}}$. Normalize the algebraized polynomial [equation \(11.2\)](#) to be monic; its unique monic factor A collecting the roots in $u_0 + \mathbf{m}_{\mathbb{C}}$ has degree m and reduces to $(U - u_0)^m$, and the coefficients of $A - (U - u_0)^m$, and of the complementary factor minus its ordinary reduction, are supported in $S^* \setminus \{0\}$.*

If θ_0 is real, both T_0 and T are real on finite real arguments, and $m = 1$, then the unique lifted zero is real. No corresponding reality conclusion is asserted for an arbitrary multiple zero.

Proof. Divide $P(U) = \sum (a_k + e_k) U^{k+n}$ by $a_n + e_n$; the inverse leading coefficient is an ordinary nonzero scalar times a geometric series in an infinitesimal, so the normalized perturbation has support in S^* by [lemma 2.2](#). Its ordinary reduction has the coprime monic factorization $P_0 = A_0 B_0$ with $A_0 = (U - u_0)^m$ and $B_0(u_0) \neq 0$. Seek monic factors $A = A_0 + \Delta A$, $B = B_0 + \Delta B$ of the same degrees with positive-support corrections. The finite-dimensional ordinary linear map

$$(X, Y) \mapsto XB_0 + A_0Y, \quad \deg X < m, \quad \deg Y < 2n - m, \quad (12.1)$$

is an isomorphism onto the polynomials of degree below $2n$: its kernel is zero by coprimality and the degree restrictions, and the dimensions agree.

At a positive exponent $\nu \in S^*$ the coefficient of $AB = P$ is a linear equation with map [equation \(12.1\)](#), whose right-hand side is the prescribed coefficient of $P - P_0$ minus the already determined coefficient of $\Delta A \Delta B$; in that product the two exponents are positive and strictly below ν . Transfinite recursion on the well-ordered monoid S^* therefore constructs the factors, each equation involving only finitely many terms by [lemma 2.2](#), and the same recursion proves uniqueness. *This is a coefficient recursion, not a convergence claim for a Newton sequence;* by [remark 2.3\(i\)](#) no valuation-improvement argument would be available.

Translate $U = u_0 + H$. Then $A(u_0 + H)$ is monic of degree m with all lower coefficients infinitesimal, and every one of its roots is infinitesimal: if a root h were appreciable or infinite, dividing its equation by h^m would give 1 plus a finite sum of infinitesimals equal to zero. Its m roots exist in an algebraically closed divisible Hahn workspace, and the complementary factor cannot vanish at $u_0 + h$ for infinitesimal h , its standard part there being $B_0(u_0) \neq 0$. The local bijection $U \mapsto \theta_0 - i \operatorname{Log}_m(U/u_0)$ transports these roots to the specified angle monad, its nonzero linear term preserving multiplicity. Finally conjugation preserves a real trigonometric equation and the monad of a real θ_0 ; when there is exactly one root, that root must be fixed by conjugation. $\square$

The support bound belongs to the factor coefficients, not to individual roots. A root may require a fractional scale such as $t^{\lambda/2}$ even when the coefficient perturbation uses only t^λ — which is exactly what the tangency of [theorem 9.2](#) exhibits.

Example 12.2 (A double zero splitting off the real angle axis). The equation $1 - \operatorname{Cos} z = -\varepsilon$, with $\varepsilon > 0$ infinitesimal, has two zeros in the monad of zero,

$$z = \pm i \operatorname{arcosh}(1 + \varepsilon), \quad \operatorname{arcosh}(1 + \varepsilon) = \log_{\mathbf{No}}(1 + \varepsilon + \sqrt{2\varepsilon + \varepsilon^2}),$$

since $\operatorname{Cos}(iy) = \cosh y$. They have valuation $\frac{1}{2}v(\varepsilon)$ and are *not* real. This is the complex continuation of the disappearance of real intersections after tangency in [theorem 9.2](#), and the strip counterpart of the negative- τ case of [theorem 9.4](#).

13 Finite Fourier calculus and positivity on the surreal circle

Nothing in this section concerns an uncontrolled infinite Fourier series. The number of frequencies and of sample points is an ordinary finite integer, while the coefficients themselves may have arbitrarily complicated set-sized surreal supports.

13.1 Exact quadrature by roots of unity

Let $M \in \mathbb{N}_{>0}$ be ordinary and $\zeta = \operatorname{cis}(2\pi/M)$, an ordinary primitive M th root of unity by [corollary 4.3](#). For every integer k ,

$$\frac{1}{M} \sum_{j=0}^{M-1} \zeta^{jk} = \begin{cases} 1, & M \mid k, \\ 0, & M \nmid k, \end{cases} \quad (13.1)$$

a finite geometric-sum identity, therefore unchanged over $\mathbf{SC}$.

Theorem 13.1 (Finite Fourier inversion, sampling, and Parseval). *Let $T(u) = \sum_{k=-N}^N c_k u^k$ with $c_k \in \mathbf{SC}$ and let $M > 2N$. For any base direction $u_0 \in \mathbb{T}(\mathbf{No})$,*

$$c_k = \frac{1}{M} \sum_{j=0}^{M-1} T(u_0 \zeta^j) (u_0 \zeta^j)^{-k}, \quad |k| \leq N, \quad (13.2)$$

$$\frac{1}{M} \sum_{j=0}^{M-1} |T(u_0 \zeta^j)|^2 = \sum_{k=-N}^N |c_k|^2. \quad (13.3)$$

More generally, for arbitrary $z_0, \dots, z_{M-1} \in \mathbf{SC}$ and $\hat{z}_k = \frac{1}{M} \sum_j z_j \zeta^{-jk}$,

$$z_j = \sum_{k=0}^{M-1} \hat{z}_k \zeta^{jk}, \quad \sum_{j=0}^{M-1} |z_j|^2 = M \sum_{k=0}^{M-1} |\hat{z}_k|^2. \quad (13.4)$$

Thus M exact samples determine every coefficient, even when the coefficients or the base direction have infinitesimal parts.

Proof. Insert the finite Laurent sum into [equation \(13.2\)](#): distinct frequency differences have absolute value at most $2N < M$, so [equation \(13.1\)](#) kills every off-diagonal term. Multiplying T by its conjugate and applying the same identity gives [equation \(13.3\)](#). For [equation \(13.4\)](#), substitute the definition of $\widehat{z}_k$, use the geometric-sum identity for inversion, and substitute the inversion formula and its conjugate into $\sum_j z_j \bar{z}_j$. Every sum is finite and $\bar{\zeta} = \zeta^{-1}$, so no support or convergence condition arises. $\square$

All nonzero terms in the norm identity are positive surreals; *Fourier cancellation cannot destroy energy at a hidden infinitesimal order*. Exact coefficient recovery is quite different from testing only the signs of the sample values: in [example 13.6](#) below, every ordinary-angle sample is positive, yet the exact sample values still contain the infinitesimal information, and the reconstructed polynomial exposes its negative windows.

13.2 Coefficientwise integration and Parseval

For an ordinary analytic f with primitive F on a real interval and finite endpoints $a = a_0 + \varepsilon$, $b = b_0 + \eta$, define the lifted integral by

$$\begin{aligned} \int_a^b f^\sharp(\theta) d\theta &= F^\sharp(b) - F^\sharp(a) \\ &= \int_{a_0}^{b_0} f(t) dt + \sum_{m \geq 1} \frac{F^{(m)}(b_0)}{m!} \eta^m - \sum_{m \geq 1} \frac{F^{(m)}(a_0)}{m!} \varepsilon^m, \end{aligned} \quad (13.5)$$

the constant of the primitive cancelling. More generally a *uniformly supported* family $F(\theta) = \sum_\gamma t^\gamma f_\gamma(\theta)$ along an ordinary parameter interval — that is, an element of the ring (R1) of [section 1.3](#), with one well-ordered support and all coefficient functions analytic on that one ordinary interval — is integrated coefficientwise, $\int_a^b F = \sum_\gamma t^\gamma \int_{a_0}^{b_0} f_\gamma$, with the same endpoint corrections. This convention is not a definition of an unrestricted Riemann integral on all surreal functions, and other surreal integration frameworks have substantially different scope and hypotheses [15]. In particular

$$\frac{1}{2\pi} \int_0^{2\pi} \text{cis}(k\theta) d\theta = \begin{cases} 1, & k = 0, \\ 0, & k \neq 0, \end{cases} \quad k \in \mathbb{Z}, \quad (13.6)$$

where the ordinary coefficient integral is the usual real or complex integral; its Hahn assembly is not an ordinary Riemann limit over the full class interval.

Theorem 13.2 (Finite Fourier identities over **SC**). *If $F(\theta) = \sum_{k=-n}^n f_k \text{cis}(k\theta)$ and $G(\theta) = \sum_{k=-m}^m g_k \text{cis}(k\theta)$, extending coefficients by zero where needed, then*

$$f_j = \frac{1}{2\pi} \int_0^{2\pi} F(\theta) \text{cis}(-j\theta) d\theta, \quad (13.7)$$

$$\frac{1}{2\pi} \int_0^{2\pi} F(\theta) \overline{G(\theta)} d\theta = \sum_k f_k \bar{g}_k, \quad \frac{1}{2\pi} \int_0^{2\pi} |F(\theta)|^2 d\theta = \sum_k |f_k|^2, \quad (13.8)$$

the last expression being positive unless $F = 0$. For a real trigonometric polynomial $P(\theta) = a_0 + \sum_{k=1}^n (a_k \cos k\theta + b_k \sin k\theta)$ the same convention gives

$$\frac{1}{2\pi} \int_0^{H 2\pi} P(\theta)^2 d\theta = a_0^2 + \frac{1}{2} \sum_{k=1}^n (a_k^2 + b_k^2). \quad (13.9)$$

Proof. Expand the finite sums, use $\overline{\text{cis}(k\theta)} = \text{cis}(-k\theta)$, and apply [equation \(13.6\)](#); every operation on the frequency indices is finite. Positivity follows from the ordering of $\mathbf{No}$ and positivity of each squared norm. Multiplication by fixed surreal constants is legitimate because Hahn products have finite convolution at each exponent. $\square$

These identities remain valid for infinite and for infinitesimal coefficients. *They do not establish convergence or completeness for an unrestricted infinite Fourier series.* In particular, infinitely many ordinary Fourier summands at the same Hahn exponent do not form a strongly summable family merely because the corresponding real Fourier series converges.

13.3 Dirichlet and Fejér kernels, and why they are not an approximation theorem

Finite geometric sums and [theorem 3.1](#) give

$$D_N(\theta) = \sum_{k=-N}^N \text{cis}(k\theta) = \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\theta/2)}, \quad (13.10)$$

$$K_N(\theta) = \frac{1}{N+1} \left| \sum_{j=0}^N \text{cis}(j\theta) \right|^2 = \frac{1}{N+1} \left(\frac{\sin((N+1)\theta/2)}{\sin(\theta/2)} \right)^2, \quad (13.11)$$

the quotient formulas being used away from $\theta \in 2\pi\mathbb{Z}$, where the finite sums give the removable values $D_N = 2N+1$ and $K_N = N+1$. In particular $K_N \geq 0$ at every finite surreal angle, with normalized mean $\mathcal{M}(K_N) = 1$, where $\mathcal{M}(T) = c_0$ is the rotation-invariant normalized mean, equal to the quadrature average whenever $M > N$ and, after expanding the finitely many coefficients into normal form, to $\frac{1}{2\pi} \int_0^{H 2\pi} T$. These are exact algebraic positivity and normalization statements. *They are not, by themselves, a fine-topological approximation theorem as N grows.* For a fixed nonzero infinitesimal θ , every ordinary finite N still has $(N+1)\theta$ infinitesimal, so $K_N(\theta) \sim N+1$, not the ordinary fixed-nonzero-real-angle behaviour. One must specify a coefficient topology or another asymptotic framework before formulating any infinite Fourier convergence theorem.

13.4 Fejér–Riesz factorization with surcomplex coefficients

The classical Fejér–Riesz theorem factors a nonnegative trigonometric polynomial as a squared modulus; see [12] for its established context. *The theorem below is a real-closed-field extension of that classical result, not a claim of a new spectral-factorization theorem.* Its relevance here is that the positivity hypothesis must range over surreal, not merely ordinary, directions, and that coefficients and zeros may carry arbitrarily fine surreal scales while the factorization stays finite and exact.

Lemma 13.3 (Nonnegative polynomials over a real closed field). *Let F be real closed, $K = F[i]$, and $p \in F[t]$ nonnegative at every $t \in F$. Then there is $q \in K[t]$ with $p(t) = q(t)\overline{q(t)}$ for all $t \in F$, and $\deg q = \frac{1}{2} \deg p$ when $p \neq 0$.*

Proof. For $p = 0$ take $q = 0$. Otherwise factor p over F into real linear factors and irreducible quadratics. Every real root has even multiplicity: an odd multiplicity would change the sign across that root, as is seen by taking a sufficiently small increment avoiding the finitely many other roots. The leading constant is positive, and each monic irreducible quadratic is $(t - a)^2 + b^2$ with $b > 0$. Take one factor $t - a - ib$ from each conjugate pair, half the multiplicity of each real linear factor, and the positive square root of the leading constant; their product is q . $\square$

Theorem 13.4 (Surcomplex Fejér–Riesz theorem). *Let $T(\theta) = \sum_{k=-n}^n c_k \operatorname{cis}(k\theta)$ with $c_{-k} = \bar{c}_k$ and $c_k \in \mathbf{SC}$. The following are equivalent.*

1. $T(\theta) \geq 0$ for every finite surreal angle θ , equivalently $T(u) \geq 0$ for every $u \in \mathbb{T}(\mathbf{No})$.
2. There is $Q(U) = \sum_{j=0}^n q_j U^j \in \mathbf{SC}[U]$ with

$$T(\theta) = |Q(\operatorname{cis} \theta)|^2 \quad \text{for every finite surreal } \theta. \quad (13.12)$$

If $T \neq 0$, then Q may be chosen with no zero in $|U| < 1$; with that choice and the normalization $Q(0) > 0$ it is unique.

Proof. (2) $\Rightarrow$ (1) is immediate. Conversely, use the rational parametrization [equation \(4.8\)](#) and form

$$p(t) = (1 + t^2)^n \sum_{k=-n}^n c_k C(t)^k. \quad (13.13)$$

Since $C(t) = (1 + it)^2/(1 + t^2)$ and $C(t)^{-1} = (1 - it)^2/(1 + t^2)$, the denominators cancel and p is a polynomial of degree at most $2n$, with real values and real coefficients by the Hermitian symmetry of the c_k . The hypothesis and the surjectivity of the rational parametrization give $p(t) \geq 0$ for every $t \in \mathbf{No}$, so [lemma 13.3](#) gives $p(t) = |q(t)|^2$ with $q \in \mathbf{SC}[t]$ of degree at most n . Define

$$Q(U) = \left(\frac{1+U}{2} \right)^n q \left(\frac{U-1}{i(U+1)} \right), \quad (13.14)$$

a polynomial of degree at most n by the degree bound on q . For $U = C(t)$ one has $(1+U)/2 = 1/(1-it)$, hence $|Q(C(t))|^2 = |q(t)|^2/(1+t^2)^n = T(C(t))$. This proves equality at every unit point except possibly $U = -1$; on the circle, replacing conjugate powers by negative powers makes the difference a Laurent polynomial which, after multiplication by U^n , is an ordinary-degree polynomial with infinitely many roots, hence identically zero.

For the normalized form, let d be the actual degree, so $c_d, c_{-d} \neq 0$, and put $P(U) = U^d T(U)$, of degree $2d$ with nonzero constant term and satisfying the self-reciprocal identity

$$P(U) = U^{2d} \overline{P(1/\bar{U})}, \quad (13.15)$$

where the right side means coefficient conjugation and reciprocal substitution. Algebraic closedness pairs every root $\varsigma \neq 0$ with $1/\bar{\varsigma}$, preserving multiplicity. A root on the unit circle has even multiplicity: at a unit root u_0 use the nonsingular rational local parameter $U = u_0 C(t)$, under which the pullback of T is a real rational function with nonvanishing denominator; if its zero order were odd, its signs at sufficiently small positive and negative surreal t would be opposite — factor out the first nonzero power of t and bound the remaining finite polynomial terms by the leading coefficient — and one of those signs would be negative, contradicting the hypothesis. *The argument explicitly permits the test values t to be infinitesimal*, which is what makes it work over $\mathbf{No}$.

Now select, for each non-unit reciprocal pair, the root outside the unit disk with its multiplicity, and half the multiplicity of every unit root: d roots in all. Let Q_0 be their monic product. The degree- $2d$ polynomials $P(U)$ and $U^d Q_0(U) \overline{Q_0(1/\bar{U})}$ have the same roots with the same multiplicities and nonzero constant terms, so they differ by a nonzero constant $\varkappa$; on the circle this says $T = \varkappa |Q_0|^2$, and evaluating at a unit point away from the finite root set, where $T > 0$ and $|Q_0|^2 > 0$, gives $\varkappa \in \mathbf{No}_{>0}$. Set $Q = \sqrt{\varkappa} Q_0$ and multiply by a unit constant to make $Q(0) > 0$. Any factor with no zeros inside the disk must select exactly the same outside roots and half of each unit-root multiplicity, so two such factors differ by a unit scalar, and their positive values at zero force that scalar to be one. $\square$

The proof is finite algebra over a real closed field: no real compactness theorem, no Hilbert-space completion, and no limit of spectral factors.

Corollary 13.5 (Fourier coefficient bounds from positivity). *Under the nonnegativity hypothesis of theorem 13.4,*

$$c_0 = \sum_{j=0}^n |q_j|^2, \quad |c_k| \leq c_0 \quad (-n \leq k \leq n),$$

and $c_0 > 0$ when $T \neq 0$. If $n \geq 1$ the highest coefficient satisfies the sharper bound $2|c_n| \leq c_0$.

Proof. Expanding equation (13.12) gives $c_k = \sum_{j=0}^{n-k} q_{j+k} \bar{q}_j$ for $k \geq 0$, and finite-dimensional Cauchy–Schwarz — proved over $\mathbf{No}$ by expanding a sum of squared norms — gives $|c_k|^2 \leq (\sum_j |q_{j+k}|^2)(\sum_j |q_j|^2) \leq c_0^2$. For $k = n$, $c_n = q_n \bar{q}_0$ and $2|q_n||q_0| \leq |q_n|^2 + |q_0|^2 \leq c_0$. $\square$

Example 13.6 (Ordinary-angle testing misses an infinitesimal negative window). Fix $0 < \varepsilon \in \mathbf{m}_{\mathbb{R}}$ and put

$$T(\theta) = (\sin \theta - \varepsilon)^2 - \varepsilon^4, \tag{13.16}$$

a real-valued trigonometric polynomial of degree two. It is strictly positive at every *ordinary* real θ : if $\sin \theta \neq 0$ is an ordinary real, the square is a positive real leading term; if $\sin \theta = 0$, the value is $\varepsilon^2 - \varepsilon^4 > 0$. Nevertheless at the finite surreal angle $\theta_* = \arcsin \varepsilon$,

$$T(\theta_*) = -\varepsilon^4 < 0,$$

so no factorization equation (13.12) exists for this T . The negative window near zero is exactly

$$\arcsin(\varepsilon - \varepsilon^2) < \theta < \arcsin(\varepsilon + \varepsilon^2),$$

of width $2\varepsilon^2(1 + \vartheta)$ with $\vartheta \in \mathbf{m}_{\mathbb{R}}$, centred to first order at an angle of scale ε ; there is a second such window near $\pi - \arcsin \varepsilon$. The four boundary angles are simple zeros and attain the bound of theorem 11.1. The same phenomenon occurs with cosine: $(\cos \theta - \varepsilon)^2 - \varepsilon^4$ is positive at every ordinary real angle and equals $-\varepsilon^4$ at $\theta = \arccos \varepsilon$.

This is a concrete reason not to replace positivity on $\mathbb{T}(\mathbf{No})$ by positivity of a real-angle trace, even at a fixed very small trigonometric degree. It does not contradict proposition 2.5: the coefficients in equation (13.16) depend on a non-real infinitesimal parameter, so that polynomial is not a single ordinary real-coefficient analytic function. For contrast,

$$|u - (1 + \varepsilon)|^2 = \varepsilon^2 + 2(1 + \varepsilon)(1 - \cos \theta) \quad (u = \operatorname{cis} \theta)$$

is positive at every surreal direction, with exact minimum ε^2 at $u = 1$: a positive trigonometric polynomial whose positivity margin is genuinely infinitesimal.

14 Spherical trigonometry over $\mathbf{No}$

The distinction between algebraic geometry and angle evaluation extends beyond the plane. This section and the next record two substantial analogues, proved from Gram identities, using only three-dimensional vectors and finite matrix operations.

Let $P, Q, R \in \mathbf{No}^3$ be linearly independent unit vectors and define the minor central side angles

$$a = \arccos(Q \cdot R), \quad b = \arccos(R \cdot P), \quad c = \arccos(P \cdot Q),$$

all in $(0, \pi)$ by [theorem 5.1](#). At P the unit tangent vectors toward Q and R are

$$T_{PQ} = \frac{Q - \cos c P}{\sin c}, \quad T_{PR} = \frac{R - \cos b P}{\sin b},$$

and $A \in (0, \pi)$ is their angle; define B, C cyclically.

Theorem 14.1 (Spherical cosine and sine laws). *With the above notation,*

$$\cos a = \cos b \cos c + \sin b \sin c \cos A, \tag{14.1}$$

$$\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c} = \frac{|\det(P, Q, R)|}{\sin a \sin b \sin c}, \tag{14.2}$$

together with the cyclic cosine laws. On a sphere of arbitrary positive surreal radius ϱ the corresponding coherent minor arc lengths are ϱa , ϱb , ϱc .

Proof. The tangent vectors are unit vectors perpendicular to P , and their dot product is

$$T_{PQ} \cdot T_{PR} = \frac{\cos a - \cos b \cos c}{\sin b \sin c},$$

which is [equation \(14.1\)](#). In the tangent plane the magnitude of the determinant of two unit tangent vectors is the sine of their angle; since P is a unit normal,

$$\sin A = |\det(P, T_{PQ}, T_{PR})| = \frac{|\det(P, Q, R)|}{\sin b \sin c},$$

the cross-product identity used here being a polynomial consequence of the Euclidean Gram determinant and therefore valid over $\mathbf{No}$. Dividing by $\sin a$ and relabelling gives [equation \(14.2\)](#). Scaling the circle containing each great-circle arc gives the arc-length statement from [theorem 16.1](#). $\square$

Linear independence excludes great-circle degeneracy and makes every sine in a denominator nonzero. The theorem allows all three points to be infinitesimally close, so it includes spherical triangles on arbitrarily small angular scales, and equally spherical triangles on a sphere of infinite radius. *It is a family of metric identities under explicit nondegeneracy hypotheses, not a complete global geometric or measure-theoretic foundation for spherical geometry over $\mathbf{No}$.*

Spatial spherical trigonometry over $\mathbf{No}^3$ — polar duality, reconstruction, the exceptional side-side-angle family, branch-correct excess, area and holonomy — is developed in `docs/surreal/euclidean-three-space/` (`e3:sec:spherical-laws` to `e3:sec:differential`); its `e3:thm:phase` is the Hahn-workspace form of [theorem 4.2](#).

15 Hyperbolic trigonometry in the full surreal disk

15.1 A disk metric including infinitesimal boundary distances

Let $\mathbb{D}_{\mathbf{No}} = \{z \in \mathbf{SC} : |z| < 1\}$. Unlike the ordinary-disk thickening used in some coherent analytic statements, this disk includes points infinitesimally inside the unit circle whose standard parts lie on its boundary. For $a \in \mathbb{D}_{\mathbf{No}}$ set

$$\varphi_a(z) = \frac{z - a}{1 - \bar{a}z}, \quad (15.1)$$

whose denominator is nonzero because $|\bar{a}z| < 1$. Direct expansion gives

$$1 - |\varphi_a(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2} > 0, \quad (15.2)$$

so φ_a is a disk automorphism with inverse $(w + a)/(1 + \bar{a}w)$; rotations are disk automorphisms as well. Define

$$\varrho(z, w) = \left| \frac{z - w}{1 - \bar{w}z} \right|, \quad d_H(z, w) = \log_{\mathbf{No}} \frac{1 + \varrho(z, w)}{1 - \varrho(z, w)} = 2 \operatorname{artanh} \varrho(z, w). \quad (15.3)$$

Proposition 15.1 (Disk invariance and metric properties). *The quantities ϱ and d_H are invariant under rotations and under the maps φ_a , and d_H is an $\mathbf{No}$ -valued metric on $\mathbb{D}_{\mathbf{No}}$.*

Proof. Invariance of ϱ follows by substituting equation (15.1), putting differences over common denominators and cancelling absolute values; invariance of d_H follows. Symmetry, positivity and separation of points follow from the definition and equation (15.2). For the triangle inequality move the middle point to zero: if $|z| = r$, $|w| = s$ and the angle between them is θ , then

$$\varrho(z, w)^2 = \frac{r^2 + s^2 - 2rs \cos \theta}{1 + r^2 s^2 - 2rs \cos \theta} \leq \left(\frac{r + s}{1 + rs} \right)^2,$$

since for fixed r, s the left-hand ratio, written as $1 - (1 - r^2)(1 - s^2)/(1 + r^2 s^2 - 2rs \cos \theta)$, is largest when $\cos \theta = -1$. Applying the increasing logarithm and using

$$\frac{1 + (r + s)/(1 + rs)}{1 - (r + s)/(1 + rs)} = \frac{1 + r}{1 - r} \cdot \frac{1 + s}{1 - s}$$

gives $d_H(z, w) \leq d_H(z, 0) + d_H(0, w)$. $\square$

At a vertex A the hyperbolic angle between the sides towards B and C is the Euclidean angle between $\varphi_A(B)$ and $\varphi_A(C)$. This is invariant under disk automorphisms: after moving the vertex to zero, the remaining automorphism fixing zero is a rotation, as direct algebra with the fractional-linear formulas shows; equivalently the nonzero derivative $\varphi'_a(z) = (1 - |a|^2)/(1 - \bar{a}z)^2$ preserves angles. *All these angles are finite, even when their side distances are infinite.*

15.2 Hyperbolic cosine and sine laws

Theorem 15.2 (Hyperbolic triangle laws over $\mathbf{No}$). *Let A, B, C be distinct points of $\mathbb{D}_{\mathbf{No}}$ not on one hyperbolic line, with $a = d_H(B, C)$, $b = d_H(C, A)$, $c = d_H(A, B)$ and hyperbolic interior angles $\alpha, \beta, \gamma \in (0, \pi)$. Then*

$$\cosh a = \cosh b \cosh c - \sinh b \sinh c \cos \alpha, \quad (15.4)$$

$$\frac{\sin \alpha}{\sinh a} = \frac{\sin \beta}{\sinh b} = \frac{\sin \gamma}{\sinh c}. \quad (15.5)$$

The distances a, b, c may be infinite.

Proof. Move A to zero and rotate B to $r > 0$, writing $C = s \operatorname{cis} \alpha$ after a possible orientation reversal. Here $r = \tanh(c/2)$ and $s = \tanh(b/2)$, so $\cosh c = (1 + r^2)/(1 - r^2)$ and $\sinh c = 2r/(1 - r^2)$, similarly for b . With $q = \varrho(B, C)$,

$$q^2 = \frac{r^2 + s^2 - 2rs \cos \alpha}{1 + r^2 s^2 - 2rs \cos \alpha}, \quad \cosh a = \frac{1 + q^2}{1 - q^2},$$

and substitution yields $\cosh a = ((1+r^2)(1+s^2)-4rs \cos \alpha)/((1-r^2)(1-s^2))$, which is [equation \(15.4\)](#); by invariance the same calculation applies at every vertex. For the sine law set $X = \cosh a$, $Y = \cosh b$, $Z = \cosh c$; the cosine law gives

$$\sin^2 \alpha = \frac{1 + 2XYZ - X^2 - Y^2 - Z^2}{(Y^2 - 1)(Z^2 - 1)}.$$

Dividing by $\sinh^2 a = X^2 - 1$ produces an expression symmetric in a, b, c , so the squares of all three ratios in [equation \(15.5\)](#) agree; the ratios being strictly positive, the unsquared equality follows. $\square$

Example 15.3 (An infinitely distant point inside the disk). For $0 < \varepsilon \in \mathfrak{m}_{\mathbb{R}}$ and $z = 1 - \varepsilon$,

$$d_H(0, z) = \log_{\mathbf{No}} \frac{2 - \varepsilon}{\varepsilon} = -\log_{\mathbf{No}} \varepsilon + \log 2 + \log_{\mathbf{No}}(1 - \varepsilon/2),$$

which is infinite, the last logarithm being a strongly summable infinitesimal series. *No value of sine or cosine at an infinite real circular angle is required to obtain either this distance or the triangle laws.*

Remark 15.4 (A second model, not restated). The same two laws can be proved in the Lorentz hyperboloid model $\mathbb{H}_{\mathbf{No}}^2 = \{P \in \mathbf{No}^3 : L(P, P) = 1, P_0 > 0\}$ with $L(P, Q) = P_0 Q_0 - P_1 Q_1 - P_2 Q_2$ and $a = \operatorname{arcosh} L(Q, R)$ etc., by the Lorentz Gram-determinant computation that proves [theorem 14.1](#): the tangent vectors $T_{PQ} = (Q - \cosh c P)/\sinh c$ have L -norm -1 , their $-L$ inner product is $(\cosh b \cosh c - \cosh a)/(\sinh b \sinh c)$, and the Lorentz Gram matrix of P, T_{PQ}, T_{PR} has determinant $\sin^2 \alpha$, giving $\sin \alpha = |\det(P, Q, R)|/(\sinh b \sinh c)$. We do not restate the theorem in that model: it is the same pair of identities, and one model suffices. What the hyperboloid form makes visible is that $L(P, Q) \geq 1$ with equality only for $P = Q$ — from $L(P, Q) \geq \sqrt{1 + p^2} \sqrt{1 + q^2} - pq \geq 1$ and $(1 + p^2)(1 + q^2) - (1 + pq)^2 = (p - q)^2$ — so the distances are genuinely positive. Unlike the circular functions, the hyperbolic functions use the ordered real surreal exponential *without* discarding a purely infinite part, so infinite distances represent genuine large hyperbolic separation while the angles between tangent directions remain finite.

16 Coherent arc length, sector area, and the limit pitfall

Theorem 16.1 (Circle length and sector area at every radius). *For $O \in \mathbf{SC}$, $R \in \mathbf{No}_{>0}$ and finite $a \leq b$, consider the unwrapped parametrized arc $z(\theta) = O + R \operatorname{cis} \theta$, $a \leq \theta \leq b$. Its coherent arc length, computed from its speed with [equation \(13.5\)](#), is $R(b - a)$, and its oriented sector area about O is $R^2(b - a)/2$. These count turns according to the chosen unwrapped interval; in particular one turn has coherent circumference $2\pi R$ and the disk has area πR^2 .*

Proof. One has $z'(\theta) = iR \operatorname{cis} \theta$, so $|z'| = R$, and the integral of this constant is $R(b - a)$, including the infinitesimal endpoint corrections. The usual parametrized sector functional has integrand $\frac{1}{2} \operatorname{Im}(\overline{z - O} z') = R^2/2$, whose integral is $R^2(b - a)/2$. All expressions are finite algebraic combinations of a fixed surreal scalar with scalar-lifted trigonometric functions, so their coefficientwise integrals have valid common support in the ring $(R1)$; the chosen primitives are $R\theta$ and $R^2\theta/2$, and a change of their one global additive constant cancels at the endpoints. $\square$

As a related contour identity, direct parametrization gives $\int_{|z-O|=R}^H \frac{dz}{z-O} = 2\pi i$. *This is a coherent contour functional, not an assertion about ordinary paths being connected in the full fine topology*, and the word “coherent” is part of the definition: no Riemann-integral theory for every fine-continuous class function is being claimed, nor any ordinary real-parameter fine-continuous traversal of a surreal circle, which [proposition 2.4](#) forbids. For an infinitesimal positive central angle δ , arc and chord are relatively equivalent, $2R \sin(\delta/2) \sim R\delta$, with quadratic relative discrepancy, independently of whether R is tiny or enormous; and for a simple arc of angular span $\delta \in [0, \pi]$ the chord comparison reads $\frac{2}{\pi}L \leq \text{chord} \leq L$.

Proposition 16.2 (Failure of fine convergence of inscribed perimeters). *Let $R > 0$ be any surreal radius and let $P_n = 2nR \sin(\pi/n)$, $n = 3, 4, \dots$, be the perimeters of the ordinary finite regular inscribed polygons of [corollary 7.5](#). The sequence does not converge to $2\pi R$ in the fine topology, and the set $\{P_n : n \geq 3\}$ has no least upper bound in $\mathbf{No}$.*

Proof. The ordinary real numbers $q_n = P_n/R = 2n \sin(\pi/n)$ satisfy $q_n < 2\pi$ and converge to 2π in $\mathbb{R}$. For any positive surreal infinitesimal η the positive ordinary gap $2\pi - q_n$ exceeds η for every n , so $2\pi R - P_n > R\eta$ for every n , ruling out fine convergence. An infinite upper bound for the q_n cannot be least, because 2π is a smaller upper bound; a finite candidate with standard part below 2π is not an upper bound by ordinary real convergence; a candidate with standard part above 2π can be decreased; and every finite candidate with standard part exactly 2π can be decreased by a positive infinitesimal and remain an upper bound, each ordinary gap still exceeding any such correction. Multiplication by R is an order isomorphism and preserves the conclusion. $\square$

Thus the circumference formula of [theorem 16.1](#) is a genuine coherent-integration theorem. *It must not be justified by importing a real supremum or a real sequence limit into $\mathbf{No}$ without specifying the interpretation.* This example identifies exactly which classical limiting step may not be transplanted.

17 Infinite real phases: a canonical choice, its classification, and two obstructions

17.1 Why the Maclaurin series is not the definition

For a nonzero infinitely large $x \in \mathbf{No}$ one has $v(x) < 0$, and the terms of the formal sine series have valuations $v((-1)^n x^{2n+1}/(2n+1)!) = (2n+1)v(x)$, a strictly *decreasing* sequence. Their supports cannot have a well-ordered union, so that family is not strongly summable and merely writing its formal expression defines nothing. At a finite argument with nonzero real standard part the direct Maclaurin family also fails to be strongly summable, its ordinary terms repeatedly contributing at exponent zero; this is exactly why [equation \(3.1\)](#)–[equation \(3.2\)](#) evaluate the real centre first in ordinary analysis and sum only the infinitesimal corrections. Failure of direct strong summability at a real argument is not a failure of the canonical finite lift.

17.2 The finite-part projection and the canonical normalization

Every real surreal decomposes uniquely as

$$x = p + r + \varepsilon, \quad p \in \Pi, \quad r \in \mathbb{R}, \quad \varepsilon \in \mathfrak{m}_{\mathbb{R}}, \quad (17.1)$$

where Π consists of zero and the normal forms supported entirely at negative t -exponents, whose nonzero elements are purely infinite. Put $\text{fin}(x) = r + \varepsilon$, an *additive* projection $\mathbf{No} \rightarrow \mathcal{O}_{\mathbb{R}}$ with kernel Π , so that $\mathbf{No} = \Pi \oplus \mathcal{O}_{\mathbb{R}}$; the finite part retains both the ordinary constant and all infinitesimal terms, and is not merely the standard part. It is *not* multiplicative: $\text{fin}(\omega \cdot \omega^{-1}) = 1$ while $\text{fin}(\omega) \text{fin}(\omega^{-1}) = 0$. Reducing phases is thus compatible with adding angles, not with reducing the factors of their products.

The omnific integers have the normal-form description $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$. They form a discrete subring: sums and products of purely infinite normal forms have no positive exponents and no constant term, and every positive element of $\mathbf{Oz}$ is at least 1. They form an integer part of $\mathbf{No}$: writing $x = p + f$ with f finite and choosing the ordinary integer n with $n \leq f < n + 1$ (taking the appropriate side when f is infinitesimally adjacent to an integer) gives $p + n \leq x < p + n + 1$. Since Π is closed under multiplication and division by nonzero ordinary real constants,

$$2\pi\mathbf{Oz} = \Pi + 2\pi\mathbb{Z}, \quad \pi\mathbf{Oz} = \Pi + \pi\mathbb{Z}. \quad (17.2)$$

Ehrlich and Kaplan's additional characterization is that $\mathbf{Oz}$ is the unique integer part of $\mathbf{No}$ which is initial in its simplicity tree [8, Lemma 11.1]. *We use that published characterization to explain the word canonical; we do not need to reproduce its tree-theoretic proof.*

Definition 17.1 (Canonical real-surreal sine and cosine). For every $x \in \mathbf{No}$ set

$$\text{Sin } x = \sin(\text{fin } x), \quad \text{Cos } x = \cos(\text{fin } x), \quad \Phi(x) = \text{Cos } x + i \text{Sin } x.$$

This is exactly the integer-part normalization of [8, §11]: reducing x modulo $2\pi\mathbf{Oz}$ leaves the same finite representative modulo $2\pi\mathbb{Z}$ as reducing $\text{fin } x$. *The infinitesimal part is retained*, so for $p \in \Pi$ and infinitesimal ε one has $\text{Sin } p = 0$, $\text{Cos } p = 1$ and $\text{Sin}(p + \varepsilon) = \varepsilon - \varepsilon^3/6 + \dots$; the operation is not standard part. The notation is consistent with [definition 10.2](#), with which it agrees wherever both apply.

Theorem 17.2 (Canonical global exponential and its period class). *Define $\text{Exp}(x + iy) = \exp_{\mathbf{No}}(x)\Phi(y)$. Then $\text{Exp} : (\mathbf{SC}, +) \rightarrow (\mathbf{SC}^\times, \cdot)$ is onto, extends the ordinary complex exponential, and satisfies*

$$\begin{aligned} \ker \text{Exp} &= 2\pi i \mathbf{Oz}, & |\text{Exp } z| &= \exp_{\mathbf{No}}(\text{Re } z), & \text{Exp}' &= \text{Exp}, \\ \text{Exp}(z + h) &= \text{Exp}(z) \text{Exp}_{\mathfrak{m}}(h) & (h \in \mathfrak{m}_{\mathbf{SC}}). \end{aligned} \quad (17.3)$$

Putting $\text{Sin } z = (\text{Exp}(iz) - \text{Exp}(-iz))/(2i)$ and $\text{Cos } z = (\text{Exp}(iz) + \text{Exp}(-iz))/2$ extends [definition 17.1](#) to all of $\mathbf{SC}$, satisfies Euler's formula $\text{Exp}(iz) = \text{Cos } z + i \text{Sin } z$, the addition identities, $\text{Sin}^2 z + \text{Cos}^2 z = 1$, $\text{Sin}' = \text{Cos}$ and $\text{Cos}' = -\text{Sin}$; and for real surreal x, y ,

$$\text{Sin}(x + iy) = \text{Sin } x \cosh y + i \text{Cos } x \sinh y, \quad \text{Cos}(x + iy) = \text{Cos } x \cosh y - i \text{Sin } x \sinh y. \quad (17.4)$$

Moreover $(\text{Cos } z + i \text{Sin } z)^n = \text{Cos}(nz) + i \text{Sin}(nz)$ for every ordinary integer n , whence $\text{Cos}(nz) = T_n(\text{Cos } z)$ and $\text{Sin}((n + 1)z) = \text{Sin } z U_n(\text{Cos } z)$ as polynomial identities, with no division problem at a sine zero.

Proof. Both fin and the finite phase map are additive-to-multiplicative in the appropriate sense, so Exp is a group homomorphism, and its modulus is immediate. For $w \neq 0$ write $w = \rho \text{cis } \theta$ with θ finite (theorem 4.2); then $w = \text{Exp}(\log_{\mathbf{No}} \rho + i\theta)$. If $\text{Exp}(x + iy) = 1$ the modulus forces $x = 0$ and theorem 4.2 forces $\text{fin}(y) \in 2\pi\mathbb{Z}$, so equation (17.1) and equation (17.2) give the kernel. For infinitesimal $h = a + ib$ one has $\text{fin}(y + b) = \text{fin}(y) + b$, so the real exponential Taylor law equation (10.1) and the finite phase law give $\text{Exp}(z + h) = \text{Exp}(z) \text{Exp}_{\mathbf{m}}(a) \text{Exp}_{\mathbf{m}}(ib) = \text{Exp}(z) \text{Exp}_{\mathbf{m}}(h)$; this is a local surcomplex power series, proving fine-local analyticity and the derivative. Substitution gives the remaining identities, and expanding products by the exponential group law gives the addition and Pythagorean identities. $\square$

This calculation identifies the map with the canonical surcomplex exponential of [8, Theorem 11.1]. *The local power-series argument establishes the fine-local analytic property needed here without asserting uniqueness from a differential equation.*

Corollary 17.3 (Zero classes and periods). *On the full surcomplex plane,*

$$\{z : \text{Sin } z = 0\} = \pi\mathbf{Oz}, \quad \{z : \text{Cos } z = 0\} = \pi/2 + \pi\mathbf{Oz}.$$

All these zeros are real surreals and are simple as fine-analytic zeros. The period class of Sin , of Cos and of the pair is $2\pi\mathbf{Oz}$. Hence 2π is the least positive period, but the period class is not the cyclic group $2\pi\mathbb{Z}$: it contains every purely infinite real surreal.

Proof. $\text{Sin } z = 0$ says $\text{Exp}(2iz) = 1$, so $2iz \in 2\pi i\mathbf{Oz}$; for cosine use $\text{Cos } z = \text{Sin}(z + \pi/2)$. At a sine zero $\text{Cos}^2 z = 1$, so the derivative is nonzero. A common period is exactly an element T with $\text{Exp}(iT) = 1$, hence $T \in 2\pi\mathbf{Oz}$; if T is a period of sine alone, evaluating at 0 and $\pi/2$ gives $\text{Sin } T = 0$ and $\text{Cos } T = 1$, so it is a common period, and similarly for cosine. $\square$

This is compatible with every finite-angle geometric theorem above, whose angle ambiguity is still only $2\pi\mathbb{Z}$. It is also the first appearance of the phenomenon that theorem 17.6 shows to be unavoidable.

17.3 Classification of all circular group-law extensions

Theorem 17.4 (All extensions of the finite phase). *For every group homomorphism $\chi : (\Pi, +) \rightarrow (\mathbb{T}(\mathbf{No}), \cdot)$ the formula*

$$\Psi_{\chi}(p + f) = \chi(p) \text{cis } f, \quad p \in \Pi, \quad f \in \mathcal{O}_{\mathbb{R}}, \quad (17.5)$$

defines a group homomorphism $\mathbf{No} \rightarrow \mathbb{T}(\mathbf{No})$ extending canonical finite-angle cis , and every such extension is uniquely of this form. Writing $\Psi_{\chi} = C_{\chi} + iS_{\chi}$ one obtains real-valued global sine and cosine satisfying the addition formulas, $C_{\chi}^2 + S_{\chi}^2 = 1$ and, in fine-local calculus, $S'_{\chi} = C_{\chi}$, $C'_{\chi} = -S_{\chi}$; for every $h \in \mathbf{m}_{\mathbb{R}}$,

$$\Psi_{\chi}(x + h) = \Psi_{\chi}(x) \text{Exp}_{\mathbf{m}}(ih). \quad (17.6)$$

Each $E_{\chi}(x + iy) = \exp_{\mathbf{No}}(x) \Psi_{\chi}(y)$ is then an onto exponential homomorphism with $E'_{\chi} = E_{\chi}$ and $|E_{\chi}(z)| = \exp_{\mathbf{No}}(\text{Re } z)$, and the corresponding global sine and cosine agree with the canonical strip functions of section 10. The canonical choice $\chi = 1$ is exactly $\Psi_1 = \Phi$ of definition 17.1.

Proof. The direct-sum decomposition $\mathbf{No} = \Pi \oplus \mathcal{O}_{\mathbb{R}}$ and the two group laws prove that equation (17.5) is a homomorphism with the desired restriction. Conversely the restriction of any extension to Π

determines χ , and its value at $p + f$ must be the product of its two restrictions. An infinitesimal increment belongs to the finite summand, which gives [equation \(17.6\)](#); expanding its right-hand side proves fine-local analyticity and the derivative identities. The proof of [theorem 10.1](#) applies verbatim to E_χ . $\square$

Example 17.5 (Any prescribed phase at ω). Let $\ell(p) \in \mathbb{R}$ be the coefficient of $\omega = t^{-1}$ in a purely infinite normal form. For any finite angle $\tau \in \mathcal{O}_{\mathbb{R}}$ the rule $\chi_\tau(p) = \text{cis}(\tau\ell(p))$ is a character, because ℓ is additive and $\tau\ell(p)$ is finite, and its extension satisfies $\Psi_{\chi_\tau}(\omega) = \text{cis } \tau$. Thus $S(\omega)$ and $C(\omega)$ can jointly be assigned *any* point of the surreal unit circle without changing the finite-angle theory, the addition laws, or the fine-local differential equations: phases $0, \pi/2, \pi$ give respectively $(C(\omega), S(\omega)) = (1, 0), (0, 1), (-1, 0)$. The corresponding complex family E_{χ_τ} , with $E_{\chi_0}(i\omega) = 1$ and $E_{\chi_\pi}(i\omega) = -1$, is the family already constructed in [\[1\]](#); the character formulation exposes the entire freedom.

The trivial character gives the explicit normal-form extension $\Psi_1(x) = \text{cis}(\text{fin } x)$. *It is a possible convention, not a consequence of the local trigonometric axioms alone:* the classification above concerns group homomorphisms with the specified finite restriction, and is not a classification of all solutions of a differential equation. In particular it does not by itself answer the separate robustness questions formulated in [\[8, §11\]](#) for more general initial fields and their integer parts. Sources 20 and 21 use it to answer one specific possibility in the first of them negatively — initiality of the kernel-multiplier ring does not force the canonical exponential ([theorem 19.33](#)) — and those questions remain open otherwise ([section 19.8](#)). Conversely, the existence of these alternatives does not negate the canonical status of the initial-integer-part construction: canonicity there refers to that additional structure, not to uniqueness under the weaker analytic hypotheses. The two statements are complementary, and both are kept.

Theorem 17.6 (Infinite periods are unavoidable). *Every homomorphic extension in [theorem 17.4](#) has a kernel element in every coset $p + \mathcal{O}_{\mathbb{R}}$ of the finite subgroup; in particular, for every nonzero purely infinite p it has an infinite period. No extension agreeing with canonical finite-angle cis can have kernel exactly $2\pi\mathbb{Z}$, and the full surreal additive quotient $\mathbf{No}/2\pi\mathbb{Z}$ cannot be identified with $\mathbb{T}(\mathbf{No})$ by any extension of the finite phase.*

Proof. Finite-angle cis is already surjective onto the whole unit circle ([theorem 4.2](#)), so choose a finite a with $\text{cis } a = \chi(p)^{-1}$; then $\Psi_\chi(p + a) = 1$. For nonzero purely infinite p , the element $p + a$ remains infinitely large in modulus. Being in the kernel of a group homomorphism makes it a period of the phase and of both coordinate functions. $\square$

This is stronger than a bare example of nonuniqueness, and stronger than the observation that one particular convention happens to create extra zeros: the ordinary statement “all periods are ordinary multiples of 2π ” is *incompatible* with extending the circular group law to all surreal real inputs while retaining its finite-angle meaning. The canonical choice of [corollary 17.3](#), whose period class is $2\pi\mathbf{Oz}$, is the instance of this phenomenon that the integer-part normalization produces; the theorem says no other choice escapes it. Each E_χ agrees with the canonical strip functions of [section 10](#), so all disagreement occurs strictly beyond that strip, and the real-period and zero sets outside it are correspondingly larger and choice-dependent. [Theorem 19.17](#) strengthens the theorem: every such extension has cofinally large periods all of whose rational multiples are periods.

17.4 Two coherence obstructions, both over the fixed-domain ring

Recall from [section 1.3](#) the ring (R1): a *common-domain Hahn-analytic* datum on an ordinary neighbourhood J of a point is a family $H = \sum_{\gamma \in S} h_\gamma t^\gamma$ with *one* well-ordered support S and ordinary real-analytic coefficients h_γ all defined on the *same* J ; evaluation at finite thickened points uses Taylor substitution, and ordinary holomorphic coefficients on a complex neighbourhood may be used instead. This is the real restriction of the coherence convention of [1], and it is strictly stronger than independently prescribing local power series on different monads. Both theorems in this subsection are statements about (R1)–(R2) *and about no other coefficient ring*; neither is asserted over the radius-free ring (R3) or the formal ring (R4).

Theorem 17.7 (No coherent infinite-frequency bounded sine). *Let S_χ be any global extension from [theorem 17.4](#) and let $L > 0$ be infinite. There is no common-domain Hahn-analytic H over the ring (R1) on an ordinary neighbourhood of zero whose evaluation satisfies $H^\sharp(x) = S_\chi(Lx)$ throughout its finite surreal thickening.*

Proof. For every ordinary real r in the neighbourhood the right-hand side lies in $[-1, 1]$, so every negative-exponent coefficient $h_\gamma(r)$ of $H(r)$ is zero by uniqueness of normal form and finiteness. This holds for all such r , so every negative-exponent coefficient function vanishes identically, and $H'(0) = \sum_\gamma h'_\gamma(0)t^\gamma$ has no negative exponents and is finite. But equality on the thickening implies equality of the fine derivatives at zero, and by [equation \(17.6\)](#), or by the chain rule with the constant scalar L ,

$$\left. \frac{d}{dx} S_\chi(Lx) \right|_{x=0} = L,$$

which is infinite. This contradiction proves the theorem. $\square$

The requirement that the coefficient functions share an ordinary neighbourhood is essential, which is to say: the theorem is about (R1)–(R2) and would be a different statement over any other ring. On a sufficiently small surreal scale every $S_\chi(Lx)$ still has its usual local power series; what fails is the compatibility of that infinite-frequency behaviour with a bounded, common-domain ordinary analytic coefficient description.

Theorem 17.8 (Why all-scale coherence forces a polynomial). *Suppose a global function f had, at every positive surreal radius ϱ , a Hahn-coherent chart over the ring (R1) for $w \mapsto f(\varrho w)$ on the infinitesimal thickening of one fixed ordinary neighbourhood of zero. Then f has only finitely many nonzero Taylor coefficients at zero. In particular neither the global exponential nor any global sine S_χ satisfies this condition.*

Proof. Write $a_n = f^{(n)}(0)/n!$. The support of every $a_n \varrho^n$ is contained in the chart's single well-ordered support, because ordinary differentiation introduces no new Hahn exponents. If infinitely many a_n are nonzero, choose a positive surreal Λ larger than four times every $|v(a_n)|$ and take $\varrho = t^{-\Lambda}$. For successive nonzero indices $m > n$,

$$v(a_m \varrho^m) - v(a_n \varrho^n) = v(a_m) - v(a_n) - (m - n)\Lambda < 0,$$

producing an infinite decreasing sequence inside the alleged chart support, a contradiction. $\square$

The stronger polynomial classification, including global propagation of that finite germ, is proved in [1]; the displayed obstruction already suffices here, and *this is a restriction on uniform analytic descriptions over (R1), not a failure of the algebraic trigonometric identities or of the*

arbitrary-radius circle geometry. A circle with enormous radius uses the map $R \operatorname{cis} \theta$ with a *finite* angle; it does not require $w \mapsto \operatorname{Sin}(Rw)$ to possess one Hahn support over an entire ordinary angular domain. It is also why the all-scale coherence category, in which entire functions are polynomials, is not the category being asserted for these global sine functions.

18 The rotation group $\operatorname{SO}(2, \mathbf{No})$

This section is the contribution of source 19, *Rotations of the Surreal Plane: The Norm-One Torus, Infinitesimal Angles, and Topology at Every Scale* (September 22, 2026). It organizes the finite-angle theory of [section 4](#) around the group

$$\operatorname{SO}(2, \mathbf{No}) = \{A \in M_2(\mathbf{No}) : A^\top A = I, \det A = 1\}, \quad (18.1)$$

whose structure theorem

$$\operatorname{SO}(2, \mathbf{No}) \cong \mathbb{T}(\mathbf{No}) \cong \mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z} \cong \mathbb{T}(\mathbb{R}) \times (\mathfrak{m}_{\mathbb{R}}, +) \quad (18.2)$$

is exactly [proposition 4.1](#) together with [theorem 4.2](#), and is printed only there. In this section the word *rotation* always means the algebraic isometry; continuity is an additional property in a specified topology, never part of an informal picture of turning a rigid object through ordinary real time. The entries of a rotation matrix lie in $[-1, 1]$, but there is a proper class of infinitesimal corrections to them: bounded coordinates do not make the group set-sized, and they do not make it compact in any ordinary sense.

18.1 What source 19 re-derives, and the conventions of this section

Source 19 was written against repository commit [dcf86662b574](#), at which the present article already stood unchanged, and it inspected only the opening of this article. It states that the finite-angle quotient, the split real/infinitesimal phase, the projective half-angle coordinates and the distinction between global normalization and arbitrary extensions were already here, and presents itself as a group-centred synthesis rather than a discovery of these. The overlap is larger than it records: the results in the following table were also already here, and each is printed once, at the place indicated, with both sources credited. The only genuinely different proof among them is kept as a marked second route ([remark 18.4](#)). For the evaluation of formal series at infinitesimals, source 19 cites the surreal analytic use of Neumann’s lemma in [\[17, §3\]](#); here that role is played by [lemma 2.2](#).

Source 19 statement	Printed as
$\mathbb{T}(F) \cong \operatorname{SO}(2, F)$ by $a + ib \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$; the group is abelian	proposition 4.1
Structure theorem; logarithmic linearization	theorem 4.2 , equations (4.4) and (4.5) ; same
$\mathbb{T}(\mathbf{No}) \cong \mathbb{T}(\mathbb{R}) \times \mathfrak{m}_{\mathbb{R}}$; finite-angle uniformization	argument (conjugation makes the logarithm
with kernel $2\pi\mathbb{Z}$; polar decomposition	purely imaginary), not a second route
Infinitesimal exponential and logarithm, inverse	equations (4.1) and (4.2) , lemmas 2.2 and 2.6
and addition laws, formal evaluation on	
infinitesimals, leading-term control	
$\operatorname{cis}(r + \varepsilon) = e^{ir} \operatorname{Exp}_{\mathfrak{m}}(i\varepsilon)$ and its addition law	equation (3.3) , theorem 3.1
Principal angle in $(-\pi, \pi]$ with endpoint correction;	corollary 4.3 , remark 4.4
a branch of Arg is not additive	

Source 19 statement	Printed as
Rational chart C , its inverse and its homogeneous group law	theorem 4.5 , in this article’s homogeneous convention (below)
Angular distance; chord $= 2\sin(\text{angle}/2)$ and its series; equal valuations of angular and chordal distance	theorem 5.4
All n th roots; no infinitesimal torsion	corollary 4.3 ; second route in remark 18.4
$v(\sin \varepsilon) = v(\varepsilon)$, $v(1 - \cos \varepsilon) = 2v(\varepsilon)$; $v(u - v) = v(\ell_{\mathbf{m}}(u) - \ell_{\mathbf{m}}(v))$ in one monad	proposition 3.3 ; equation (5.10)
Displacement $ uz - vz = z u - v $ at any radius	equation (5.13) ; its valuation form is equation (18.12)
Set-sized subsets closed and discrete; convergent set-indexed nets eventually constant; no nonconstant ordinary real paths	proposition 2.4 ; the Cauchy-net clause is added in section 18.9
fin, $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$, the normalized global phase, kernel $2\pi\mathbf{Oz}$, the Ehrlich–Kaplan exponential	definition 17.1 , theorem 17.2 , corollary 17.3
All extensions of the finite phase; explicit extensions with any value at ω ; unavoidable infinite periods	theorem 17.4 , example 17.5 (a finite τ there covers source 19’s real parameter), theorem 17.6 (its coset form is source 19’s “for every infinite x a finite θ with $x - \theta$ in the kernel”)
$(\text{cis})' = i \text{ cis}$; every global phase obeys the same local law; Taylor data on each monad select cis	theorem 3.1 , equation (17.6) , proposition 3.4
A derivation on $\mathbf{No}$ is not a fine derivative	the closing paragraph of section 2

Notation. Source 19’s symbols are converted once, here, and not used again. Its $\mathcal{O}, \mathbf{m}$ are $\mathcal{O}_{\mathbb{R}}, \mathbf{m}_{\mathbb{R}}$. Its Exp, Log on infinitesimals are $\text{Exp}_{\mathbf{m}}, \text{Log}_{\mathbf{m}}$, because Exp in this report is the global exponential of [theorem 17.2](#). Its H_0, H_{χ} and purely infinite summand $\mathcal{J}$ are Φ, Ψ_{χ} and Π ; its angular distance d_{ang} is $\angle$; its coefficient $a_{\omega}(p)$ is the $\ell(p)$ of [example 17.5](#). Its infinitesimal angle $\ell(u)$ is written $\ell_{\mathbf{m}}(u)$ below, since the bare letter ℓ is taken by that coefficient; its identity monad U is $\mathbb{T}_{\mathbf{m}}$, since U is used for Chebyshev polynomials and for $E(iz)$; its rotation matrix $R(a, b)$ is $\text{Rot}(a, b)$, since R is a circumradius; its generator J is $\mathbf{J}$, since J is an interval in (R1); its conjugation κ is $\mathbf{c}$, since $\kappa = v(f'(a))$ in [theorem 9.7](#); its dual-number variable δ , double-angle map p_2 , Clifford bivector E , invariant differential α , rescaling automorphism F_{λ} and circles S_r are $\mathbf{d}, \text{sq}, e_{12}, \varpi, \Xi_a$ and $\mathcal{S}_{\rho}$. Source 19 writes the homogeneous chart as $[s : t] \mapsto C(t/s)$ with law $[s : t] * [p : q] = [sp - tq : sq + tp]$; in this article’s convention $[p : q] \mapsto C(p/q)$ this is exactly [equation \(4.9\)](#). Throughout this section F denotes a real closed field and $L = F[i]$; in the topological subsections F is a *set* field, and in the algebraic ones the field $\mathbf{No}$ itself may be taken, all arguments being finite algebra.

For $u \in \mathbb{T}(\mathbf{No})$ with $u_0 = \text{st}(u)$ put

$$\ell_{\mathbf{m}}(u) = -i \text{Log}_{\mathbf{m}}(u/u_0) \in \mathbf{m}_{\mathbb{R}}, \quad u = u_0 \text{Exp}_{\mathbf{m}}(i\ell_{\mathbf{m}}(u)), \quad \mathbb{T}_{\mathbf{m}} = \{u \in \mathbb{T}(\mathbf{No}) : \text{st}(u) = 1\}. \quad (18.3)$$

By [theorem 4.2](#), $\ell_{\mathbf{m}}$ is additive, $\mathbb{T}_{\mathbf{m}} \rightarrow (\mathbf{m}_{\mathbb{R}}, +)$ is an isomorphism, and

$$1 \longrightarrow \mathbb{T}_{\mathbf{m}} \longrightarrow \mathbb{T}(\mathbf{No}) \xrightarrow{\text{st}} \mathbb{T}(\mathbb{R}) \longrightarrow 1 \quad (18.4)$$

is split by the literal inclusion of the ordinary unit circle. The splitting uses the embedded real field, coefficientwise standard part and formal Taylor evaluation; once these are fixed no basis or arbitrary section is chosen, and the infinitesimal angle of a unit is recovered exactly, not up to an infinitesimal error.

Size convention. **No** is a proper class, so statements about $\mathrm{SO}(2, \mathbf{No})$ are class-group statements, readable in a class theory supporting the surreal construction or relative to a universe. No proper-class Hahn sum is formed, no quantifier ranges over a set of all class functions, and the group is never treated as a set-sized manifold. Topological statements about the full class are made with ball neighbourhoods and explicit quantifiers, not with a set of all open subclasses.

Remark 18.1 (Infinitesimal angles are not a surreal vector space). $\mathfrak{m}_{\mathbb{R}}$ is an $\mathcal{O}_{\mathbb{R}}$ -module and an $\mathbb{R}$ -vector space, but not an **No**-vector space: $\omega^{-1} \in \mathfrak{m}_{\mathbb{R}}$ while $\omega \cdot \omega^{-1} = 1 \notin \mathfrak{m}_{\mathbb{R}}$. Accordingly $\mathbb{T}_{\mathfrak{m}}$ is torsion-free, uniquely divisible, and an $\mathbb{R}$ -vector group through $r \odot \mathrm{Exp}_{\mathfrak{m}}(i\varepsilon) = \mathrm{Exp}_{\mathfrak{m}}(ir\varepsilon)$, $r \in \mathbb{R}$; the same formula does not extend to all surreal scalars while staying in $\mathbb{T}_{\mathfrak{m}}$. That the algebraic Lie algebra of [section 18.7](#) is one-dimensional over **No** is a different statement, about tangent vectors.

18.2 Isometries fixing the origin

[Proposition 4.1](#) identifies the orientation-preserving *linear* isometries. Linearity need not be assumed.

Theorem 18.2 (Isometry rigidity). *Let F be a real closed field, $\|x\| = \sqrt{x_1^2 + x_2^2}$ on F^2 , and let $f : F^2 \rightarrow F^2$ satisfy $f(0) = 0$ and $\|f(x) - f(y)\| = \|x - y\|$ for all x, y . Then f is F -linear and orthogonal. Neither continuity nor surjectivity is assumed. The maps of this kind with positive orientation are exactly $\mathrm{SO}(2, F)$.*

Proof. Polarization $2\langle x, y \rangle = \|x\|^2 + \|y\|^2 - \|x - y\|^2$ shows that f preserves inner products. Put $u = f(e_1)$, $w = f(e_2)$; their Gram matrix is the identity, so they form an orthonormal basis. The inner products of $f(x)$ with u and w are x_1 and x_2 , hence $f(x) = x_1u + x_2w$. This is linear and orthogonal, and an orthogonal matrix has determinant ± 1 . $\square$

Preserving only $\|f(x)\| = \|x\|$ would not suffice: circles of different radii can be rotated by different units, destroying pairwise distances. A field automorphism that changes surreal distances is not an isometry in this exact surreal-valued sense.

Every origin-fixing isometry is therefore uniquely $z \mapsto uz$ or $z \mapsto u\bar{z}$ with $u \in \mathbb{T}(F)$. With $\mathbf{c}(z) = \bar{z}$ one has $\mathbf{c} u \mathbf{c} = u^{-1}$, so

$$\mathrm{O}(2, F) \cong \mathbb{T}(F) \rtimes \{1, \mathbf{c}\}, \quad (18.5)$$

the nontrivial element acting by inversion, and subtracting the image of the origin gives the Euclidean group $F^2 \rtimes \mathrm{O}(2, F)$. *These are algebraic semidirect products, not assertions about connected components in an unspecified topology.*

18.3 The rational chart: exceptional cases and the logarithmic bridge

In the affine coordinate of [theorem 4.5](#) the exceptional cases of the law $t \oplus s = (t + s)/(1 - ts)$ are

$$t \oplus s = \infty \quad (ts = 1), \quad \infty \oplus t = -1/t \quad (t \neq 0), \quad \infty \oplus 0 = \infty, \quad \infty \oplus \infty = 0;$$

the identity is 0 and the inverse is $t \mapsto -t$, with ∞ fixed. For instance $C(1) = i$, and $1 \oplus 1 = \infty$ records $i^2 = -1$. In homogeneous coordinates the inverse of $[p : q]$ is $[-p : q]$, and $[p : q] = [r : s]$ exactly when $ps - qr = 0$. Near the missing half-turn,

$$C(t) + 1 = \frac{2(1 + it)}{1 + t^2} : \quad (18.6)$$

an infinite chart parameter is a finite angle infinitesimally close to π , not an infinite angle, which is one reason the projective formula is more robust than an unqualified use of $\tan(\theta/2)$.

Remark 18.3 (A parametrization of points, not an isomorphism of varieties). [Theorem 4.5](#) concerns F -points and semialgebraic maps. It is *not* an isomorphism of the complete variety $\mathbb{P}_F^1$ with the affine algebraic torus: over $F[i]$ the denominator $p^2 + q^2$ vanishes at nonzero pairs, and identifying the two statements would erase exactly the complex points at which the rational formula has poles.

For infinitesimal $t \in \mathfrak{m}_{\mathbb{R}}$ the chart value lies in $\mathbb{T}_{\mathfrak{m}}$, and

$$\ell_{\mathfrak{m}}(C(t)) = -i \operatorname{Log}_{\mathfrak{m}} C(t) = 2 \arctan t = 2 \sum_{n \geq 0} \frac{(-1)^n t^{2n+1}}{2n+1}, \quad (18.7)$$

because $\operatorname{Log}_{\mathfrak{m}} C(t) = \operatorname{Log}_{\mathfrak{m}}(1 + it) - \operatorname{Log}_{\mathfrak{m}}(1 - it)$ and the even powers cancel. This is a direct bridge between exact rational arithmetic and the infinitesimal angle, consistent with $t = \tan(\theta/2)$ in [theorem 4.5](#) and with [equation \(5.5\)](#). On infinitesimal parameters the law $t \oplus s$ stays infinitesimal and is linearized by $2 \arctan$; no branch issue arises there.

18.4 Powers, roots, torsion and finite subgroups

Remark 18.4 (Second route to all roots, via the splitting). Write $u = u_0 \operatorname{Exp}_{\mathfrak{m}}(i\varepsilon)$ with $u_0 = e^{ir} \in \mathbb{T}(\mathbb{R})$ and $\varepsilon = \ell_{\mathfrak{m}}(u)$. For every ordinary integer n the direct-product law gives $u^n = u_0^n \operatorname{Exp}_{\mathfrak{m}}(in\varepsilon)$. For ordinary $n \geq 1$ the solutions of $v^n = u$ are exactly

$$v_k = e^{i(r+2\pi k)/n} \operatorname{Exp}_{\mathfrak{m}}(i\varepsilon/n), \quad 0 \leq k < n. \quad (18.8)$$

Indeed, writing $v = d \operatorname{Exp}_{\mathfrak{m}}(i\eta)$, uniqueness of the split factors in [equation \(4.5\)](#) gives $d^n = u_0$ and $n\eta = \varepsilon$; the first equation has the displayed n ordinary solutions and the second exactly one, and the v_k are distinct because their standard parts are. This is source 19's proof. The count in [corollary 4.3](#) uses instead the kernel of cis and the degree bound for $X^n - u$; the route here uses neither, only the splitting and ordinary complex roots.

Corollary 18.5 (No new finite-order rotations). *The torsion subgroup of $\mathbb{T}(\mathbf{No})$ consists exactly of the ordinary complex roots of unity and is isomorphic to $\mathbb{Q}/\mathbb{Z}$. Every finite subgroup of $\mathbb{T}(\mathbf{No}) \cong \operatorname{SO}(2, \mathbf{No})$ is the literal standard group $\mu_n = \{e^{2\pi i k/n} : 0 \leq k < n\}$ for one ordinary $n \geq 1$.*

Proof. If $u^n = 1$ then $n\ell_{\mathfrak{m}}(u) = 0$, so $\ell_{\mathfrak{m}}(u) = 0$ and u is an ordinary root of unity ([corollary 4.3](#)). A finite subgroup has a finite exponent N , hence lies in the cyclic group μ_N , and is μ_n for a divisor n of N . $\square$

Thus infinitesimal rotations produce new elements of *infinite* order, not new regular polygons with a finite nonstandard symmetry order; ordinary finite order is the only meaning of torsion used.

Corollary 18.6 (Divisibility). *$\operatorname{SO}(2, \mathbf{No})$ is divisible, has no nontrivial homomorphism to a finite group, and has no proper subgroup of finite index.*

Proof. Divisibility is [equation \(18.8\)](#). If f maps onto a finite group of order N , write $u = v^N$; then $f(u) = f(v)^N = 1$. A finite-index subgroup of an abelian group gives a finite quotient. $\square$

Square roots without transcendental evaluation. For $u = a + ib \neq -1$ in $\mathbb{T}(F)$ the unique square root with positive real part is

$$w = \sqrt{\frac{1+a}{2}} + i \frac{b}{2\sqrt{(1+a)/2}}, \quad (18.9)$$

since $b^2 = (1 - a)(1 + a)$ gives $w^2 = u$; the other root is $-w$, and the roots of -1 are $\pm i$. Only ordered-field square roots are used. This branch suits a half-angle near the identity and is *not* a globally multiplicative choice: no multiplicative section of $u \mapsto u^2$ exists on the whole circle, since it would send -1 , of order two, to an element whose square is -1 but whose order divides two. The obstruction is elementary and persists over the surreals.

The double-angle map and Spin(2). The map $\text{sq} : \mathbb{T}(F) \rightarrow \mathbb{T}(F)$, $a + ib \mapsto (a^2 - b^2) + 2iab$, is onto with kernel $\{\pm 1\}$. In the Clifford algebra with $e_1^2 = e_2^2 = -1$ put $e_{12} = e_1 e_2$, so $e_{12}^2 = -1$; then

$$(a + be_{12}) e_1 (a - be_{12}) = (a^2 - b^2)e_1 + 2ab e_2, \quad (18.10)$$

with the corresponding expression for e_2 , so conjugation by the spinor $a + be_{12}$, $a^2 + b^2 = 1$, induces the rotation $\text{sq}(a + ib)$. This is a finite algebraic calculation and does not depend on ordinary path lifting in the fine topology.

Proposition 18.7 (Finite orthogonal subgroups). *A finite subgroup of $\text{O}(2, \mathbf{No})$ is either some μ_n or conjugate, by a rotation, to the standard dihedral group $\mu_n \rtimes \{1, c\}$.*

Proof. Its rotation subgroup is some μ_n by [corollary 18.5](#). If it contains a reflection $z \mapsto u\bar{z}$, choose v with $v^2 = u$. Conjugating by multiplication by v sends $z \mapsto u\bar{z}$ to $z \mapsto v^{-1}u\bar{v}\bar{z} = \bar{z}$ and leaves all rotations unchanged, so the group becomes $\mu_n \cup \mu_n c$. $\square$

The conjugacy is stronger than saying the abstract finite groups have familiar names, and weaker than saying all their matrices are literally real: before conjugation a reflection axis may have a genuinely nonstandard direction.

18.5 Valuation scales and precision

For $\gamma > 0$ in the value group put $\mathbb{T}_m^{\geq \gamma} = \{u \in \mathbb{T}_m : v(u - 1) \geq \gamma\}$ and $\mathbb{T}_m^{> \gamma} = \{u \in \mathbb{T}_m : v(u - 1) > \gamma\}$. Since $v(\text{Exp}_m(i\varepsilon) - 1) = v(\varepsilon)$ ([equation \(4.2\)](#)), these are subgroups, identified by ℓ_m with the valuation subgroups of $\mathbb{m}_\mathbb{R}$.

Theorem 18.8 (One real angular coordinate at each scale). *For every surreal $\gamma > 0$ there is a canonical isomorphism*

$$\mathbb{T}_m^{\geq \gamma} / \mathbb{T}_m^{> \gamma} \cong (\mathbb{R}, +), \quad (18.11)$$

sending the class of u to the coefficient of t^γ in $\ell_m(u)$.

Proof. On angles of valuation at least γ , extraction of the coefficient at t^γ is additive, onto (by the angles rt^γ), and has kernel the angles of valuation greater than γ . Transport through ℓ_m . $\square$

This is a scale *filtration*, not a decomposition of $\mathbb{T}_m$ as a direct sum of one copy of $\mathbb{R}$ per surreal exponent: individual angles can have infinite well-ordered supports, and neither an unrestricted product nor the finite-support vectors captures the normal-form condition. Measuring the cosine defect alone loses first-order information, since $v(1 - \cos \varepsilon) = 2v(\varepsilon)$ ([proposition 3.3](#)): a cosine error of scale $t^{2\gamma}$ can encode an angular change of scale t^γ .

Displacement at any radius. For $u, v \in \mathbb{T}(\mathbf{No})$ and $z \in \mathbf{SC}$ one has exactly $|uz - vz| = |z||u - v|$, the same amplification in every direction, with no supremum over vectors. Hence for $z \neq 0$ and a nonzero infinitesimal angle ε , by [equation \(5.13\)](#),

$$v(\text{Exp}_m(i\varepsilon)z - z) = v(z) + v(\varepsilon), \quad (18.12)$$

so an infinitesimal angle at an infinite radius can produce an infinitesimal, appreciable or infinite displacement according to the sign of $v(z) + v(\varepsilon)$. Here and below $x \sim y$ is the algebraic relation of [section 1.2](#), not a sequence limit.

Example 18.9 (A finite displacement from an infinitesimal angle). Take $t = \omega^{-1}$ and $u = C(t)$. The rational formula gives exactly

$$u\omega = \omega - \frac{2\omega^{-1}}{1 + \omega^{-2}} + i \frac{2}{1 + \omega^{-2}}.$$

The angle $\ell_{\mathfrak{m}}(u) = 2 \arctan \omega^{-1}$ is infinitesimal, yet $u\omega - \omega$ is finite with standard part $2i$: the longitudinal change is infinitesimal and the transverse displacement appreciable. Likewise the single angle $\varepsilon = \omega^{-2}$ displaces the points ω , ω^2 , ω^3 by amounts $\sim \omega^{-1}$, ~ 1 , $\sim \omega$, while the rotation matrix, the same in all three cases, is infinitesimally close to the identity.

Precision under products, powers and roots. If $u_j = c_j \text{Exp}_{\mathfrak{m}}(i\varepsilon_j)$ and approximations $\hat{u}_j = c_j \text{Exp}_{\mathfrak{m}}(i\hat{\varepsilon}_j)$ have the same standard phases, then for an ordinary finite number of factors

$$\ell_{\mathfrak{m}}\left(\prod_j u_j / \prod_j \hat{u}_j\right) = \sum_j (\varepsilon_j - \hat{\varepsilon}_j), \quad (18.13)$$

whose valuation is at least the least individual error valuation and may be larger through cancellation. Multiplication by a fixed ordinary nonzero integer, or division by it when a fixed root branch is selected, preserves a nonzero infinitesimal valuation, so ordinary-degree powers and matched root branches neither lose nor gain valuation precision; a mismatched standard root branch causes an appreciable error.

Hahn products. If a set-indexed family $(\varepsilon_j)_{j \in I}$ of infinitesimal angles is strongly summable, its sum is infinitesimal and

$$\prod_{j \in I}^{\text{Hahn}} \text{Exp}_{\mathfrak{m}}(i\varepsilon_j) = \text{Exp}_{\mathfrak{m}}\left(i \sum_{j \in I} \varepsilon_j\right). \quad (18.14)$$

To justify the notation, expand the left side over finite-support choices of powers of the factors. Each exponent of a resulting term is a finite word in the well-ordered positive union of the supports, which has only finitely many representations by [lemma 2.2](#), and each letter comes from only finitely many members of the family by strong summability; so each exponent receives finitely many terms, each coefficient identity reduces to a finite exponential identity, and the value is independent of ordering. For example, with $t = \omega^{-1}$, $\prod_{n \geq 1}^{\text{Hahn}} \text{Exp}_{\mathfrak{m}}(it^n) = \text{Exp}_{\mathfrak{m}}(it/(1-t))$. *The finite partial products do not converge to this value in the fine topology* ([section 18.9](#)), and repeating one fixed nonzero infinitesimal angle infinitely often is not a strongly summable family, so [equation \(18.14\)](#) assigns no canonical value to an “infinite number of identical turns.”

Surreal exponents of a rotation. For an ordinary integer n , u^n is a group operation. For a nonordinary surreal s the group axioms supply nothing analogous. On $\mathbb{T}_{\mathfrak{m}}$ the unique logarithm gives the partial definition

$$(\text{Exp}_{\mathfrak{m}}(i\varepsilon))^s := \text{cis}(s\varepsilon) \quad \text{whenever } s\varepsilon \in \mathcal{O}_{\mathbb{R}}, \quad (18.15)$$

an analytic scaling of a uniquely chosen infinitesimal angle, *not* a transfinite iteration of the group operation. Beyond that domain a global phase extension ([theorem 17.4](#)) supplies additional choices; for general u even the choice of a finite argument is a branch, and changing it by $2\pi n$ can change the result after multiplication by s .

18.6 The action on the plane and its polynomial invariants

For $\rho > 0$ let $\mathcal{S}_\rho = \{z \in \mathbf{SC} : |z| = \rho\}$. The action of $\mathrm{SO}(2, \mathbf{No})$ on $\mathcal{S}_\rho$ is free and transitive: the unique rotation taking z to w is multiplication by w/z , and $uz = z$ with $z \neq 0$ forces $u = 1$. The orbits are the origin and one circle for each positive surreal radius. The absence of nonzero fixed points of a nonidentity rotation is an algebraic fact and needs no continuous motion from the identity. The products $\langle z, w \rangle$ and $[z, w]$ of equation (2.2) are invariant under simultaneous rotation, and the cyclic order of directions is defined by ordering the two semicircles with determinant signs, or by finite arguments modulo 2π — not by ordinary real paths in the fine topology.

Proposition 18.10 (Polynomial invariants). *For a real closed field F , the polynomial invariants of the algebraic rotation group acting on one plane vector are*

$$F[X, Y]^{\mathrm{SO}(2)} = F[X^2 + Y^2], \quad (18.16)$$

and the same holds for invariance under all F -valued rotations, in particular with surreal coefficients.

Proof. Over $L = F[i]$ put $Z = X + iY$, $W = X - iY$; a unit u acts by $Z \mapsto uZ$, $W \mapsto u^{-1}W$, so $Z^a W^b$ has weight $a - b$. Independence of distinct Laurent powers of u forces an invariant to contain only terms with $a = b$, so it lies in $L[ZW] = L[X^2 + Y^2]$, and conjugation fixes exactly $F[X^2 + Y^2]$. For invariance under F -points, the rational chart supplies infinitely many u , and a Laurent polynomial in one variable vanishing at infinitely many points is zero. $\square$

This is a statement about polynomial functions, not a classification of all class functions on orbits; arbitrary functions of the radius are nonpolynomial invariants.

18.7 The norm-one torus and its Lie algebra

The coordinate ring of the rotation group over F is $A_F = F[a, b]/(a^2 + b^2 - 1)$. The gradient (2a, 2b) vanishes nowhere on the circle, so the group is smooth of dimension one. Over L put $z = a + ib$, with inverse $a - ib$; then $a = (z + z^{-1})/2$, $b = (z - z^{-1})/(2i)$ and

$$A_F \otimes_F L \cong L[z, z^{-1}]. \quad (18.17)$$

So the rotation group is the norm-one torus $\ker(\mathrm{Res}_{L/F} \mathbb{G}_m \rightarrow \mathbb{G}_m)$, split over L and anisotropic over F : conjugation acts on its character lattice $\mathbb{Z}$ by $n \mapsto -n$, and there is no nontrivial F -defined character to $\mathbb{G}_m$. These are the standard algebraic-group interpretations [18, Chapters 10 and 12]. For $\mathbf{No}$ this fixed algebraic group is applied at surreal points, rather than treating the class $\mathbf{No}$ as a set-sized coefficient field of scheme theory; each finite construction lives in a real closed Hahn workspace.

Tangent vectors. With dual numbers $F[d]/(d^2)$, a tangent vector $I + dX$ satisfies $(I + dX)^T(I + dX) = I + d(X^T + X)$, so X is skew-symmetric, hence traceless, and the determinant adds no condition:

$$\mathfrak{so}(2, F) = FJ, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad [aJ, bJ] = 0. \quad (18.18)$$

This is the precise sense in which the surreal tangent space is one-dimensional over $\mathbf{No}$. From $J^2 = -I$,

$$\exp(TJ) = I \sum_{n \geq 0} \frac{(-1)^n T^{2n}}{(2n)!} + J \sum_{n \geq 0} \frac{(-1)^n T^{2n+1}}{(2n+1)!}, \quad (18.19)$$

Hahn-evaluable at infinitesimal T and equal to $\text{Rot}(\cos T, \sin T)$ at finite T by the real-plus-infinitesimal prescription, where $\text{Rot}(a, b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. At an arbitrary infinite T the displayed series is not strongly summable, its J-component being the sine family of [section 17](#): the Lie algebra has all surreal scalars, but the canonical finite-angle exponential is defined only on $\mathcal{O}_{\mathbb{R}}J$.

Proposition 18.11 (No algebraic additive exponential). *Every morphism of algebraic groups $\mathbb{G}_a \rightarrow \text{SO}(2)$ over a real closed field is trivial.*

Proof. Over L the target is $\mathbb{G}_m$, and a regular map $\mathbb{A}_L^1 \rightarrow \mathbb{G}_m$ is a unit of $L[T]$, a constant, which must be 1. Triviality after scalar extension gives triviality over F . $\square$

The analytic phase is therefore not a missing algebraic morphism: even over $\mathbb{R}$ the exponential is transcendental, and over the surreals its domain and summation convention must also be specified.

The invariant differential. On the circle $\varpi = a db - b da$ is invariant, $a da + b db = 0$ by differentiating the equation, and the rational chart pulls it back to

$$C^* \varpi = \frac{2 dt}{1 + t^2}, \quad (18.20)$$

locally $d(2 \arctan t) = d\theta$. This is an identity of algebraic differential forms whose local primitive is evaluated by the infinitesimal rule; *it does not by itself define integration over arbitrary full-class contours*. The tangent space, the algebraic dimension and this form survive even though the fine topology has no nonconstant ordinary real paths; a connected-component argument cannot replace the algebraic tangent calculation.

18.8 Algebraic representations and nonalgebraic symmetries

Theorem 18.12 (Algebraic endomorphisms). *For the norm-one torus over a real closed field F ,*

$$\text{End}_{F, \text{alg}}(\text{SO}(2)) \cong \mathbb{Z}, \quad n \longleftrightarrow (u \mapsto u^n), \quad (18.21)$$

and the algebraic automorphisms are exactly the identity and inversion.

Proof. Over L a regular homomorphism $\mathbb{G}_m \rightarrow \mathbb{G}_m$ is an invertible Laurent polynomial cz^n , and the identity value forces $c = 1$. Each z^n commutes with the descent involution $z \mapsto z^{-1}$, so it is defined over F . An inverse would have exponent m with $mn = 1$. $\square$

For $n \neq 0$ the power maps are isogenies of degree $|n|$, whose surreal fibres [equation \(18.8\)](#) describes; they multiply the infinitesimal angle exactly by n .

Over L a finite-dimensional algebraic representation of $\mathbb{G}_m$ is a finite direct sum of integer-weight spaces: write its coaction $v \mapsto \sum_n v_n \otimes z^n$ (finitely many terms for each v); coassociativity makes each v_n a weight- n vector, the counit axiom gives $v = \sum_n v_n$, and independence of Laurent monomials makes the sum direct. Over F conjugation pairs weight n with weight $-n$, so the irreducible algebraic representations are the trivial one and, for each ordinary $n \geq 1$, the two-dimensional

$$u \longmapsto \text{Rot}(\text{Re}(u^n), \text{Im}(u^n)), \quad (18.22)$$

and every finite-dimensional algebraic representation is a direct sum of these. This is the rank-one case of torus representation theory [18, §12h]; the weight calculation shows why *no new weight labels occur: the weights are ordinary integers*, not omnific integers or surreal frequencies. Finite linear

combinations of these matrix coefficients are algebraic trigonometric polynomials ([theorem 11.1](#)). *Nothing here supplies a Fourier convergence theorem, a Haar measure on the full class, or a classification of arbitrary abstract representations.*

For each nonzero ordinary real a define

$$\Xi_a(u_0 \text{Exp}_{\mathfrak{m}}(i\varepsilon)) = u_0 \text{Exp}_{\mathfrak{m}}(ia\varepsilon), \quad u_0 \in \mathbb{T}(\mathbb{R}), \varepsilon \in \mathfrak{m}_{\mathbb{R}}. \quad (18.23)$$

It is a group automorphism with inverse $\Xi_{1/a}$, fixes every ordinary rotation, rescales every infinitesimal angle, and is continuous in the fine topology by [theorem 18.14](#). For $a \neq 1$ it is not algebraic: an algebraic endomorphism fixing all ordinary units has exponent 1 and is the identity. No basis of a large vector space is chosen. Likewise

$$u \mapsto \begin{pmatrix} 1 & \ell_{\mathfrak{m}}(u) \\ 0 & 1 \end{pmatrix} \in \text{GL}_2(\mathbf{No}) \quad (18.24)$$

is a homomorphism, nontrivial on infinitesimal rotations and trivial on ordinary ones. The line spanned by $(1, 0)$ is invariant and has no invariant complement, since a vector (x, y) with $y \neq 0$ is moved by $\ell_{\mathfrak{m}}(u)y$ in its first coordinate; so this fine-continuous representation is not semisimple, consistent with the semisimplicity of the algebraic representations above.

18.9 The fine topology of the circle

The fine topology is that of [proposition 2.4](#): balls $|z - a| < \rho$ with ρ any positive surreal, the circle carrying the induced topology. Coordinatewise order neighbourhoods give the same topology, because $\max\{|x|, |y|\} \leq |x + iy| \leq |x| + |y|$; addition, multiplication, inversion away from zero and conjugation are continuous by their elementary difference identities, so the circle is a class topological group in this explicit neighbourhood sense. *All positive surreal radii are allowed*: restricting to real radii, or to radii in one set-sized Hahn subfield, gives a different topology; in particular a set-sized Hahn field K_{Γ} has its own intrinsic order topology, while the topology it inherits from $\mathbf{SC}$ is discrete by [proposition 2.4](#). Enlarging the admissible radii changes the topology but no algebraic identity.

The existence of a positive surreal below any set of positive surreals, used in the proof of [proposition 2.4](#), also gives: *every set-indexed Cauchy net is eventually constant*. Indeed, take a positive lower bound for all its nonzero pairwise distances; on a tail all pairwise distances are smaller than it, hence zero. *This is special to the full surreal class*: it is not a theorem about a non-Archimedean real closed set field with its intrinsic topology, it does not forbid a properly class-indexed net of radii from approaching zero, and the restriction on index size is essential.

The subgroup $\mathfrak{m}_{\mathbb{R}} \subset \mathbf{No}$ is open, since it contains $(-\omega^{-1}, \omega^{-1})$ and its translates by elements of $\mathfrak{m}_{\mathbb{R}}$; an open subgroup is closed, its complement being a union of open cosets. The same holds for $\mathfrak{m}_{\mathbb{C}}$ and every nonzero multiple $\rho \mathfrak{m}_{\mathbb{C}}$. For $a \in \mathbf{SC}$ and $\rho > 0$ the set $a + \rho \mathfrak{m}_{\mathbb{C}}$ is therefore clopen and lies in the ball of radius ρ ; and if $a \neq b$, taking $\rho = |a - b|$ separates them, since $(b - a)/\rho$ has norm one and is not infinitesimal.

Proposition 18.13 (Total separation without isolated points). *The fine circle $\mathbb{T}(\mathbf{No})$ is totally separated and has a basis of clopen neighbourhoods, but has no isolated points. In particular topological connectedness is not the meaning of the algebraic connectedness of the norm-one torus.*

Proof. The clopen sets above separate any two points. Given u and $\rho > 0$, choose a positive infinitesimal $\eta < \rho/4$; then $0 < |\text{Exp}_{\mathfrak{m}}(i\eta) - 1| = 2\sin(\eta/2) < \eta < \rho$, so $u \text{Exp}_{\mathfrak{m}}(i\eta)$ is a point of the ball other than u . $\square$

So the circle is not discrete as a class, although every set-sized subset is; in the universe-relative reading, small subsets are discrete while the whole larger-universe carrier is not. No contradiction is involved.

Theorem 18.14 (The splitting is topological only with a discrete real factor). *With the fine topology on $\mathbb{T}(\mathbf{No})$ and on $\mathfrak{m}_{\mathbb{R}}$, the splitting [equation \(4.5\)](#) is a homeomorphism*

$$\mathbb{T}(\mathbf{No})_{\text{fine}} \cong \mathbb{T}(\mathbb{R})_{\text{discrete}} \times (\mathfrak{m}_{\mathbb{R}})_{\text{fine}}, \quad (18.25)$$

and the fine quotient topology on $\mathbb{T}(\mathbf{No})/\mathbb{T}_{\mathfrak{m}}$ is discrete, not the usual topology of the real circle.

Proof. The monad $\mathbb{T}_{\mathfrak{m}} = \mathbb{T}(\mathbf{No}) \cap (1 + \mathfrak{m}_{\mathbb{C}})$ is open, and its cosets are the fibres of st . Within one fibre, for infinitesimal ε, η , $\frac{1}{2}|\varepsilon - \eta| \leq |\text{Exp}_{\mathfrak{m}}(i\varepsilon) - \text{Exp}_{\mathfrak{m}}(i\eta)| \leq 2|\varepsilon - \eta|$, since the ratio is 1 plus an infinitesimal when $\varepsilon \neq \eta$ ([corollary 5.5](#)). So $\text{Exp}_{\mathfrak{m}}(i \cdot)$ and $\ell_{\mathfrak{m}}$ are mutually inverse continuous maps on each monad. All monads being open, the circle is their topological disjoint union, which is the product with a discrete real factor; every union of fibres is open, which gives the quotient statement. $\square$

The embedded ordinary circle is itself a closed discrete subspace; its usual topology is recovered only by a coarser choice ([section 18.10](#)), not as a subspace topology.

Corollary 18.15 (Small subgroups). *Every subgroup of the fine circle generated by a set is closed and discrete, and none is dense.*

Proof. Such a subgroup consists of finite words in the generators and their inverses, hence is a set, and [proposition 2.4](#) applies. It is not the whole circle, which is a proper class, the rational chart embedding all of $\mathbf{No}$. $\square$

In particular an ordinary irrational rotation has a countable orbit that is closed and discrete in the fine topology, although its orbit is dense in the usual real circle: pigeonholing $N + 1$ phases into N equal arcs gives a nonzero angular step below $2\pi/N$. Arbitrarily small *real* steps suffice for ordinary density but do not approach every surreal scale. There is consequently no ordinary real Lie-group interpretation of the fine circle: a set-sized compact subspace would be compact and discrete, hence finite, while no neighbourhood is finite. *Classical compactness arguments and Haar integration cannot be imported from the familiar picture of $\text{SO}(2)$.*

Strong sums are not fine limits. With $t = \omega^{-1}$, $\text{Exp}_{\mathfrak{m}}(it)$ is well defined, but the error of its degree- N Taylor polynomial has leading exponent $N + 1$, so for every ordinary N it exceeds the fixed positive surreal $t^{\omega} = \omega^{-\omega}$ in absolute value; the partial sums do not converge finely. Similarly $\text{Exp}_{\mathfrak{m}}(it^n)$ does not converge finely to 1, all its nonzero distances having finite valuation $n < \omega$. In the intrinsic topology of the field $\mathbb{R}((t^{\mathbb{Q}}))$, by contrast, the integers are cofinal in the value group, so $t^n \rightarrow 0$ and $\text{Exp}_{\mathfrak{m}}(it^n) \rightarrow 1$ there; in a higher-rank fixed field even this may fail if some positive value exceeds every $n v(t)$. These examples separate three notions: exact Hahn summation, convergence in a fixed workspace, and convergence at all surreal scales.

18.10 The standard-part topology and semialgebraic paths

The *standard-part topology* on $\mathbb{T}(\mathbf{No})$ has open sets $\text{st}^{-1}(V)$, V open in the usual topology of $\mathbb{T}(\mathbb{R})$. It deliberately forgets infinitesimal directions. It is a group topology, since st is a homomorphism, but it is not T_0 : distinct units with the same standard part have the same neighbourhoods, and

identifying topologically indistinguishable points yields the usual real circle. The map st from the fine circle to the usual circle is continuous, preimages of real open sets being unions of open monads, but it is not a quotient map: the quotient topology induced from the fine source is discrete ([theorem 18.14](#)). *These two uses of the same set-theoretic map must not be conflated.*

Let F now be a fixed real closed *set* field. In the semialgebraic category a path is a continuous semialgebraic map on $[0, 1]_F$, not an ordinary real-parameter path. A unit $u \neq -1$ is joined to 1 by

$$\gamma_u(s) = \frac{(1-s) + su}{|(1-s) + su|}, \quad 0 \leq s \leq 1, \quad (18.26)$$

whose denominator never vanishes, since a zero would make u a negative real unit, that is -1 ; 1 is joined to -1 by two such arcs through i , and multiplying by a starting unit joins any two points. The nonnegative square root is semialgebraic, so this proves semialgebraic path connectedness directly; the circle is closed and bounded, hence definably compact. The semialgebraic fundamental group is the usual one,

$$\pi_1^{\text{sa}}(\mathbb{T}(F), 1) \cong \mathbb{Z}; \quad (18.27)$$

this is imported, not proved here: it invokes the standard comparison and scalar-extension theorems of semialgebraic homotopy [19, 20], applicable because the circle is defined without parameters. There is no conflict with [proposition 2.4](#): the topology is intrinsic to F and the parameter interval is $[0, 1]_F$. The same formula on $[0, 1]_{\mathbf{No}}$ with the fine topology is a nonconstant continuous class map, but that domain is not connected in the ordinary sense; and $[0, 1]_{\mathbb{R}}$ with the topology induced from $\mathbf{No}$ is discrete, not the usual interval.

Four statements therefore hold simultaneously and answer different questions: the algebraic torus is geometrically connected of dimension one; its semialgebraic F -points are definably path connected; its full fine surreal points are totally separated; its standard-part quotient is the ordinary connected circle. Correspondingly the exact sequence $2\pi\mathbb{Z} \rightarrow \mathcal{O}_{\mathbb{R}} \rightarrow \mathbb{T}(\mathbf{No})$ is an algebraic group statement with local analytic charts, *not* a claim that a connected real line is a universal covering space of the fine circle; semialgebraic and ordinary covering theories have their own categories and hypotheses.

18.11 Local angular velocities

The fine derivative ([section 2](#)), which is local Taylor calculus and not differentiation of a convergent sequence of partial sums [17, §3], gives $\frac{d}{d\theta} \text{cis } \theta = i \text{cis } \theta$ and $\frac{d}{d\theta} \text{Rot}(\cos \theta, \sin \theta) = J \text{Rot}(\cos \theta, \sin \theta)$ on finite angles. For a fixed $a \in \mathbf{No}$ the canonical local flow is

$$\text{Rot}_a(x) = \text{Rot}(\cos(ax), \sin(ax)), \quad x \in D_a = \{x \in \mathbf{No} : ax \in \mathcal{O}_{\mathbb{R}}\}, \quad (18.28)$$

with $D_a = \mathbf{No}$ if $a = 0$ and otherwise $D_a = a^{-1}\mathcal{O}_{\mathbb{R}}$, an open additive subgroup; on D_a , $\text{Rot}_a(x+y) = \text{Rot}_a(x) \text{Rot}_a(y)$, $\text{Rot}'_a(x) = aJ \text{Rot}_a(x)$ and $\text{Rot}_a(0) = I$. Every surreal angular velocity is allowed locally, but an infinite velocity requires a correspondingly small canonical time domain.

Every extension Ψ_χ of [theorem 17.4](#) obeys the same local law [equation \(17.6\)](#), is fine-continuous, and has $\Psi'_\chi = i\Psi_\chi$ and $\Psi_\chi^{(n)} = i^n \Psi_\chi$. The extensions of [example 17.5](#) therefore have identical finite behaviour and identical local differential laws but can disagree at ω : *solving the local rotation equation and imposing $\Psi(0) = 1$ does not choose a global phase normalization*. A sharper warning concerns solutions not assumed to be homomorphisms.

Example 18.16 (A locally constant factor). Let $g(x) = 1$ for $x \in \mathfrak{m}_{\mathbb{R}}$ and $g(x) = -1$ otherwise. Both regions are fine-clopen, so g is locally constant with zero fine derivative everywhere, and is not

constant. On $\mathcal{O}_{\mathbb{R}}$ the function $f(x) = g(x) \operatorname{cis} x$ satisfies $f' = if$ and $f(0) = 1$, yet differs from $\operatorname{cis} x$ off the identity monad.

So even on a finite surreal interval the ordinary mean-value and initial-value uniqueness arguments cannot be used without further hypotheses. A coherent Taylor specification on every real-centred monad together with the ordinary real values does select cis ([proposition 3.4](#)); that is additional global data, not a consequence of fine differentiability, and the classification of [theorem 17.4](#) is not a classification of solutions of the fine differential equation. A derivation of the scalar field, such as a Berarducci–Mantova derivation [16], is yet another operation; no compatibility between it and a globally normalized value such as $\operatorname{Sin} \omega = 0$ is assumed.

18.12 Exact computation with rotations

Substantial rotation algebra can be implemented before arbitrary surreals are.

Representation	Best operations	Necessary caution
$(a, b), a^2 + b^2 = 1$	Action on coordinates, dot products, exact matrix operations	Normalizing an arbitrary pair needs a positive square root
Projective $[p : q]$	Composition equation (4.9) and inverse $[-p : q]$ by field arithmetic alone	Keep the point at infinity; equality is $ps - qr = 0$, not coordinatewise
$(u_0, \varepsilon), u_0 \in \mathbb{T}(\mathbb{R}), \varepsilon \in \mathfrak{m}_{\mathbb{R}}$	Products, matched roots, valuations, infinitesimal error accounting	Exact real phase data and Hahn support information must be represented honestly

In split form $(u_0, \varepsilon)(w_0, \eta) = (u_0 w_0, \varepsilon + \eta)$ and $(u_0, \varepsilon)^{-1} = (u_0^{-1}, -\varepsilon)$, exactly; a chosen N th-root branch divides ε by N and takes the specified ordinary root of u_0 , and [equation \(18.13\)](#) gives the valuation of propagated error in a finite-support approximation. *No representation of arbitrary surreal class elements by finite strings is asserted, and equality of effective Hahn-series names is not automatically decidable*; these concern representations and algorithms, not the definition of the group.

Source 19 proposes a layered formal development: finite algebra (matrix/unit-circle isomorphism, isometry rigidity, projective chart, action, invariants) over an ordered field with square roots; a valuation layer ($\mathcal{O}_{\mathbb{R}}$, $\mathfrak{m}_{\mathbb{R}}$, standard part, conjugation); a strong-series layer (exp/log inversion, purely imaginary logarithm of a near-one unit), which yields the splitting; ordinary complex phase for the finite-angle uniformization and kernel; a separate topology layer resting on the lower-bound cut; a semialgebraic layer; and a global layer resting on $\mathbf{No} = \Pi \oplus \mathcal{O}_{\mathbb{R}}$ rather than on a Taylor series at an infinite argument. It names the exact class-group splitting, with its fine-topological version, as a next formal target, and says this order is compatible with the repository’s division between generic finite algebra, Hahn evaluation and concrete surreal foundations. The collection’s formalization ledger `docs/FORMALIZATION.md` is not among the files source 19 lists as inspected. Already at its pin, and still now, that ledger records Lean proofs, on the actual surcomplex field, of several of these layers as stated in the analysis report — the infinitesimal exponential/logarithm group isomorphism, the purely imaginary logarithm of a unit, and surjectivity of the finite phase onto the unit circle with kernel exactly $2\pi\mathbb{Z}$ — and of the affine and projective circle charts over any ordered field (mapped, as prerequisites, to [theorem 4.5](#)). The splitting [equation \(4.5\)](#) itself, its fine-topological version [theorem 18.14](#), and every label of this section have no Lean mapping; nothing in this section is formally verified.

18.13 Comparison with the classical circle

Question	Ordinary real plane	Surreal plane
Origin-fixing rotations	$\mathrm{SO}(2, \mathbb{R})$	$\mathrm{SO}(2, \mathbf{No})$, a class group
Matrix form	$\mathrm{Rot}(a, b)$, $a^2 + b^2 = 1$	The same equations
Geometric phase domain	$\mathbb{R}/2\pi\mathbb{Z}$	$\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z}$
Infinitesimal rotations	Trivial	$\mathbf{T}_{\mathbf{m}} \cong (\mathbf{m}_{\mathbb{R}}, +)$
Canonical factorization	Ordinary phase	Ordinary phase times exact infinitesimal phase
Torsion; finite subgroups	Roots of unity; cyclic	The same literal roots of unity and μ_n
Roots of degree $n \geq 1$	Exactly n	Exactly n , one infinitesimal lift per standard root
Algebraic Lie algebra	$\mathbb{R}\mathbf{J}$	$\mathbf{No}\mathbf{J}$, in the universe-relative reading
Algebraic endomorphisms; weights	Integer power maps; integers	The same; still ordinary integers
Fine topology	The usual topology	Totally separated, no isolated points, set-sized subsets discrete
Ordinary real paths in the fine target	Nonconstant ones exist	All constant
Standard-part quotient	The circle itself	The usual circle, only for the coarser standard-part topology
Semialgebraic π_1	$\mathbb{Z}$	$\mathbb{Z}$ over a fixed real closed set field (imported)
Global canonical phase	Periodic exponential	$\Phi(x) = \mathrm{cis}(\mathrm{fin} x)$, kernel $2\pi\mathbf{Oz}$
Infinitely many repeated turns	Not a finite group operation	Still requires an explicit summation or phase convention

The resulting picture is neither “just the usual circle” nor “trigonometry breaks down”: the group has the familiar finite algebra, a very large exact infinitesimal angular factor, and several useful but inequivalent geometries, and which one is relevant depends on whether a question concerns finite equations, asymptotic scales, ordinary observations or definable paths. Higher-dimensional orthogonal groups are a natural extension, where infinitesimal logarithms are noncommutative and Baker–Campbell–Hausdorff terms appear; source 19 leaves them aside, and the collection now contains a separate report on the three-dimensional case, [docs/surreal/euclidean-three-space](#).

19 Period arithmetic and noncanonical surcomplex exponentials

This section is the contribution of sources 20 and 21. It continues [section 17](#): [theorem 17.4](#) says that every extension of the finite phase to all real surreal arguments is Ψ_{χ} for a character χ of Π , and [theorem 17.6](#) says that every such extension has infinite periods. Here we study the *arithmetic* of those periods: which of them remain periods after division by every ordinary integer, which scalars preserve all of them under multiplication, and whether a choice of χ can be made invariant under automorphisms of the underlying valued field. The main application is a partial negative answer to the first open question of [8, §11.1]: a surcomplex exponential with the canonical strip values, the modulus law and all local Taylor identities can have kernel-multiplier ring exactly $\mathbb{Z}$, which is initial in the surreal simplicity tree but is not an integer part ([theorem 19.33](#)). Only that specific sufficiency possibility is refuted; the question itself stays open ([section 19.8](#)).

19.1 What sources 20 and 21 contribute, and the conventions of this section

The two manuscripts. Source 20 is *Exponential Kernels and Arithmetic Rigidity in the Surcomplex Field: Minimal Multiplier Rings, Nonsplit Periods, and the Limits of Canonical Extension*; source 21 is *Period Arithmetic and Noncanonical Surcomplex Exponentials: Profinite Defects, Minimal Multiplier Rings, and a Partial Answer to an Ehrlich–Kaplan Question*. Both are dated September 22, 2026, and both were written against repository commit 465a54b479a1, at which this article stood exactly as it did before they were merged here (Sections 1–18 and the concluding section were unchanged between that commit and this merge). Both take [theorems 17.2, 17.4](#) and [17.6](#) and the local law ([17.6](#)) as prerequisites, restate [theorem 17.4](#) in turn coordinates, and do not claim these results; they are cited here and not proved again. Source 21 also records that no profinite-defect or minimal-multiplier construction was found in this article at that commit, which is correct.

One spine, two bases of hypothesis. The two manuscripts were written independently and prove the same central fact: a normalized surcomplex exponential can have kernel-multiplier ring $\mathbb{Z}$. Source 21 is the base text of this section, because it proves more under weaker hypotheses: its set-sized theorem ([theorem 19.26](#)) is proved in ZFC on a fixed continuum-sized field such as $\mathbf{No}(\varepsilon_0)$ or $F_{\mathbb{Q}} = \mathbb{R}((t^{\mathbb{Q}}))$; it prescribes an arbitrary ordinary character on all of Π and an arbitrary profinite defect; and it keeps the standard part of the phase fixed at every argument. Source 20’s own set-sized examples live on arbitrarily large, unspecified fragments $\mathbf{No}(\lambda)$ obtained by reflecting a construction on all of $\mathbf{No}$ made with global choice ([theorem 19.36](#)). Source 20 contributes what source 21 lacks: a relative construction with values in the ordinary circle, the reflection theorem, the external invariant $\mathrm{Hom}(\mathcal{I}, \mathbb{Z})$, the first-order sentence $\mathrm{RingKer}$, definability of $\mathbb{Z}$ with an omitted type, and the example of a non-initial multiplier ring. Source 21 contributes what source 20 lacks: the profinite defect and its realization, the cardinal count on $F_{\mathbb{Q}}$, cofinal divisible periods on $\mathbf{No}$, and the free action of valued-field translations with the shadow formula.

Printed once. The following results are proved in both manuscripts; each is printed once, at the place indicated, with both credited. The proofs agree in substance except where a second route is marked.

Result	Source 20	Source 21	Printed as
Finite phase in turns, kernel $\mathbb{Z}$	Prop. 3.1	Prop. 2.1	theorem 4.2 (radians); not reprinted
Character coordinates of a phase	Prop. 3.3	Prop. 3.2	theorem 17.4 ; coordinates in proposition 19.2
Phases $\leftrightarrow$ exponentials; modulus, conjugation, kernel	Prop. 3.4	Prop. 8.1	proposition 19.5
Kernels are exactly the additive integer parts	Thm. 4.1	Thm. 3.4	theorem 19.3
Local law and all fine derivatives	Prop. 10.1	Props. 3.6, 8.3	equation (17.6) ; proposition 19.6
Split period sequence $\Leftrightarrow$ additive angle lift $\Leftrightarrow$ zero defect	Thm. 7.1	Thm. 4.5	theorem 19.10
Explicit nonliftable profinite character	Lemma 7.2 (3-adic)	§10 (2-adic)	example 19.16 , both kept
Multiplier ring and the integer-part criterion	Lemma 5.1, Thm. 5.2	Props. 7.1, 8.2, eq. (8.7)	proposition 19.19 and theorem 19.20
Minimal multiplier ring $\mathbb{Z}$ by two-vector avoidance	Lemma 6.1, Thms. 6.2, 6.4	Lemmas 7.2–7.6, Thms. 7.5, 7.7	theorems 19.26 and 19.28 ; source 20’s different route is theorem 19.31

Result	Source 20	Source 21	Printed as
Initial multiplier ring that is not an integer part	Cor. 6.5	Thm. 8.4	theorem 19.33
Strong shift automorphisms	Thm. 11.1	Thm. 9.2	theorem 19.43 , which is [23, Theorem 5.2]
No phase equivariant under valued-field automorphisms	Thm. 11.2 ($c = 1/2$)	Thm. 9.3, Cor. 9.4	theorem 19.44 ; source 20's case in remark 19.46

Credits the sources owe. The shift automorphisms of [theorem 19.43](#) are the fixed-shift flows of the collection's report on field automorphisms of **SC** [23, Theorem 5.2, label `saut:thm:shiftflow`], with its parameters $\delta = 1$ and $\lambda = [\cdot]_0$ and the sign of the flow parameter reversed; source 21 cites that theorem and [22, §4], source 20 cites neither the repository report nor the flows. The residue map, the maximal divisible subgroup and the splitting and homomorphism facts behind [proposition 19.8](#) and [theorems 19.10](#) and [19.11](#) are classical facts about $\mathbb{Z}$ -groups [21, §§2–4]; source 21 says so, source 20 proves its versions without this citation. Neither source claims them as new, and neither is credited with them here.

Where the merge had to choose. (i) Source 21's text is the base, and source 20's statements are inserted where it has no counterpart. (ii) Where both prove one result, one proof is printed. (iii) Source 20's minimal-multiplier construction is not the same theorem as source 21's: it changes the ordinary character (outside a prescribed set-sized subspace) and keeps the infinitesimal character zero, so that all phase values lie in the ordinary circle, which is what its reflection theorem needs; source 21 fixes the ordinary character everywhere and changes only the infinitesimal character. Both are printed ([theorem 19.31](#) and [theorems 19.26](#) and [19.28](#)). (iv) Source 20's non-equivariance theorem is the special case $c = 1/2$ of source 21's free action; source 20's shorter proof is kept as a remark. (v) Two consequences are obtained by combining the sources and are stated by neither: source 20's first-order sentence separates source 21's example on **No**(ε_0) from the canonical exponential, and source 21's defects give three pairwise nonisomorphic exponentials on **No**(ε_0)[i] in ZFC ([corollary 19.40](#)). Both are marked where they are stated.

Set theory, kept attached to each statement. Statements about a set-sized field are theorems of ZFC. Statements about a class map on all of **No** are made in NBG, reading “all phases” as a scheme with a class parameter rather than as a class whose elements are proper classes; every normal form, coefficient family and recursion stage is a set, and a quotient of a proper class is never treated as a set of classes. *Global choice* (NBG with global choice, written GBC) is used exactly where it is named: in [theorem 19.28](#), in [theorem 19.31](#) and its variants, in the **No**[i] case of [theorem 19.33](#), and to build the class maps that [theorem 19.36](#) reflects. The realization theorem ([theorem 19.14](#)) uses only a choice between set-sized vector spaces, even for a defect defined on all of Π . The set-sized consequences of the global-choice constructions rest on the conservativity of GBC over ZFC for statements about sets; no large cardinal, saturation hypothesis or continuum hypothesis is used anywhere in this section.

Radians, turns, and the symbols of this section. This article measures angles in radians: the finite phase is cis with kernel $2\pi\mathbb{Z}$, and Φ , Ψ_χ , Exp and E_χ are exactly the objects of [section 17](#). Both sources measure phases in *turns*, with finite phase $a \mapsto e^{2\pi ia}$ and kernel $\mathbb{Z}$; a turn-phase at x is here the radian phase at $2\pi x$. Because the arithmetic concerns the ordinary integers, periods are divided by 2π once and for all: the *normalized period group* of a phase is $\mathcal{I}(\Psi) = \{a : \Psi(2\pi a) = 1\}$ ([definition 19.1](#)). The remaining conversions are:

- Purely infinite part: the sources' $\mathcal{P}$ is Π (the letter $\mathbb{P}$ is used for the projective line). Principal part $\text{pr}(x) = x - \text{fin } x \in \Pi$ (source 21's pr ; source 20 writes $p(x)$).
- Phases: source 21's admissible phase Φ and canonical phase Φ_0 , and source 20's normalized phase c and canonical phase c_0 , are here Ψ and Φ ; the letter Φ keeps its meaning from [definition 17.1](#) and never denotes an arbitrary phase (other reports of the collection use Φ for unrelated objects, none of which occurs here). Source 20's circle $\mathcal{U}_F$ is $\mathbb{T}(F)$, and its $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ is always written $\mathbb{R}/\mathbb{Z}$ here, since $\mathbb{T}(F)$ is the unit circle.
- Characters: the character $\chi : \Pi \rightarrow \mathbb{T}(\mathbf{No})$ of [theorem 17.4](#) has coordinates $(\psi, \mathcal{L})$ ([proposition 19.2](#)). Source 21's pair (h, L) is $(\psi, \mathcal{L})$; source 20's ordinary-circle character $\chi : \mathcal{P} \rightarrow \mathbb{R}/\mathbb{Z}$ is ψ with $\mathcal{L} = 0$, and its general character η is our χ . The letter h keeps its meaning as a generic infinitesimal increment, and L its meaning in [theorem 17.7](#).
- Exponentials: $E_\chi(x + iy) = \exp_{\mathbf{No}}(x)\Psi_\chi(y)$ as in [theorem 17.4](#); source 20's E_χ with $y/(2\pi)$ and a turn-valued χ is our E_χ for the character with coordinates $(\chi, 0)$. Source 20's finite phase e_0 and circle map $e(\theta)$ are $f \mapsto \text{cis}(2\pi f)$ and $\theta \mapsto e^{2\pi i\theta}$; source 21's $e_{\mathcal{O}}$ and $e_{\mathbb{R}}$ are the same two maps. (In their sources one macro name stands for $\exp_{\mathbf{No}}$ in source 20 and for $r \mapsto e^{2\pi ir}$ in source 21; the printed symbols differ, and neither is used here.) The strip exponential of [theorem 10.1](#) keeps the letter E without subscript.
- Periods and multipliers: source 20's normalized kernel D and source 21's I_Φ are $\mathcal{I}$; source 21's maximal divisible subgroup D_Φ or $D(I)$ is $\mathcal{I}^{\text{div}}$ (so neither source's D survives); source 20's $M(E)$ and source 21's Z_E are Z_E ; Mult is unchanged.
- Profinite objects: source 21's defect δ_Φ is pd_Ψ (the letter δ is the side gap $b + c - a$, and “defect” alone means the angle defect $d(\alpha)$ or the triangle-inequality defect, so this one is always called the *profinite defect*); its residue map ρ_I is $\text{res}_{\mathcal{I}}$; its $\mathcal{V} = \widehat{\mathbb{Z}}/\mathbb{Z}$ is written $\widehat{\mathbb{Z}}/\mathbb{Z}$; its Σ, ∂, s are $\mathbb{Q}^\vee, \text{cl}, \varsigma$; its continuum $\mathfrak{c}$ is $2^{\aleph_0}$; its representative $\theta(p)$ is $\bar{\psi}(p)$; its retraction r is $\mathfrak{r}$.
- Fields: source 21's $K_\Gamma = \mathbb{R}((t^\Gamma))$ is F_Γ of [section 1.3](#), since K_Γ denotes $\mathbb{C}((t^\Gamma))$ in this article and in the collection's notation guide; its K_0 and K_ε are $F_{\mathbb{Q}}$ and $\mathbf{No}(\varepsilon_0)$. Source 20's F keeps its name here; birthdays are written out in words.
- Automorphisms: source 21's T_c and source 20's σ_a are Sh_c (T_n is a Chebyshev polynomial, σ a valuation threshold, and T_s in [23] has the opposite sign); their exponent map $\ell(g)$ is $[g]_0$, the real coefficient of ω^0 in the exponent g , since $\ell(p)$ is the coefficient of ω in [example 17.5](#). Source 21's $[X]_p$ is that $\ell(p)$. Source 20's witness monomial h_x is μ_x .

19.2 Phases on a field, periods, and exponentials

Fields. In this section F is one of the following: $\mathbf{No}$; a set-sized initial trigonometric-exponential subfield $\mathbf{No}(\lambda) = \{x \in \mathbf{No} : \text{the birthday of } x \text{ is less than } \lambda\}$ for an epsilon number λ , in particular $\mathbf{No}(\varepsilon_0)$ [8, §11.2, Theorem 11.1], [7]; or, where no real exponential is needed, a Hahn field $F_\Gamma = \mathbb{R}((t^\Gamma))$, in particular $F_{\mathbb{Q}}$, or more generally a non-Archimedean truncation-closed subfield of $\mathbf{No}$ containing $\mathbb{R}$ on which the finite phase is available. Initial subfields are truncation closed. $\mathbf{No}(\varepsilon_0)$ has cardinality $2^{\aleph_0}$, since its elements have countable sign sequences, of countably many lengths, and it contains $\mathbb{R}$; it is *not* asserted to be a full Hahn field. Write $\mathcal{O}_F, \mathfrak{m}_F, \Pi_F$ for the finite, infinitesimal and purely infinite elements of F ; truncation closure gives $F = \Pi_F \oplus \mathcal{O}_F$ and $\mathcal{O}_F = \mathbb{R} \oplus \mathfrak{m}_F$. The group Π_F is a $\mathbb{Q}$ -vector space and a nonunital subring, and its nonzero elements are infinite. The finite phase $\text{cis} : (\mathcal{O}_F, +) \rightarrow \mathbb{T}(F)$ is onto with kernel $2\pi\mathbb{Z}$, and $\mathbb{T}(F) \cong \mathbb{T}(\mathbb{R}) \times (\mathfrak{m}_F, +)$ through $\ell_{\mathfrak{m}}$

of equation (18.3): this is theorem 4.2 for **No**, and the same proof for these F , as both sources note. The occurrence of the ordinary $\mathbb{Z}$, not **Oz**, in this kernel is essential.

Definition 19.1 (Phases and normalized periods). A *phase* on F is a homomorphism $\Psi : (F, +) \rightarrow \mathbb{T}(F)$ whose restriction to $\mathcal{O}_F$ is *cis*. Its *normalized period group* is $\mathcal{I}(\Psi) = \{a \in F : \Psi(2\pi a) = 1\}$, so that $\ker \Psi = 2\pi\mathcal{I}(\Psi)$. An *additive integer part* of F is an additive subgroup $\mathcal{I} \subseteq F$ containing 1, with least positive element 1, such that every $x \in F$ has an $a \in \mathcal{I}$ with $a \leq x < a + 1$; no multiplicative closure is included. An *integer part* is an additive integer part that is a subring.

The canonical phase of definition 17.1 has $\mathcal{I}(\Phi) = \mathbf{Oz}$; on an initial subfield F it has $\mathcal{I} = \mathbf{Oz} \cap F$, the unique initial integer part of F [8, Lemma 11.1], and on F_Γ the phase $\text{cis} \circ \text{fin}$ has $\mathcal{I} = \Pi_F \oplus \mathbb{Z}$. *Initial* means closed under proper sign-sequence prefixes; it does not mean small, ordered or definable.

Proposition 19.2 (Coordinates of a phase; restating theorem 17.4). *Every phase on F is $\Psi_\chi(p+f) = \chi(p) \text{cis } f$ for a unique homomorphism $\chi : \Pi_F \rightarrow \mathbb{T}(F)$, and there are unique maps $\psi : \Pi_F \rightarrow \mathbb{R}/\mathbb{Z}$, additive, and $\mathcal{L} : \Pi_F \rightarrow \mathfrak{m}_F$, $\mathbb{Q}$ -linear, with*

$$\chi(2\pi p) = e^{2\pi i \psi(p)} \text{Exp}_m(2\pi i \mathcal{L}(p)) \quad (p \in \Pi_F). \quad (19.1)$$

Every such pair occurs; we write $\Psi_{\psi, \mathcal{L}}$ for the phase. The canonical phase is $\Psi_{0,0} = \Phi$.

Proof. The first assertion is theorem 17.4, whose proof applies verbatim on F (sources 20 and 21 state it so). Since $\Pi_F = 2\pi\Pi_F$, compose $p \mapsto \chi(2\pi p)$ with the splitting $\mathbb{T}(F) \cong \mathbb{T}(\mathbb{R}) \times \mathfrak{m}_F$: the first coordinate is an additive map to $\mathbb{T}(\mathbb{R}) \cong \mathbb{R}/\mathbb{Z}$, the second is additive between uniquely divisible groups, hence $\mathbb{Q}$ -linear. $\square$

Source 21 uses these coordinates to separate the ordinary phase from the infinitesimal one; source 20 works mainly with $\mathcal{L} = 0$, where every value of χ lies in the ordinary circle.

Theorem 19.3 (Kernel dictionary). *The map $\Psi \mapsto \mathcal{I}(\Psi)$ is a bijection from phases on F onto additive integer parts of F . Every such $\mathcal{I}$ has unique floors, satisfies $\mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$, and sits in an exact sequence*

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{I} \xrightarrow{\text{pr}} \Pi_F \longrightarrow 0. \quad (19.2)$$

In particular a phase is determined by its kernel. If $\Pi_F \neq 0$, then $\mathcal{I}$ contains infinite elements and is not cyclic.

Proof. Let Ψ be a phase. The finite restriction gives $\mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$. For $x \in F$ choose finite f with $\text{cis}(2\pi f) = \Psi(2\pi x)$, so $x - f \in \mathcal{I}$. Every finite $f = r + \varepsilon$ has a unique ordinary integer n with $n \leq f < n+1$: the ordinary floor of r , lowered by one when $r \in \mathbb{Z}$ and $\varepsilon < 0$. Then $a = x - f + n \in \mathcal{I}$ is a floor of x ; two floors differ by an element of $\mathcal{I}$ of absolute value less than one, hence by 0, and 1 is least positive since $(0, 1)$ is finite and meets $\mathcal{I}$ in nothing. Applying the same correction to $p \in \Pi_F$ proves that $\text{pr} : \mathcal{I} \rightarrow \Pi_F$ is onto; its kernel is $\mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$.

Conversely let $\mathcal{I}$ be an additive integer part. A finite element of $\mathcal{I}$ minus its ordinary floor lies in $\mathcal{I} \cap [0, 1) = \{0\}$, so $\mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$, and floors are unique. Put $\Psi_{\mathcal{I}}(y) = \text{cis}(y - 2\pi \lfloor y/2\pi \rfloor_{\mathcal{I}})$. For two arguments the floors of the sum and of the summands differ by 0 or 1, which $\text{cis}(2\pi \cdot)$ ignores, so $\Psi_{\mathcal{I}}$ is a homomorphism; it restricts to *cis* on $\mathcal{O}_F$ and has normalized period group $\mathcal{I}$. Two phases with the same $\mathcal{I}$ agree at $2\pi x$, because both take there the finite phase of x minus its common floor.

For $p \neq 0$ in Π_F every element of the fibre of pr over p is infinite. If $\mathcal{I} = \tau\mathbb{Z}$ were cyclic, then $1 = n\tau$ for an ordinary $n \neq 0$, and every element of $\mathcal{I}$ would be finite. $\square$

The dictionary is source 20's Theorem 4.1 with Corollary 4.2 and source 21's Theorem 3.4; the proof printed is source 21's. That $\mathcal{I}$ meets every coset $p + \mathcal{O}_F$ is [theorem 17.6](#) in normalized form; the floor formulation isolates the additional arithmetic used below. For $F = \mathbf{No}$ and $\Psi = \Phi$ the floor is Ehrlich and Kaplan's integer-part floor [8, §11].

Proposition 19.4 (Division with remainder; source 20). *For every ordinary $n \geq 1$ and $d \in \mathcal{I}$ there are unique $e \in \mathcal{I}$ and $r \in \{0, \dots, n-1\}$ with $d = ne + r$. Hence $\mathcal{I}/n\mathcal{I} \cong \mathbb{Z}/n\mathbb{Z}$, induced by the inclusion of $\mathbb{Z}$.*

Proof. Let e be the $\mathcal{I}$ -floor of d/n ; then $r = d - ne \in \mathcal{I} \cap [0, n)$ is an ordinary integer. If $ne + r = ne' + r'$, then $e - e' = (r' - r)/n \in \mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$ has absolute value less than one. $\square$

This arithmetic is additive; it does not allow arbitrary elements of $\mathcal{I}$ to be multiplied inside $\mathcal{I}$, and that is exactly where the counterexamples below enter. For set-sized $\mathcal{I}$ the ordered group $\mathcal{I}$ is discretely ordered with divisible quotient $\mathcal{I}/\mathbb{Z} \cong \Pi_F$, that is, a $\mathbb{Z}$ -group, a model of Presburger arithmetic in its ordered-group presentation [21, §2]; this does not make it a model of arithmetic with multiplication.

Related (batch 33). For $\mathcal{I} = \mathbf{Oz}$ the collection's report on exponential relations over omnific integers (`docs/foundations-and-computation/exponential-relations-over-omnific-integers`) re-derives this proposition independently (label `exr:prop:division`), with $x \equiv y \pmod{n}$ exactly when the constant terms of x and y are congruent, and reads the class-sized $\mathbf{Oz}$ formula by formula as a $\mathbb{Z}$ -group. It adds the zero predicates of finite exponential sums with algebraic coefficients and real algebraic slopes. With rational slopes the result is a definitional expansion of Presburger arithmetic, and $\mathbb{Z}$ is an elementary substructure (`exr:thm:rational`); with algebraic slopes the theory is still decidable (`exr:thm:algdecision`). The boundary is the one stated above: adjoining the graph of multiplication, as the relation $\exp(xy) = \exp(z)$, makes the theory undecidable (`exr:thm:undecidable`).

Proposition 19.5 (Phases and exponentials). *Let F carry a real exponential $\exp_F$, i.e. $F = \mathbf{No}$ or $F = \mathbf{No}(\lambda)$ with λ an epsilon number. For every phase Ψ_χ on F , $E_\chi(x + iy) = \exp_F(x)\Psi_\chi(y)$ is a surjective homomorphism $(F[i], +) \rightarrow (F[i]^\times, \cdot)$ extending $\exp_F$ and the strip exponential of [theorem 10.1](#), with*

$$|E_\chi(z)| = \exp_F(\operatorname{Re} z), \quad E_\chi(\bar{z}) = \overline{E_\chi(z)}, \quad \ker E_\chi = 2\pi i \mathcal{I}(\Psi_\chi). \quad (19.3)$$

Conversely every additive-to-multiplicative homomorphism $E : F[i] \rightarrow F[i]^\times$ that agrees with the strip exponential at finite imaginary arguments and satisfies $|E(z)| = \exp_F(\operatorname{Re} z)$ is E_χ for exactly one χ , namely the restriction of $y \mapsto E(iy)$ to Π_F .

Proof. The forward assertions other than conjugation and the kernel are [theorem 17.4](#). Conjugation follows from $\Psi(-y) = \Psi(y)^{-1} = \overline{\Psi(y)}$. A kernel element has modulus $\exp_F(x) = 1$, so $x = 0$, and then $\Psi_\chi(y) = 1$ means $y \in 2\pi\mathcal{I}$. Conversely $\Psi(y) = E(iy)$ has modulus one, is a homomorphism, and is cis on finite y ; on the real axis $E = \exp_F$; the group law gives $E(x + iy) = \exp_F(x)\Psi(y)$. $\square$

Ehrlich and Kaplan establish the integer-part case [8, Proposition 11.2]; here the formula is used for additive integer parts that are not rings. The proposition is source 20's Proposition 3.4 and source 21's Proposition 8.1.

Related (batch 33). For $F = \mathbf{No}$, on the additive domain $\overline{\mathbb{Q}} + \Pi$, whose imaginary parts are finite algebraic numbers, every E_χ restricts to $a + p \mapsto e^a \exp_{\mathbf{No}}(p)$, the exponential of the report

on exponential relations over omnific integers (`exr:def:domain`). There the values at finitely many distinct points are linearly independent over $\overline{\mathbb{Q}}$ (`exr:thm:independence`, from Gonshor's monomial theorem and the Lindemann–Weierstrass theorem). So all the E_χ agree on that domain and are injective there; the choice of phase classified here does not enter.

Proposition 19.6 (The local law, for every phase). *For every phase Ψ on F , every $x \in F$, $h \in \mathfrak{m}_F$, and, when F has a real exponential, every $z \in F[i]$ and $w \in \mathfrak{m}_F[i]$,*

$$\Psi(x+h) = \Psi(x) \operatorname{Exp}_{\mathfrak{m}}(ih), \quad E_\chi(z+w) = E_\chi(z) \operatorname{Exp}_{\mathfrak{m}}(w) = E_\chi(z) \sum_{n \geq 0} \frac{w^n}{n!}.$$

In the fine sense, with tolerances ranging over the positive elements of F , $\Psi^{(n)} = i^n \Psi$ and $E_\chi^{(n)} = E_\chi$ for every ordinary n , and Ψ is uniformly continuous for the fine additive uniformity.

Proof. The identities are [equation \(17.6\)](#) and its exponential form, on F . For the fine derivative write $\operatorname{Exp}_{\mathfrak{m}}(w) = 1 + w + w^2 R(w)$; for infinitesimal w the factor $R(w)$ is finite with standard part $1/2$, so $|R(w)| < 1$. Given a positive tolerance $\epsilon \in F$, choose a positive infinitesimal $\varrho \in F$; then $0 < |w| < \min\{\varrho, \epsilon/(1 + |E_\chi(z)|)\}$ gives $|(E_\chi(z+w) - E_\chi(z))/w - E_\chi(z)| \leq |E_\chi(z)||w| < \epsilon$, and induction gives the higher derivatives; the phase is treated in the same way. Finally $|\Psi(x+h) - \Psi(x)| = |\operatorname{cis} h - 1|$ does not depend on x , and $(\operatorname{cis} h - 1)/h$ is finite with standard part i , so $|h| < \min\{\varrho, \epsilon/2\}$ forces $|\Psi(x+h) - \Psi(x)| < \epsilon$. $\square$

This is source 20's Proposition 10.1 and source 21's Propositions 3.6 and 8.3; it is inherited, not a new uniqueness theorem. The sums are strong Hahn sums, not limits of finite partial sums in the fine topology. *The local law does not remove the freedom:* for two phases Ψ_1, Ψ_2 the quotient $\Psi_1 \Psi_2^{-1}$ is a homomorphism trivial on $\mathcal{O}_F$, hence constant on every coset of $\mathcal{O}_F$; those cosets are fine-open, so the quotient is locally constant and invisible to differentiation (source 20). The derivative rules are the same for all extensions; they do not say that the values of distinct exponentials agree at infinite points.

19.3 Profinite residues and the profinite defect

Let $\widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$, with $\mathbb{Z}$ embedded diagonally. Throughout, $\widehat{\mathbb{Z}}/\mathbb{Z}$ is an *abstract* additive group, not a separated topological quotient ($\mathbb{Z}$ is dense in $\widehat{\mathbb{Z}}$).

Lemma 19.7. *$\widehat{\mathbb{Z}}/\mathbb{Z}$ is uniquely divisible, hence a $\mathbb{Q}$ -vector space, of cardinality and $\mathbb{Q}$ -dimension $2^{\aleph_0}$.*

Proof. Multiplication by $n > 0$ is injective on $\widehat{\mathbb{Z}}$ with image the kernel of reduction modulo n ; for $\alpha \in \widehat{\mathbb{Z}}$ choose $r \in \mathbb{Z}$ with $\alpha \equiv r$ modulo n , so $(\alpha - r)/n \in \widehat{\mathbb{Z}}$. If $n\alpha = m \in \mathbb{Z}$ then $n \mid m$ and $\alpha = m/n \in \mathbb{Z}$, so the quotient is torsion free. $\widehat{\mathbb{Z}}$ has cardinality $2^{\aleph_0}$, a countable quotient leaves it unchanged, and an uncountable $\mathbb{Q}$ -vector space has cardinality equal to its dimension. $\square$

Let $\mathcal{I} = \mathcal{I}(\Psi)$. For $a \in \mathcal{I}$, $\Psi(2\pi a/n)$ is an n th root of unity, hence one of the ordinary ones, $\Psi(2\pi a/n) = e^{2\pi i r_n(a)/n}$ with $r_n(a) \in \mathbb{Z}/n\mathbb{Z}$. These residues are compatible under divisibility, so $\operatorname{res}_{\mathcal{I}}(a) = (r_n(a))_n \in \widehat{\mathbb{Z}}$, and

$$a - r \in n\mathcal{I} \iff r_n(a) \equiv r \pmod{n}. \quad (19.4)$$

Proposition 19.8 (The maximal divisible subgroup; classical). $\text{res}_{\mathcal{I}}$ is additive and restricts to the diagonal embedding on $\mathbb{Z}$, and

$$\mathcal{I}^{\text{div}} := \ker \text{res}_{\mathcal{I}} = \bigcap_{n \geq 1} n\mathcal{I} = \{a \in \mathcal{I} : \mathbb{Q}a \subseteq \mathcal{I}\}$$

is the maximal divisible subgroup of $\mathcal{I}$.

Proof. Additivity is immediate and [equation \(19.4\)](#) gives the intersection formula. In the uniquely divisible ambient group, $a \in n\mathcal{I}$ for every n says $a/n \in \mathcal{I}$ for every n , i.e. $\mathbb{Q}a \subseteq \mathcal{I}$; every divisible subgroup of $\mathcal{I}$ has this property. $\square$

This is the residue map and divisible kernel of the $\mathbb{Z}$ -group $\mathcal{I}$ [21, Proposition 4.1 and Remark 4.2], read off from a phase; source 21 presents it so.

Theorem 19.9 (The profinite defect and its kernel; source 21). For $p \in \Pi_F$ choose $a \in \mathcal{I}$ with $\text{pr}(a) = p$ and put $\text{pd}_{\Psi}(p) = \text{res}_{\mathcal{I}}(a) + \mathbb{Z} \in \widehat{\mathbb{Z}}/\mathbb{Z}$. This is well defined and $\mathbb{Q}$ -linear, and pr restricts to an isomorphism $\mathcal{I}^{\text{div}} \cong \ker \text{pd}_{\Psi}$. Writing $q : \widehat{\mathbb{Z}} \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$,

$$\text{res}_{\mathcal{I}}(\mathcal{I}) = q^{-1}(\text{pd}_{\Psi}(\Pi_F)), \quad \mathcal{I}/\mathcal{I}^{\text{div}} \cong q^{-1}(\text{pd}_{\Psi}(\Pi_F)) \subseteq \widehat{\mathbb{Z}},$$

so the nondivisible quotient is represented in a set of cardinality at most $2^{\aleph_0}$. For set-sized F , $\Pi_F/\text{pr}(\mathcal{I}^{\text{div}}) \cong \text{pd}_{\Psi}(\Pi_F)$ has $\mathbb{Q}$ -dimension at most $2^{\aleph_0}$; for $F = \mathbf{No}$ the same quotient is represented by the class map pd_{Ψ} with set-sized range, without forming a set of proper classes.

Proof. Two lifts differ by an element of $\mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$, whose residue is diagonal; lifts of sums may be added; source and target are uniquely divisible. A divisible subgroup meets $\mathbb{Z}$ trivially, so pr is injective on $\mathcal{I}^{\text{div}}$, and its image lies in $\ker \text{pd}_{\Psi}$. If $\text{pd}_{\Psi}(p) = 0$, a lift a has $\text{res}_{\mathcal{I}}(a) = k \in \mathbb{Z}$ and $a - k \in \mathcal{I}^{\text{div}}$ has principal part p . If $q(\alpha) = \text{pd}_{\Psi}(p)$ and a lifts p , then $\alpha = \text{res}_{\mathcal{I}}(a + k)$ for some $k \in \mathbb{Z}$; the isomorphism is the first isomorphism theorem, read in the class case as the displayed residue map, whose fibres are the cosets of $\mathcal{I}^{\text{div}}$. $\square$

Theorem 19.10 (When the period sequence splits). For a phase $\Psi = \Psi_{\psi, \mathcal{L}}$ with $\chi = \Psi|_{\Pi_F}$ the following are equivalent:

- (1) $\text{pd}_{\Psi} = 0$;
- (2) the sequence [\(19.2\)](#) splits as additive groups;
- (3) there is an additive retraction $\mathfrak{r} : F \rightarrow \mathcal{O}_F$, the identity on $\mathcal{O}_F$, with $\Psi(2\pi x) = \text{cis}(2\pi \mathfrak{r}(x))$ for all x ;
- (4) there is an additive $H : \Pi_F \rightarrow \mathcal{O}_F$ with $\chi(2\pi p) = \text{cis}(2\pi H(p))$;
- (5) the ordinary character ψ has a $\mathbb{Q}$ -linear lift $\Pi_F \rightarrow \mathbb{R}$ through $\mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$.

The retraction in (3) is unique when it exists, $\mathcal{I} = \ker \mathfrak{r} \oplus \mathbb{Z}$, and $\mathcal{I}^{\text{div}} = \ker \mathfrak{r}$; every additive H as in (4) is $\mathbb{Q}$ -linear.

Proof. (1) $\Rightarrow$ (3): the isomorphism of [theorem 19.9](#) gives a section $\iota : \Pi_F \rightarrow \mathcal{I}^{\text{div}}$; $\mathfrak{r}(x) = x - \iota(\text{pr } x)$ is finite, additive, fixes $\mathcal{O}_F$, and differs from x by a period. (3) $\Rightarrow$ (2): $p \mapsto p - \mathfrak{r}(p)$ is a section. (2) $\Rightarrow$ (1): an additive section has divisible image, hence lies in $\mathcal{I}^{\text{div}}$, and it meets every principal part. If $\mathfrak{r}_1, \mathfrak{r}_2$ give the same phase, $\mathfrak{r}_1 - \mathfrak{r}_2$ is an additive map from the divisible group F to $\mathbb{Z}$, hence zero; $\mathfrak{r}(x) \in \mathbb{Z}$ characterizes $\mathcal{I}$, and all rational divisions of x lie in $\mathcal{I}$ exactly when $\mathfrak{r}(x) = 0$.

(2) $\Leftrightarrow$ (4) (source 20's route): given H , $p \mapsto p - H(p)$ is an additive section into $\mathcal{I}$; given a section ι , $H(p) = p - \iota(p)$ is finite and additive with $\chi(2\pi p) = \text{cis}(2\pi H(p))$. Additive maps between torsion-free divisible groups are $\mathbb{Q}$ -linear. (4) $\Leftrightarrow$ (5): write $H = A + \mathcal{E}$ with A real and $\mathcal{E}$ infinitesimal; by the splitting of $\mathbb{T}(F)$, [equation \(19.1\)](#) and $\chi(2\pi p) = \text{cis}(2\pi H(p))$ are equivalent to $A + \mathbb{Z} = \psi$ and $\mathcal{E} = \mathcal{L}$. The infinitesimal coordinate always lifts; all obstruction lies in ψ . $\square$

This is source 21's Theorem 4.5 (conditions (1)–(3)) and source 20's Theorem 7.1 (conditions (2), (4), (5)). A real argument for each single value of χ always exists; condition (5) asks for them *simultaneously and additively*. Replacing an arbitrary character by a real-valued linear functional therefore discards every nonzero profinite defect, while a zero defect says nothing about multiplicative closure ([section 19.6](#)).

Theorem 19.11 (Homomorphisms to $\mathbb{Z}$; source 20). *The sequence [\(19.2\)](#) splits if and only if there is a nonzero homomorphism $\mathcal{I} \rightarrow \mathbb{Z}$. For set-sized F , $\text{Hom}(\mathcal{I}, \mathbb{Z}) \cong \mathbb{Z}$ if it splits and $\text{Hom}(\mathcal{I}, \mathbb{Z}) = 0$ if it does not. For $F = \mathbf{No}$ the statement concerns existence and pointwise description of class homomorphisms, not a set of all of them.*

Proof. If it splits, $\mathcal{I} \cong \mathbb{Z} \oplus \Pi_F$, and every homomorphism from the divisible group Π_F to $\mathbb{Z}$ vanishes, so the homomorphisms are the multiples of the first projection. Conversely let $\varphi : \mathcal{I} \rightarrow \mathbb{Z}$ be nonzero and $m = \varphi(1)$; if $m = 0$, φ factors through $\mathcal{I}/\mathbb{Z} \cong \Pi_F$ and vanishes. By [proposition 19.4](#) with $n = |m|$, $d = ne + r$ gives $\varphi(d) = n\varphi(e) + rm \in m\mathbb{Z}$, so φ/m is a retraction of $\mathbb{Z} \hookrightarrow \mathcal{I}$, and pr maps its kernel bijectively onto Π_F . $\square$

The group facts in [theorems 19.10](#) and [19.11](#) are classical for $\mathbb{Z}$ -groups [[21](#), §§2–4]; source 20 proves them without that citation.

19.4 Realizing every profinite defect

Let $\mathbb{Q}^\vee = \text{Hom}(\mathbb{Q}, \mathbb{R}/\mathbb{Z})$, an abstract group (the character group behind the solenoid; no topology is used), with rational scalars $(r \cdot f)(q) = f(rq)$.

Lemma 19.12 (A rational-line exact sequence). *There is an exact sequence of $\mathbb{Q}$ -vector spaces $0 \rightarrow \mathbb{R} \xrightarrow{j} \mathbb{Q}^\vee \xrightarrow{\text{cl}} \widehat{\mathbb{Z}}/\mathbb{Z} \rightarrow 0$ with $j(a)(q) = qa + \mathbb{Z}$.*

Proof. j is injective. For $f \in \mathbb{Q}^\vee$ choose $\theta \in \mathbb{R}$ representing $f(1)$ and $g = f - j(\theta)$; then $g(1/n) = a_n/n + \mathbb{Z}$ with compatible $a_n \in \mathbb{Z}/n\mathbb{Z}$, i.e. $\alpha = (a_n) \in \widehat{\mathbb{Z}}$; put $\text{cl } f = \alpha + \mathbb{Z}$, independent of θ and additive, hence linear. If $\text{cl } f = 0$ then $\alpha = k \in \mathbb{Z}$ and $f = j(\theta + k)$. For any $\alpha \in \widehat{\mathbb{Z}}$ the rule

$$f_\alpha(k/n) = \frac{ka_n}{n} + \mathbb{Z} \tag{19.5}$$

is well defined and additive (common denominators use exactly the compatibility), with $f_\alpha(1) = 0$ and $\text{cl } f_\alpha = \alpha + \mathbb{Z}$. $\square$

By choice between sets fix a linear section $\varsigma : \widehat{\mathbb{Z}}/\mathbb{Z} \rightarrow \mathbb{Q}^\vee$ of cl ; this is a choice between set-sized vector spaces even when the defect below is defined on all of Π .

Lemma 19.13 (The defect is read from the ordinary character). *For $\psi : \Pi_F \rightarrow \mathbb{R}/\mathbb{Z}$ put $H_\psi(p)(q) = \psi(qp)$. Then $H_\psi : \Pi_F \rightarrow \mathbb{Q}^\vee$ is $\mathbb{Q}$ -linear and $\text{pd}_{\Psi_{\psi, \mathcal{L}}} = \text{cl} \circ H_\psi$ for every $\mathcal{L}$; in particular the profinite defect does not depend on $\mathcal{L}$.*

Proof. Linearity is additivity of ψ . Choose $\theta \in \mathbb{R}$ representing $\psi(p)$; then $a = p - \theta - \mathcal{L}(p) \in \mathcal{I}$ with $\text{pr}(a) = p$, and for every n , since $\mathcal{L}$ is linear, $\Psi(2\pi a/n) = e^{2\pi i(\psi(p/n) - \theta/n)}$: the infinitesimal factors cancel. These are the residues defining $\text{cl}(H_\psi(p))$. $\square$

Theorem 19.14 (Prescribed-defect realization; source 21). *Every $\mathbb{Q}$ -linear $d : \Pi_F \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$ is the profinite defect of a phase, also for a given class-linear d on Π when $F = \mathbf{No}$, without a class Hamel basis. After fixing a realizing ordinary character ψ , the infinitesimal character $\mathcal{L}$ may be chosen freely.*

Proof. Put $H = \varsigma \circ d$ and $\psi(p) = H(p)(1)$. Then $\psi(qp) = H(qp)(1) = (q \cdot H(p))(1) = H(p)(q)$, so $H_\psi = H$ and $\text{pd}_{\Psi_\psi, \mathcal{L}} = \text{cl } \varsigma d = d$ for every linear $\mathcal{L}$, by lemma 19.13. The only choice is the set map ς . $\square$

Theorem 19.15 (Counts on $F_\mathbb{Q}$; source 21, ZFC). *On $F_\mathbb{Q} = \mathbb{R}((t^\mathbb{Q}))$: every cardinal $\kappa \leq 2^{\aleph_0}$ is $\dim_\mathbb{Q} \text{pd}_\Psi(\Pi_F)$ for some phase; there are exactly $2^{2^{\aleph_0}}$ phases with reduced period group ($\mathcal{I}^{\text{div}} = 0$) and exactly $2^{2^{\aleph_0}}$ with split period group; the same counts hold for their distinct normalized period groups.*

Proof. All supports lie in the countable $\mathbb{Q}$, so $|F_\mathbb{Q}| = 2^{\aleph_0}$; $\Pi_F \supseteq \mathbb{R}\omega$ gives $\dim_\mathbb{Q} \Pi_F = 2^{\aleph_0} = \dim_\mathbb{Q} \widehat{\mathbb{Z}}/\mathbb{Z}$ (lemma 19.7), so linear maps of every rank exist and are realized by theorem 19.14. A linear isomorphism d gives a reduced group by theorem 19.9; multiplying the images of a basis independently by 1 or 2 gives $2^{2^{\aleph_0}}$ of them, with distinct defects, phases and (by theorem 19.3) groups. For split phases take $\mathbb{Q}$ -linear $A : \Pi_F \rightarrow \mathbb{R}$ and $\Psi_A(2\pi(p+f)) = \text{cis}(2\pi(f + A(p)))$; distinct A give distinct phases, their difference mapping the divisible Π_F into $\mathbb{Z}$. There are at most $(2^{\aleph_0})^{2^{\aleph_0}} = 2^{2^{\aleph_0}}$ maps $F_\mathbb{Q} \rightarrow F_\mathbb{Q}[i]$. $\square$

No continuum hypothesis or regularity of $2^{\aleph_0}$ is used. A reduced group still has infinite elements (every additive integer part of a non-Archimedean field has them), and these are counts of embedded period groups in a fixed field, not a claim that their abstract groups are pairwise nonisomorphic.

Example 19.16 (Two explicit nonliftable profinite characters). *Source 21.* Let $\alpha_2 \in \widehat{\mathbb{Z}}$ have 2-adic component 0 and every odd component 1; its residue a_n for $n = 2^e m$, m odd, solves $a_n \equiv 0 \pmod{2^e}$, $a_n \equiv 1 \pmod{m}$: $a_2 = 0$, $a_3 = 1$, $a_6 = 4$, $a_{12} = 4$, $a_{24} = 16$, $a_{40} = 16$. It is not a diagonal integer, so $[\alpha_2] \neq 0$, and f_{α_2} vanishes on integers and dyadic rationals but not at $1/3$. On $\mathbf{No}$, or on F_Γ with $1 \in \Gamma$, choose a $\mathbb{Q}$ -linear $\nu : \mathbb{R} \rightarrow \mathbb{Q}$ fixing $\mathbb{Q}$ (a Hamel-basis choice on which nothing displayed depends) and put $\psi(p) = f_{\alpha_2}(\nu(\ell(p)))$, $\mathcal{L} = 0$. Then

$$\Psi(2\pi\omega) = 1, \quad \Psi(2\pi\omega/2^e) = 1, \quad \Psi(2\pi\omega/3) = e^{2\pi i/3}, \quad \text{pd}_\Psi(\omega) = [\alpha_2] \neq 0,$$

so a period need not bring its rational divisions with it. With $u = (\omega - 1)/3$ one has $\Psi(2\pi u) = \Psi(2\pi\omega) = 1$ but $\Psi(2\pi u\omega) = e^{-2\pi i/3} \neq 1$, since $u\omega = (\omega^2 - \omega)/3$: this period group is not a ring, and it is not split. The example establishes neither a minimal multiplier ring nor reducedness.

Source 20. Let $\alpha_3 \in \widehat{\mathbb{Z}}$ have 3-adic component 1 and every other component 0: for $n = 3^k m$ with $3 \nmid m$, $a_n \equiv 0 \pmod{m}$ and $a_n \equiv 1 \pmod{3^k}$. The first residues are

n	2	3	4	6	8	9	12	18	24	27
a_n	0	1	0	4	0	1	4	10	16	1

and $f_{\alpha_3}(1) = 0$, $f_{\alpha_3}(1/2^k) = 0$, $f_{\alpha_3}(1/3^k) = 1/3^k + \mathbb{Z}$. It does not lift to an additive map $\mathbb{Q} \rightarrow \mathbb{R}$: a lift is $q \mapsto qf(1)$ with $f(1) \in \mathbb{Z}$ divisible by every 2^k , hence zero, which cannot represent $f_{\alpha_3}(1/3)$. (In general f_α lifts exactly when $[\alpha] = 0$, by lemma 19.12.) The roots are chosen coherently — the cube of the value at $1/3$ is the value at 1 — but not by dividing one additive angle.

19.5 Divisible periods are unavoidable and cofinal on $\mathbf{No}$

Theorem 19.17 (Cofinal rationally divisible periods; source 21). *Let Ψ be any phase on $\mathbf{No}$. For every $B \in \mathbf{No}$ there is $a > B$ with $\Psi(2\pi qa) = 1$ for every $q \in \mathbb{Q}$. Equivalently, $\mathcal{I}(\Psi)^{\text{div}}$ is cofinal in $\mathbf{No}$.*

Proof. We may take $B \geq 0$. Choose $g > 0$ with $\omega^g > 2B + \omega$, let $\kappa = (2^{\aleph_0})^+$ and $p_\beta = \omega^{g+\beta}$ for ordinals $\beta < \kappa$. The values $\text{pd}_\Psi(p_\beta)$ lie in the set $\widehat{\mathbb{Z}}/\mathbb{Z}$ of cardinality $2^{\aleph_0}$, so $\text{pd}_\Psi(p_\beta) = \text{pd}_\Psi(p_\gamma)$ for some $\beta < \gamma$. Since $\gamma - \beta \geq 1$, $p_\gamma \geq \omega p_\beta$ and $p = p_\gamma - p_\beta \geq (\omega - 1)\omega^g > B + \omega$. As $p \in \ker \text{pd}_\Psi$, [theorem 19.9](#) gives $a \in \mathcal{I}^{\text{div}}$ with $\text{pr}(a) = p$; $a - p$ is finite, so $a > B$. $\square$

The argument is one set-sized pigeonhole, not cardinal arithmetic with a purported cardinality of $\mathbf{No}$. It strengthens [theorem 17.6](#): not only does every coset of $\mathcal{O}_{\mathbb{R}}$ contain a period, but whole rational lines of periods reach beyond every bound. It does not assert that strong sums are fine limits.

Corollary 19.18. *No phase on $\mathbf{No}$ has a reduced period group. The reduced examples on $F_{\mathbb{Q}}$ of [theorem 19.15](#) therefore cannot stay reduced in any extension to $\mathbf{No}$ (this is conditional only on an extension being given; nothing is asserted about extending an arbitrary class character).*

The defect rank is nevertheless unrestricted on $\mathbf{No}$: for $\kappa \leq 2^{\aleph_0}$ take a rank- κ linear $A : \mathbb{R} \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$ and $d(p) = A(\ell(p))$; [theorem 19.14](#) realizes it, and its divisible periods are still cofinal. A *set/class contrast*: on the set field $F_{\mathbb{Q}}$ period groups can be reduced; on the class $\mathbf{No}$ they must contain cofinally large rational lines.

19.6 Multiplier rings and the integer-part criterion

For an additive integer part $\mathcal{I}$ put $\text{Mult}(\mathcal{I}) = \{a \in F : a\mathcal{I} \subseteq \mathcal{I}\}$. It is a unital subring containing $\mathbb{Z}$ and contained in $\mathcal{I}$ (apply it to 1), hence discretely ordered with least positive element 1. For an exponential E as in [proposition 19.5](#) the *kernel-multiplier ring* of Ehrlich and Kaplan is

$$Z_E = \{z \in F[i] : E(y) = 1 \Rightarrow E(yz) = 1 \text{ for every } y \in F[i]\}. \quad (19.6)$$

Proposition 19.19 (The kernel-multiplier ring). *For $E = E_\chi$ with $\mathcal{I} = \mathcal{I}(\Psi_\chi)$, $Z_E = \text{Mult}(\mathcal{I}) \subseteq F$.*

Proof. $2\pi i \in \ker E$, so $z = u + iv \in Z_E$ gives $E(2\pi iz) = 1$, whose modulus $\exp_F(-2\pi v)$ forces $v = 0$; for real u , preserving every $2\pi ia$ with $a \in \mathcal{I}$ is $u\mathcal{I} \subseteq \mathcal{I}$. $\square$

Theorem 19.20 (The integer-part criterion). *For an additive integer part $\mathcal{I}$ of F the following are equivalent: (i) $\text{Mult}(\mathcal{I})$ is an integer part of F ; (ii) $\text{Mult}(\mathcal{I}) = \mathcal{I}$; (iii) $\mathcal{I}$ is a subring. If $\mathcal{I} = \mathcal{I}(\Psi_\chi)$ and F has a real exponential, they are also equivalent to (iv) Z_{E_χ} is an integer part, and to (v) the binary kernel condition*

$$\forall u, v \in F \quad (E_\chi(2\pi iu) = E_\chi(2\pi iv) = 1) \implies E_\chi(2\pi iuv) = 1. \quad (19.7)$$

If F is initial in $\mathbf{No}$ ($F = \mathbf{No}$ or $\mathbf{No}(\lambda)$) and Z_{E_χ} is initial, these conditions hold if and only if E_χ is the canonical exponential Exp on $F[i]$.

Proof. If (i), take $a \in \mathcal{I}$ and a $\text{Mult}(\mathcal{I})$ -floor m of a ; $m \in \mathcal{I}$ gives $a - m \in \mathcal{I} \cap [0, 1) = \{0\}$, so $\mathcal{I} \subseteq \text{Mult}(\mathcal{I})$, which is (ii). A ring equal to $\mathcal{I}$ makes $\mathcal{I}$ a ring. If $\mathcal{I}$ is a ring, each of its elements

preserves it, so $\text{Mult}(\mathcal{I}) = \mathcal{I}$, an integer part by the floor property. (iv) is (i) by [proposition 19.19](#), and (v) is (iii) written with E_χ . If F and Z_{E_χ} are initial and the conditions hold, the uniqueness of the initial integer part [8, Lemma 11.1] gives $\mathcal{I} = Z_{E_\chi} = \mathbf{Oz} \cap F$, and the kernel dictionary gives $\Psi_\chi = \Phi$. Conversely Exp has $\mathcal{I} = \mathbf{Oz} \cap F$, an initial integer part equal to its multiplier ring. $\square$

This is source 20's Lemma 5.1 and Theorem 5.2 and source 21's Propositions 7.1, 8.2 and equation (8.7); the printed proof is source 20's. In source 21's formulation: with $P_E = \{a \in F : E(2\pi ia) = 1\}$, Z_E is an integer part if and only if P_E is closed under multiplication. *The criterion is an exact reduction, not a classification of every multiplier ring, and not a substitute for independent analytic or geometric hypotheses implying its right-hand side*; merely assuming that side would restate the missing content rather than explain it.

Remark 19.21 (Ring defects of a linear phase; source 20). For a $\mathbb{Q}$ -linear $A : \Pi_F \rightarrow \mathbb{R}$ and the split phase with coordinates $(A + \mathbb{Z}, 0)$ the period group is $\mathcal{I}_A = \{p - A(p) + n : p \in \Pi_F, n \in \mathbb{Z}\}$. For $d_p = p - A(p)$ and $d_q = q - A(q)$ the product has principal part $pq - A(p)q - A(q)p$ and constant part $A(p)A(q)$, so

$$d_p d_q \in \mathcal{I}_A \iff A(pq - A(p)q - A(q)p) + A(p)A(q) \in \mathbb{Z}.$$

This holds for merely $\mathbb{Q}$ -linear A ; the generally irrational numbers $A(p), A(q)$ must not be pulled through A .

Example 19.22 (A non-ring kernel whose multiplier ring is not initial; source 20). Let $A_2(p)$ be half the real coefficient of ω^2 in p , and take the phase of [remark 19.21](#). Then $\Psi(2\pi\omega) = 1$ and $\Psi(2\pi\omega^2) = -1$, so $\mathcal{I}_{A_2} \ni \omega$ but $\omega^2 \notin \mathcal{I}_{A_2}$: the kernel is not a ring. Multiplication by ω^3 sends every element $p - A_2(p) + n$ to an element with no finite part and zero coefficient at ω^2 , which lies in $\mathcal{I}_{A_2}$, so $\omega^3 \in \text{Mult}(\mathcal{I}_{A_2})$ while $\omega^2 \notin \mathcal{I}_{A_2} \supseteq \text{Mult}(\mathcal{I}_{A_2})$. The ordinal ω^2 is a sign prefix of ω^3 , so this multiplier ring is *not initial*. Initiality cannot be tested from one infinite-angle example; the constructions below prevent every nonintegral multiplier.

19.7 Minimal multiplier rings

Source 21's route: fix the ordinary character, perturb the infinitesimal one. Fix an additive $\psi : \Pi_F \rightarrow \mathbb{R}/\mathbb{Z}$ and let $\bar{\psi}(p) \in [0, 1)$ be the ordinary representative of $\psi(p)$ (not additive). The infinitesimal character $\mathcal{L}$ is built linearly.

Lemma 19.23 (One extension step). *Let $U \subseteq \Pi_F$ be a $\mathbb{Q}$ -subspace, $\mathcal{L}_U : U \rightarrow \mathfrak{m}_F$ linear, and $a \in F$ positive infinite. Suppose $p \in \Pi_F$ is such that p and $y := \text{pr}(a(p - \bar{\psi}(p)))$ are linearly independent modulo U . Then $\mathcal{L}_U$ extends to $U + \mathbb{Q}p + \mathbb{Q}y$ so that for every later linear extension $\mathcal{L}$ the phase $\Psi_{\psi, \mathcal{L}}$ has a normalized period b with ab not a normalized period.*

Proof. Fix $0 \neq \varepsilon_* \in \mathfrak{m}_F$ (F is non-Archimedean), let $b = p - \bar{\psi}(p)$ and $f = \text{fin}(ab)$, and prescribe $\mathcal{L}(p) = 0$, $\mathcal{L}(y) = \varepsilon_* - (f - \text{st } f)$, consistent by independence. For every later extension, $\Psi(2\pi b) = e^{2\pi i(\psi(p) - \bar{\psi}(p))} = 1$, while $ab = y + f$ gives $\Psi(2\pi ab) = e^{2\pi i(\psi(y) + \text{st } f)} \text{Exp}_{\mathfrak{m}}(2\pi i\varepsilon_*)$, whose infinitesimal angle $2\pi\varepsilon_*$ is nonzero, so it is not 1. Both conclusions depend only on the two prescribed values. $\square$

Lemma 19.24 (Disjoint blocks). *If a $\mathbb{Q}$ -vector space contains two-dimensional subspaces $(W_j)_{j \in J}$ whose sum is direct and $\dim_{\mathbb{Q}} U < |J|$, then some W_j meets U only in 0.*

Proof. Otherwise nonzero $u_j \in W_j \cap U$ would be $|J|$ independent vectors of U . $\square$

Lemma 19.25 (Fresh pairs in a continuum-sized field). *Let F be a non-Archimedean truncation-closed subfield of $\mathbf{No}$ containing $\mathbb{R}$ with $|F| = 2^{\aleph_0}$, let $a \in F$ be positive infinite, and let $U \subseteq \Pi_F$ have $\dim_{\mathbb{Q}} U < 2^{\aleph_0}$. Then some p satisfies [lemma 19.23](#).*

Proof. Truncation closure gives a positive infinite monomial $X \in F$. Put $Y = \text{pr}(aX)$ and $P_a = \text{pr}(a)$; if the leading ω -exponents of a and X are $\alpha, \gamma > 0$, that of Y is $\alpha + \gamma$, above those of P_a and X . For a $\mathbb{Q}$ -basis $(r_j)_{j \in J}$ of $\mathbb{R}$, $|J| = 2^{\aleph_0}$, put $p_j = r_j X$ and $y_j = \text{pr}(a(p_j - \bar{\psi}(p_j))) = r_j Y - \bar{\psi}(p_j) P_a$. In a finite relation $(\sum v_j r_j)Y - (\sum v_j \bar{\psi}(p_j))P_a + (\sum u_j r_j)X = 0$ the leading exponent of Y forces $\sum v_j r_j = 0$, so all $v_j = 0$, and then all $u_j = 0$. Apply [lemma 19.24](#) to $W_j = \mathbb{Q}p_j + \mathbb{Q}y_j$. $\square$

Theorem 19.26 (Minimal multiplier ring on a continuum-sized field; source 21, ZFC). *Let F be as in [lemma 19.25](#), with the finite phase. For every additive $\psi : \Pi_F \rightarrow \mathbb{R}/\mathbb{Z}$ there is a $\mathbb{Q}$ -linear $\mathcal{L} : \Pi_F \rightarrow \mathfrak{m}_F$ with $\text{Mult}(\mathcal{I}(\Psi_{\psi, \mathcal{L}})) = \mathbb{Z}$. This applies to $F_{\mathbb{Q}}$ and to $\mathbf{No}(\varepsilon_0)$.*

Proof. Let λ_c be the initial ordinal of $2^{\aleph_0}$; enumerate Π_F as $(u_{\xi})_{\xi < \lambda_c}$ and the positive infinite elements of F as $(a_{\xi})_{\xi < \lambda_c}$, with repetitions allowed. Build increasing partial linear maps $\mathcal{L}_{\xi} : U_{\xi} \rightarrow \mathfrak{m}_F$: at stage ξ add u_{ξ} with value 0 if it is new (the domain then has dimension at most $\max(\aleph_0, |\xi|) < 2^{\aleph_0}$), find a pair for a_{ξ} by [lemma 19.25](#), extend by [lemma 19.23](#); at limits take unions. The union $\mathcal{L}$ is total. Every positive infinite a has a period b with ab not a period; a negative infinite multiplier would give a positive one; a finite multiplier lies in $\mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$; and $\mathbb{Z}$ always multiplies $\mathcal{I}$ into itself. Below λ_c every stage has $|\xi| < 2^{\aleph_0}$ even if $2^{\aleph_0}$ is singular: no regularity is needed. $\square$

Lemma 19.27 (Fresh pairs outside a set-sized domain). *Let $a > \mathbb{R}$ be surreal, $\psi : \Pi \rightarrow \mathbb{R}/\mathbb{Z}$ a class character, and $U \subseteq \Pi$ a set-sized subspace. Then a pair as in [lemma 19.23](#) exists.*

Proof. Let $\alpha > 0$ be the leading exponent of a and $\kappa > \max(|U|, \aleph_0)$ a set cardinal. The exponents $\gamma_{\beta} = 1 + \alpha/(4(\beta + 1))$, $\beta < \kappa$ (ordinals read as surreals), are distinct and lie in $(1, 1 + \alpha/2)$. Put $p_{\beta} = \omega^{\gamma_{\beta}}$ and $y_{\beta} = \text{pr}(a(p_{\beta} - \bar{\psi}(p_{\beta})))$, whose leading exponent is $\alpha + \gamma_{\beta}$, since $a\bar{\psi}(p_{\beta})$ has exponents at most α . These leading exponents are distinct and exceed every $\gamma_{\beta'}$ and α , so the blocks $\mathbb{Q}p_{\beta} + \mathbb{Q}y_{\beta}$ are in direct sum; [lemma 19.24](#) finishes. $\square$

Theorem 19.28 (Minimal multiplier ring on $\mathbf{No}$; source 21, GBC). *In NBG with global choice, for every class character $\psi : \Pi \rightarrow \mathbb{R}/\mathbb{Z}$ there is a class-linear $\mathcal{L} : \Pi \rightarrow \mathfrak{m}_{\mathbb{R}}$ with $\text{Mult}(\mathcal{I}(\Psi_{\psi, \mathcal{L}})) = \mathbb{Z}$.*

Proof. Global choice gives set-like ordinal enumerations of Π and of the positive infinite surreals. Run the recursion of [theorem 19.26](#) along the ordinals with [lemma 19.27](#). Every partial domain is a set (finitely many vectors at a successor, a set of earlier stages at a limit); the choices of sets and set functions use the fixed global well-order, so this is set-valued transfinite recursion with a class parameter; no class-valued stage or class Hamel basis occurs. The finite and negative cases are as before. $\square$

Corollary 19.29 (Any defect together with the minimal ring; source 21). *On the fields of [theorem 19.26](#) (ZFC) and on $\mathbf{No}$ (GBC), every $\mathbb{Q}$ -linear $d : \Pi_F \rightarrow \mathbb{Z}/\mathbb{Z}$ is the profinite defect of a phase with multiplier ring $\mathbb{Z}$; in particular $\text{pd}_{\Psi} = 0$ and $\text{Mult}(\mathcal{I}(\Psi)) = \mathbb{Z}$ occur together. Throughout the construction the standard part $\text{st } \Psi(2\pi(p + f)) = e^{2\pi i(\psi(p) + \text{st } f)}$ of the phase is fixed at every argument.*

Proof. Realize d by ψ ([theorem 19.14](#)) and apply the minimal-multiplier theorem to that ψ ; the defect does not depend on $\mathcal{L}$ ([lemma 19.13](#)), nor does the displayed standard part. Take $\psi = 0$ for the last example. $\square$

For $\psi = 0$ the phase is $\text{cis} \circ 2\pi \mathbf{r}_{\mathcal{L}}$ with $\mathbf{r}_{\mathcal{L}}(p + f) = f + \mathcal{L}(p)$, and its period group is the split group $\{p - \mathcal{L}(p) + n\}$, whose divisible part is the whole graph of $-\mathcal{L}$; yet no nonintegral scalar preserves it. Division-compatible periods are abundant while universal multiplicative symmetries are minimal.

Source 20's route: prescribe a set-sized part of the ordinary character, keep $\mathcal{L} = 0$. This route works on $\mathbf{No}$ and gives ordinary-circle values, which [section 19.9](#) needs.

Lemma 19.30 (Fresh ordinal monomial). *Let $W \subseteq \Pi$ be a set-sized $\mathbb{Q}$ -subspace and $x \in \mathbf{No} \setminus \mathbb{Q}$. There is an ordinal β such that $\mu = \omega^\beta$ satisfies $\mu, x\mu \in \Pi$ with $q\mu + rx\mu \in W$ ($q, r \in \mathbb{Q}$) only for $q = r = 0$.*

Proof. Let S be the union of the supports of elements of W and $G = \text{supp}(x) \cup \{0\}$; the set $\{s - g : s \in S \cup \{0\}, g \in G\}$ is bounded above by an ordinal β (a set of birthdays has an ordinal bound). Then each $\beta + g$ (surreal addition) is positive and above all of S , so μ and $x\mu$ are purely infinite with all exponents above S . If $q\mu + rx\mu \in W$, uniqueness of normal forms makes both sides vanish, so $q + rx = 0$, and $x \notin \mathbb{Q}$ gives $r = q = 0$. $\square$

No bound on the exponents of x and no finiteness of its support is assumed; only that single supports and the union of the supports of a set are sets.

Theorem 19.31 (Relative minimal-multiplier extension; source 20, GBC). *Let $W_0 \subseteq \Pi$ be a set-sized $\mathbb{Q}$ -subspace and $\psi_0 : W_0 \rightarrow \mathbb{R}/\mathbb{Z}$ a homomorphism. There is a class homomorphism $\psi : \Pi \rightarrow \mathbb{R}/\mathbb{Z}$ extending ψ_0 such that the phase $\Psi = \Psi_{\psi,0}$ has, for every $x \in \mathbf{No} \setminus \mathbb{Q}$, a positive purely infinite monomial μ_x with*

$$\Psi(2\pi\mu_x) = 1, \quad \Psi(2\pi x\mu_x) = -1.$$

Hence $\text{Mult}(\mathcal{I}(\Psi)) = \mathbb{Z}$. Likewise, a $\mathbb{Q}$ -linear $A_0 : W_0 \rightarrow \mathbb{R}$ extends to a $\mathbb{Q}$ -linear $A : \Pi \rightarrow \mathbb{R}$ whose split phase (coordinates $(A + \mathbb{Z}, 0)$) has $\text{Mult} = \mathbb{Z}$ and period group $\{p - A(p) + n\}$.

Proof. Fix a set-like global enumeration $(x_\alpha)_{\alpha \in \text{On}}$ of $\mathbf{No}$ and build set-sized subspaces W_α with compatible characters ψ_α . At stage α adjoin $\text{pr}(x_\alpha)$ if new, with value 0 on its rational line. If $x_\alpha \notin \mathbb{Q}$, apply [lemma 19.30](#) to obtain μ_α , adjoin the direct summand $\mathbb{Q}\mu_\alpha \oplus \mathbb{Q}x_\alpha\mu_\alpha$ and set $\psi(w + q\mu_\alpha + rx_\alpha\mu_\alpha) = \psi(w) + r/2 + \mathbb{Z}$. At limits take unions (set many stages). The union is defined on all of Π and keeps every stage assignment. If $x \notin \mathbb{Q}$ then $\mu_x \in \mathcal{I}$ and $x\mu_x \notin \mathcal{I}$, so $\text{Mult}(\mathcal{I}) \subseteq \mathbb{Q} \cap \mathcal{I} = \mathbb{Z}$. For the split variant run the same recursion with real values $A(\mu_\alpha) = 0$, $A(x_\alpha\mu_\alpha) = 1/2$; the phase at $2\pi(p + r + \varepsilon)$ is $e^{2\pi i(A(p)+r)} \text{Exp}_m(2\pi i\varepsilon)$, which is 1 exactly when $\varepsilon = 0$ and $A(p) + r \in \mathbb{Z}$, and $p \mapsto p - A(p)$ is a section. $\square$

The character on a new rational direction is specified on *all* its rational multiples, and the whole prescribed ψ_0 is retained; this is decisive below. Source 20 states the first part as its Theorem 6.2 and the split variant as Theorem 6.4.

Theorem 19.32 (A nonsplit minimal example; source 20, GBC). *There is a phase Ψ_n on $\mathbf{No}$ with ordinary-circle values, multiplier ring $\mathbb{Z}$, and nonsplit period sequence, satisfying*

$$\Psi_n(2\pi\omega) = 1, \quad \Psi_n(2\pi\omega/2^k) = 1, \quad \Psi_n(2\pi\omega/3^k) = e^{2\pi i/3^k} \quad (k \geq 1).$$

Proof. Apply [theorem 19.31](#) to $W_0 = \mathbb{Q}\omega$ and $\psi_0(q\omega) = f_{\alpha_3}(q)$ ([example 19.16](#)). A splitting would give, by [theorem 19.10\(5\)](#), a real additive lift of f_{α_3} , which does not exist. $\square$

Comparison of the routes. Source 21's theorems keep ψ fixed on all of Π_F , hence the profinite defect and the standard part of the phase at every argument, and hold in ZFC on $\mathbf{No}(\varepsilon_0)$ and $F_{\mathbb{Q}}$; with [corollary 19.29](#) they also give split ($d = 0$) and nonsplit ($d \neq 0$) minimal examples. Their phase values have nonzero infinitesimal angles, and nothing is claimed about their restriction to subfields. Source 20's theorem prescribes ψ only on a set-sized subspace, changes it elsewhere, keeps every value in the ordinary circle, and excludes every nonrational multiplier, finite ones included, by an explicit witness; that is what makes the restriction and reflection of [section 19.9](#) work.

19.8 Surcomplex exponentials and the Ehrlich–Kaplan question

Theorem 19.33 (Initial kernel-multiplier ring $\mathbb{Z}$). (a) (Source 21, ZFC.) *There is an exponential E on $\mathbf{No}(\varepsilon_0)[i]$ with all properties of [propositions 19.5](#) and [19.6](#) and $Z_E = \mathbb{Z}$. It can be chosen with*

$$\frac{E(z)}{\text{Exp}(z)} \in \mathbb{T}(\mathbf{No}(\varepsilon_0)) \cap (1 + \mathfrak{m}[i]) \quad \text{for every } z, \quad (19.8)$$

and with zero profinite defect.

(b) (Source 21, GBC.) *The same holds on $\mathbf{No}[i]$.*

(c) (Source 20, GBC.) *On $\mathbf{No}[i]$, for every set-sized $W_0 \subseteq \Pi$ and homomorphism $\psi_0 : W_0 \rightarrow \mathbb{R}/\mathbb{Z}$, there is $E = E_\chi$ with ordinary-circle phase values, coordinates extending $(\psi_0, 0)$, and $Z_E = \mathbb{Z}$; examples of this kind with split and with nonsplit period sequence exist ([theorems 19.31](#) and [19.32](#)).*

In every case $Z_E = \mathbb{Z}$ is an initial subring of the field but not an integer part.

Proof. For (a) and (b) apply [theorem 19.26](#) or [theorem 19.28](#) with $\psi = 0$ and take $E = E_\chi$ for $\Psi_\chi = \Psi_{0,\mathcal{L}}$. Then $Z_E = \mathbb{Z}$ by [proposition 19.19](#); the canonical phase is $\Psi_{0,0}$ because $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$, so $E(a + ib)/\text{Exp}(a + ib) = \text{Exp}_m(2\pi i \mathcal{L}(\text{pr}(b/2\pi)))$, of modulus one and standard part one; the defect vanishes by [lemma 19.13](#). Part (c) is [theorems 19.31](#) and [19.32](#) with [proposition 19.19](#). Every proper sign prefix of an ordinary integer is an ordinary integer, so $\mathbb{Z}$ is initial; no ordinary integer lies in $[\omega - 1, \omega]$, so it is not an integer part. $\square$

This is source 21's Theorem 8.4 and source 20's Theorem 1.1(i) with Corollary 6.5. The obstruction is not a failure of the group law, of the finite-angle normalization, of the modulus or conjugation laws, or of differentiability: all are built in.

What this answers. Ehrlich and Kaplan ask, for an initial trigonometric-exponential subfield $K \subseteq \mathbf{No}$ and any exponential on $K[i]$ extending their strip map and satisfying the modulus law, under which conditions — including initiality of Z_E — Z_E is an integer part, equivalently the exponential is the canonical one [8, §11.1, first open question; arXiv v3, printed p. 34]. [Theorem 19.33](#) shows that initiality of Z_E is *not* sufficient, even together with the strip values, the modulus and conjugation laws, the local Taylor identities of every order, and agreement of the standard part of the phase at every argument; Z_E can be as small as possible. With [theorem 19.20](#) this identifies the missing algebraic content exactly: multiplicative closure of the normalized periods.

What it does not answer. It is a partial negative answer to that one sufficiency possibility. It does not classify the additional conditions that would suffice; it does not assert that Ehrlich and Kaplan conjectured initiality to be sufficient; and it does not touch their separate second question, on how different the exponentials induced by different initial embeddings of one trigonometric-exponential field can be, whether they are isomorphic, and whether they have the same first-order theory [8,

§11.1, second open question]. The first-order separation of [section 19.9](#) is proved in the broader class of all exponentials with the strip and modulus properties, whose members need not arise from initial embeddings; it answers nothing about that second question. The robustness question therefore remains open, re-scoped to conditions beyond initiality.

Remark 19.34 (The strip map in the source). Source 21 observes that the strip map displayed on printed p. 33 of arXiv v3 of [8] is $E(x + iy) = (\exp x)(\sin y + i \cos y)$, with sine and cosine interchanged, whereas the global formula immediately after it uses $\cos b + i \sin b$; the former gives $E(0) = i$ and cannot literally be the asserted group isomorphism. We checked this against the arXiv v3 PDF for this merge. Throughout, “strip exponential” means $\exp(x)(\cos y + i \sin y)$, as in [theorem 10.1](#), consistent with the group law and the adjacent formula; no argument uses the typographical discrepancy.

$E' = E$ here is the fine derivative of a function of z , not the Berarducci–Mantova derivation of the scalar field [16]. No claim is made that the noncanonical exponentials of [theorem 19.33\(a\),\(b\)](#) are isomorphic or nonisomorphic to each other as unrestricted exponential fields (source 21); their relation to the canonical one is settled by [corollary 19.40](#) below.

19.9 Reflection to set-sized fields and first-order separation

This subsection is source 20’s. Restricting a class exponential with $Z_E = \mathbb{Z}$ to a subfield need not keep $Z_E = \mathbb{Z}$: a nonintegral element might fail to be a multiplier only because of a witness outside the subfield. The witnesses of [theorem 19.31](#) let one close the field under them, and ordinary-circle values let the map itself restrict.

Lemma 19.35 (Restriction to epsilon fragments). *Let $\Psi = \Psi_{\psi,0}$ be a phase on $\mathbf{No}$ with ordinary-circle character values. For every epsilon number λ , E_χ maps $\mathbf{No}(\lambda)[i]$ onto $\mathbf{No}(\lambda)[i]^\times$ and is an exponential there with the properties of [proposition 19.5](#).*

Proof. For $x \in F = \mathbf{No}(\lambda)$, initiality and truncation closure put $\text{pr}(x), \text{fin}(x)$ in F ; $e^{2\pi i \psi(p)}$ is an ordinary complex number; cis of a finite element of F lies in $F[i]$ by trigonometric closure, and $\exp_{\mathbf{No}}$ preserves F by exponential closure. Surjectivity follows as in [proposition 19.5](#) from the real logarithm of F and surjectivity of cis onto $\mathbb{T}(F)$. $\square$

Theorem 19.36 (Unbounded set-sized reflection). *Let E_s and E_n be the split and nonsplit examples of [theorems 19.31](#) and [19.32](#) (constructed in GBC), with witness maps $x \mapsto \mu_x^s, \mu_x^n$. For every ordinal α there is an epsilon number $\lambda > \alpha$ of cofinality ω such that on $F = \mathbf{No}(\lambda)$ both restrictions have $Z_E = \mathbb{Z}$, the first restricted period sequence splits, the second does not, and their period groups are not isomorphic.*

Proof. Take an epsilon number $\lambda_0 > \max\{\alpha, \omega\}$; given λ_n , the set of witnesses μ_x^s, μ_x^n for $x \in \mathbf{No}(\lambda_n) \setminus \mathbb{Q}$ (a set by replacement) has birthdays bounded by an ordinal; choose an epsilon number $\lambda_{n+1} > \lambda_n$ above that bound, and let $\lambda = \sup_n \lambda_n$, an epsilon number of cofinality ω . For $x \in F \setminus \mathbb{Q}$ both witnesses and their products with x lie in F , so x is not a multiplier in F ; nonintegral rationals are excluded by $\mathcal{I} \cap \mathcal{O}_F = \mathbb{Z}$; the maps restrict by [lemma 19.35](#). The split section $p \mapsto p - A(p)$ stays in F . For the nonsplit example, $\mathbb{Q}\omega \subseteq F$ and its displayed values persist, so an additive angle lift on Π_F would restrict to a forbidden lift on $\mathbb{Q}\omega$. [Theorem 19.11](#) gives nonisomorphism. $\square$

Set-theoretic strength. The reflection uses only arbitrarily large epsilon numbers and ordinal bounds for sets of birthdays; no large cardinal is hidden. Global choice organizes the construction

on **No**, and conservativity of GBC over ZFC for statements about sets means the resulting set-sized existence statements need no additional strength. The theorem gives unboundedly many suitable fragments, *not* that every fragment works for a fixed constructed phase (closure under the witnesses was part of the proof), and no computable description of the least suitable λ .

Consider the language of fields with one unary function symbol E , naming neither i , π , the real axis, the order, standard part, nor the normal-form projections, and the sentence

$$\text{RingKer} : \quad \exists \tau \left[\tau \neq 0 \wedge E(\tau) = 1 \wedge \forall x \forall y ((E(x) = 1 \wedge E(y) = 1) \Rightarrow \exists w (xy = \tau w \wedge E(w) = 1)) \right]. \quad (19.9)$$

Lemma 19.37. *If an exponential field satisfies RingKer with witness τ , then $\tau^{-1} \ker E$ is a unital subring contained in Z_E .*

Proof. $\tau^{-1} \ker E$ is additive and contains 1. For $x, y \in \ker E$, $xy/\tau \in \ker E$, so $(x/\tau)(y/\tau) \in \tau^{-1} \ker E$ and $(x/\tau)y \in \ker E$. $\square$

Theorem 19.38 (First-order separation). *Let F be **No** or **No**(λ) with $\Pi_F \neq 0$. The canonical exponential satisfies RingKer; every exponential as in [proposition 19.5](#) with $Z_E = \mathbb{Z}$ fails it. Consequently, on each field of [theorem 19.36](#) the canonical exponential field is not elementarily equivalent to either minimal example, even in the bare language of fields with E .*

Proof. For Exp take $\tau = 2\pi i$: its kernel is $2\pi i(\mathbf{Oz} \cap F)$ and $\mathbf{Oz} \cap F$ is a ring. If $Z_E = \mathbb{Z}$ and τ witnesses RingKer, [lemma 19.37](#) gives a unital ring $\tau^{-1} \ker E \subseteq \mathbb{Z}$, hence equal to $\mathbb{Z}$, so $\ker E = \tau \mathbb{Z}$ is cyclic, contrary to [theorem 19.3](#). $\square$

Corollary 19.39 (Three structures; source 20). *For arbitrarily large epsilon numbers λ of countable cofinality, **No**(λ)[i] carries at least three pairwise nonisomorphic exponentials agreeing on the real exponential and on every finite-imaginary strip; RingKer separates the canonical one from the other two, and [theorem 19.11](#) separates those two.*

Corollary 19.40 (Combining the sources; stated in neither). *In ZFC: (i) the exponential E of [theorem 19.33\(a\)](#) on **No**(ε_0)[i] fails RingKer, so $(\mathbf{No}(\varepsilon_0)[i], E)$ is not elementarily equivalent, in particular not isomorphic, to $(\mathbf{No}(\varepsilon_0)[i], \text{Exp})$; (ii) **No**(ε_0)[i] carries three pairwise nonisomorphic exponentials agreeing on the real exponential and every finite-imaginary strip: Exp, one with $Z_E = \mathbb{Z}$ and zero profinite defect, and one with $Z_E = \mathbb{Z}$ and nonzero profinite defect.*

Proof. (i) **No**(ε_0) is initial with $\Pi_F \neq 0$, and $Z_E = \mathbb{Z}$, so [theorem 19.38](#) applies; its proof is elementary. (ii) By [corollary 19.29](#) on **No**(ε_0) take $d = 0$ and some $d \neq 0$ (possible since $\dim_{\mathbb{Q}} \widehat{\mathbb{Z}}/\mathbb{Z} = 2^{\aleph_0}$); both have $Z_E = \mathbb{Z}$, so both fail RingKer; their period sequences split and do not split ([theorem 19.10](#)), so [theorem 19.11](#) distinguishes their kernels, and an isomorphism of exponential fields restricts to an isomorphism of kernels. $\square$

Part (i) settles source 21's non-claim only as far as comparison with the canonical exponential goes; whether distinct noncanonical examples of source 21 are isomorphic to one another remains open.

Corollary 19.41 (Definable integers and an omitted type; source 20). *In each set-sized example with $Z_E = \mathbb{Z}$, the parameter-free formula $\varphi_{\mathbb{Z}}(z) : \forall y (E(y) = 1 \Rightarrow E(yz) = 1)$ defines exactly the ordinary integers, with their inherited ring operations, and the exponential field is not countably saturated: the type $\{\varphi_{\mathbb{Z}}(x)\} \cup \{x \neq n \cdot 1 : n \in \mathbb{Z}\}$ is finitely satisfiable and omitted.*

The failure of saturation concerns the exponential expansion, not the underlying field. By compactness an elementary extension realizes the type, and there the definable multiplier ring has elements beyond the numerals: $Z_E = \mathbb{Z}$ does not axiomatize standardness across the elementary class. `RingKer` gives non-elementary-equivalence; the invariant $\text{Hom}(\mathcal{I}, \mathbb{Z})$ gives nonisomorphism of the split and nonsplit examples but *no* claim about their elementary equivalence, since existence of an external homomorphism to $\mathbb{Z}$ is not a first-order sentence of the exponential field.

19.10 Valued-field automorphisms act freely on phases

In Hahn notation $t^g = \omega^{-g}$ with well-ordered supports, let $[g]_0$ be the real coefficient of ω^0 in the normal form of the exponent g ; it is additive, also on infinite exponents, with $[1]_0 = 1$, and it is not a standard part of an infinite number. On $F_{\mathbb{Q}}$ it is the identity. For real c put

$$\text{Sh}_c(t^g) = t^g(1 + ct)^{-[g]_0}, \quad \text{Sh}_c\left(\sum_g a_g t^g\right) = \sum_g a_g \text{Sh}_c(t^g), \quad (19.10)$$

the powers of $1 + ct$ being their binomial series.

Lemma 19.42 (Support control). *Equation (19.10) is well defined on every Hahn series and every set-supported normal form, and it preserves strongly summable families.*

Proof. The new support lies in $S + \mathbb{N}$, which is well ordered; at a fixed exponent only finitely many $(g, n) \in S \times \mathbb{N}$ contribute, since infinitely many would give a decreasing sequence in S . For a strongly summable family apply this to the union of supports. $\square$

Theorem 19.43 (The shift family). *The maps Sh_c are strongly $\mathbb{R}$ -linear field automorphisms of $F_{\mathbb{Q}}$ and of $\mathbf{No}$ that fix every valuation and leading coefficient, with*

$$\text{Sh}_c \text{Sh}_d = \text{Sh}_{c+d}, \quad \text{Sh}_c(t) = \frac{t}{1 + ct}, \quad \text{Sh}_c(\omega) = \omega + c.$$

They preserve order, $\mathcal{O}$, $\mathfrak{m}$, standard part and the finite phase: $\text{Sh}_c(\text{cis } f) = \text{cis}(\text{Sh}_c f)$ for finite f , with $\text{Sh}_c(i) = i$.

Proof. Additivity follows from lemma 19.42; multiplicativity on monomials from additivity of $[\cdot]_0$, and in general by strong products. Each monomial image is the monomial times a unit of standard part one, so leading terms, order, $\mathcal{O}$, $\mathfrak{m}$ and standard parts are kept. Strong substitution gives $\text{Sh}_c \text{Sh}_d(t^g) = t^g(1 + ct)^{-[g]_0}(1 + dt/(1 + ct))^{-[g]_0} = t^g(1 + (c + d)t)^{-[g]_0}$, identities among units $1 + \text{infinitesimal}$, with no branch choice. The maps fix real coefficients and preserve the strong sums defining $\text{Exp}_{\mathfrak{m}}$, hence the finite phase. $\square$

These are the fixed-shift flows of [23, Theorem 5.2] with $\delta = 1$, $\lambda = [\cdot]_0$ (its ℓ of equation `saut:eq:ell`) and $\text{Sh}_c = T_{-c}$ there; they are strong 1-automorphisms in the sense of [22, §4]. Source 21 (Theorem 9.2) credits both and re-proves them to keep the action theorem independent of any decomposition theory; source 20 (Theorem 11.1) proves them without either citation. They are *not* asserted to preserve $\text{exp}_{\mathbf{No}}$, Conway’s omega map or surreal simplicity; “value-preserving” means $v(\text{Sh}_c x) = v(x)$ in the fixed value-group copy, not that the automorphism fixes the surreals labelling it.

The shifts act on phases by $(\text{Sh}_c \cdot \Psi)(x) = \text{Sh}_c(\Psi(\text{Sh}_{-c} x))$, which preserves the finite restriction.

Theorem 19.44 (Free action; source 21). *On $F_{\mathbb{Q}}$ or $\mathbf{No}$, $\text{Sh}_c \cdot \Psi = \Psi$ for a phase Ψ implies $c = 0$. The real translations act freely on phases and on their normalized period groups.*

Proof. Invariance means $\text{Sh}_c(\Psi(x)) = \Psi(\text{Sh}_c x)$. At $x = 2\pi\omega/n$, since Sh_c fixes reals, $\text{Sh}_c x = x + 2\pi c/n$, and taking standard parts gives $\text{st } \Psi(x) = \text{st } \Psi(x) e^{2\pi ic/n}$. The standard part is nonzero, so $c/n \in \mathbb{Z}$ for every n , whence $c = 0$. The acted phase has period group $\text{Sh}_c(\mathcal{I})$, and the kernel dictionary is injective. $\square$

Corollary 19.45 (Valued-field naturality fails). *No phase on $F_{\mathbb{Q}}$ or on $\mathbf{No}$ is invariant under all strongly $\mathbb{R}$ -linear valued-field automorphisms; no global phase can be singled out by an automorphism-invariant characterization using only that structure and the real coefficients.*

Remark 19.46 (Source 20's form). Source 20 proves the case $c = 1/2$ directly: equivariance under $\text{Sh}_{1/2}$ would give $\text{Sh}_{1/2}(\Psi(2\pi\omega)) = \Psi(2\pi\omega + \pi) = -\Psi(2\pi\omega)$, and since $\text{Sh}_{1/2}$ fixes standard parts of finite elements, $\text{st } \Psi(2\pi\omega) = -\text{st } \Psi(2\pi\omega)$, impossible for a unit. The finite phase itself commutes with every Sh_c ; the conflict arises only in extending it to infinite arguments. The obstruction concerns the *weaker* valued-field data. It does not contradict the canonical construction: initiality, truncation and the monomial section are additional structure, and $\text{Sh}_{1/2}$ moves the monomial ω to $\omega + 1/2$. Nor is a failure of invariance under exponential-field automorphisms asserted; that is a strictly stronger preservation requirement which the present witnesses do not satisfy in general.

The orbit of the canonical phase. Let $\Phi_c = \text{Sh}_c \cdot \Phi$ on $F_{\mathbb{Q}}$ or $\mathbf{No}$ (on $F_{\mathbb{Q}}$, Φ means $\text{cis} \circ \text{fin}$). Its period group $\text{Sh}_c(\Pi_F \oplus \mathbb{Z})$ is an integer-part ring with zero profinite defect, the groups are distinct for distinct c , and the retraction is $\text{Sh}_c \circ \text{fin} \circ \text{Sh}_{-c}$.

Proposition 19.47 (The standard-part shadow; source 21). *For a normal form $x = \sum_{\alpha} a_{\alpha} \omega^{\alpha}$ in $F_{\mathbb{Q}}$ or $\mathbf{No}$,*

$$\text{st } \Phi_c(2\pi x) = \exp\left(2\pi i \sum_{n \in \mathbb{N}} a_n (-c)^n\right),$$

a finite sum.

Proof. The coefficient of t^0 in $\text{Sh}_{-c}(x)$ comes only from terms $t^{-n} = \omega^n$ with $n \in \mathbb{N}$, since the binomial corrections raise t -exponents by nonnegative integers; $\text{Sh}_{-c}(t^{-n}) = t^{-n}(1 - ct)^n$ contributes $(-c)^n$. A reverse well-ordered support contains only finitely many nonnegative integers. Applying Sh_c after the finite phase does not change the standard part. $\square$

Translating the canonical phase changes its ordinary shadow while keeping a ring kernel; the avoidance construction of [theorem 19.33\(a\),\(b\)](#) keeps the shadow while destroying every nonintegral multiplier.

19.11 What each test detects

Test (source 21)	Detects	Does not force
Finite values and local Taylor identities	The canonical germ on every infinitesimal neighbourhood	The global phase or its period ring
Standard part of the phase at every input	The ordinary character ψ	The infinitesimal character $\mathcal{L}$ or the multiplier ring
Rational-division residues	The profinite defect and $\mathcal{I}^{\text{div}}$	Multiplicative closure of periods
Closure of normalized periods under products	$\text{Mult}(\mathcal{I}) = \mathcal{I}$	Initiality of this integer part
Initiality of the multiplier ring alone	A simplicity-closed discrete subring	The integer-part property
Invariance under all valued translations	Impossible for every phase	Nothing against extra simplicity structure

Invariant (source 20; F reflected)	Canonical	Minimal split	Minimal nonsplit
Z_E	$\mathbf{Oz} \cap F$	$\mathbb{Z}$	$\mathbb{Z}$
Normalized kernel a ring	yes	no	no
Period sequence splits	yes	yes	no
$\text{Hom}(\mathcal{I}, \mathbb{Z})$	$\mathbb{Z}$	$\mathbb{Z}$	0
RingKer	true	false	false
Finite-angle agreement; $E' = E$	yes	yes	yes

The second table does not equate the first-order theories of the two minimal examples; that is not proved. Three kinds of splitting must not be conflated (source 20): the domain always splits as $F = \Pi_F \oplus \mathcal{O}_F$; the circle always splits as $\mathbb{T}(F) \cong \mathbb{T}(\mathbb{R}) \times \mathfrak{m}_F$; neither makes the period sequence split, and a profinite character sits exactly between them. Conversely a split kernel need not be a ring: the minimal split example is additively isomorphic to the canonical kernel, yet its multiplier ring is $\mathbb{Z}$; ring closure depends on the embedding, not on the abstract group. Two failures are logically independent (source 21): [example 19.16](#) has nonzero defect and no ring closure; [theorem 19.33\(a\),\(b\)](#) has zero defect and no ring closure; the canonical orbit has zero defect and ring kernels. “All periods split into a divisible part plus integers” does not imply “the periods form a ring.”

On $F_{\mathbb{Q}}$ a reduced period group embeds in $\widehat{\mathbb{Z}}$ by its residue map, so distinct periods are separated by congruences: the classical “Leibnizian” phenomenon for $\mathbb{Z}$ -groups [21, Theorem 3.3], not a new theory of Presburger arithmetic. Nothing is claimed about tameness, decidability, o-minimality or elementary equivalence of the phase expansion with the ordinary complex exponential field. The avoidance constructions are existence arguments using linear choices; they give no finite names, computable coefficient oracles, algorithms for the phases, or effective recognition of periods, and finitely many rational divisions cannot determine an element of $\widehat{\mathbb{Z}}$.

19.12 Checks, formal status, and remaining questions

Source 20’s script (`code/20-exponential-kernels-verify.py`) passed 47,939 exact finite checks: compatible CRT residues and additivity of f_{α_3} , finite portions of the 2- and 3-power root towers, coefficient identities for $\text{Sh}_a \text{Sh}_b = \text{Sh}_{a+b}$ through a fixed order, and finite Laurent-polynomial instances of [lemma 19.30](#) including a pair of periods whose product is not a period. Source 21’s script (`code/21-period-arithmetic-verify_examples.py`) passed checks on 128 CRT moduli, 645 residue-compatibility pairs, 58,081 additivity pairs of f_{α_2} , 15 displayed example assertions,

1,575 translation-composition coefficients, 24 one-step multiplier witnesses and 20 shadow checks. All arithmetic is exact. *These checks do not establish the transfinite recursions, arbitrary surreal supports, the reflection theorem, the abstract vector-space sections or novelty.*

Neither source was verified in Lean, and neither built or modified the repository. Both propose a formalization order — the kernel dictionary over an ordered field with a surjective phase of kernel $\mathbb{Z}$; Mult and the criterion; the splitting, the CRT characters and $\widehat{\mathbb{Z}}/\mathbb{Z}$ as abstract groups; the one-step or fresh-support lemma with an explicit set-sized parameter; then the set-sized recursion, with the class statements in a universe-relative form — and identify the delicate obligations: support control for [equation \(19.10\)](#), compatibility of residues, and the cardinal bookkeeping of the recursion. These are proposed architectures, not existing declarations.

Remaining questions, as the sources state them: which discrete subrings occur as Z_E on a fixed initial field, and which rings between $\mathbb{Z}$ and an additive integer part occur for a fixed ordinary character and profinite defect (sources 20, 21; source 21 realizes $\mathbb{Z}$ for every defect and exhibits maximal rings on the canonical orbit, and classifies no intermediate ring); whether an independent compatibility with simplicity, truncation or multiplication forces the normalized periods to be a ring (source 21); whether the split and nonsplit minimal examples can have the same first-order theory (source 20); which definability requirements in a specified expansion exclude the diagonal constructions without naming the canonical projection (source 20); the orbits of phases under the full valued automorphism group, which the free action does not classify (source 21); and the isomorphism types of the resulting exponential fields when no base structure is fixed (source 21; the comparison with the canonical exponential is [corollary 19.40](#)). The second Ehrlich–Kaplan question is untouched.

20 Conclusions and dependency audit

20.1 What the theory establishes

The central geometric conclusion is exact: *arbitrary surreal lengths do not require infinite circular angles*. The finite-angle quotient $\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z} \cong \mathbb{T}(\mathbf{No}) \cong \mathbb{T}(\mathbb{R}) \times \mathfrak{m}_{\mathbb{R}}$ describes every unit direction and remembers both an ordinary direction and its full infinitesimal displacement, and ordinary finite triangle constructions give the same identities at all scales, in the plane, on the sphere and in the hyperbolic disk. The infinitesimal part is not negligible data: it controls chord precision, nearly straight angles, the order of the triangle-inequality slack, the relative flatness threshold, and tangency splitting. Valuations then extract information with no ordinary numerical counterpart — angle hierarchies, flattening thresholds, and condition loss near collisions.

A second conclusion concerns the complex plane. The canonical strip $\mathcal{V}$ is large enough for a surjective complex sine with completely determined fibres and for the complete algebraic solution theory of every finite trigonometric polynomial; the passage $U = E(iz)$ turns periodic root questions into finite-degree polynomial questions. Positive-support coefficient lifting distinguishes stable finite algebras from unstable individual roots, and the three stability certificates of [section 9.5](#) cover three genuinely different questions. Positivity, too, has to be tested on the full surreal circle and not only on its ordinary real shadow.

Finally, full real-argument oscillation is a different problem. The Ehrlich–Kaplan integer-part normalization supplies one explicit answer; the character classification shows it is one member of a family all of whose members are fine-locally analytic, obey the same addition laws and the same differential equations, necessarily acquire infinite periods, and fail common-domain coherence at

infinite frequency. *Thus neither nonexistence nor canonical uniqueness is an appropriate blanket claim about “surreal sine.”* The answer depends on the domain and on the analytic structure specified.

Read through its rotation group, the circle is a class group with the classical finite algebra and an exact, very large infinitesimal factor (section 18). Isometries fixing the origin are linear; torsion and finite subgroups are the standard ones; each valuation layer of infinitesimal rotations is one real line; algebraic endomorphisms and weights are ordinary integers while fine-continuous nonalgebraic symmetries exist; and the fine, standard-part and semialgebraic geometries of the same circle answer different questions and must not be interchanged.

The periods of the global extensions have an arithmetic of their own (section 19). Normalized period groups are exactly the additive integer parts; their rational-division residues form a profinite defect, which can be prescribed arbitrarily and vanishes exactly when a linear finite-part choice suffices; on **No** divisible periods are cofinal, on $\mathbb{R}((t^{\mathbb{Q}}))$ they can be absent. Neither abundance of divisible periods nor perfect local exponential behaviour forces multiplicative closure: the kernel-multiplier ring can be the initial ring $\mathbb{Z}$, so initiality alone does not select the Ehrlich–Kaplan exponential, while closure of the normalized periods under products, together with initiality, does. The canonical choice therefore depends on global structure beyond the local identities and beyond valued-field symmetry, which acts freely on all phases.

20.2 A dependency audit

Result family	Inputs actually used
Finite sine, cosine, circle logarithm	Ordinary analytic germs, normal forms, positive-support Neumann lemma
Triangle, circle and concurrence geometry	Ordered-field algebra, positive square roots, finite-angle circle parametrization
Valuation estimates, defect and flatness	Exact algebraic identities and unit-leading infinitesimal series
Tangency, cosine fold, coupled collision	Strong composition of square-root and inverse-sine germs at infinitesimal arguments
Conditioned root existence and stability	A finite rational tangent chart, polynomial intermediate values, a nonsingular residue root; no Newton-sequence convergence
Cluster lifting	Coprime polynomial factorization, a finite Sylvester map, transfinite coefficient recursion over S^*
Strip functions, hyperbolic disk, spherical laws	Gonshor exponential and logarithm, Gram determinants, plus the finite-angle theory
Trigonometric root bounds	Algebraic closedness and the strip exponential quotient
Fejér–Riesz factorization	Real closedness, rational circle parametrization, finite polynomial factorization
Finite Fourier identities	Finite geometric sums; the coefficientwise integral of the ring (R1) where an integral is used
Arc and sector formulas	The explicit primitive-based coefficientwise integral convention, ring (R1)
Infinite phase classification	The additive normal-form split $\mathbf{No} = \Pi \oplus \mathcal{O}_{\mathbb{R}}$ and circle surjectivity
Coherence obstructions	Ring (R1) only; normal-form uniqueness and one well-ordered support
Rotation group: isometries, torsion, dihedral subgroups, invariants	Polarization, the splitting of theorem 4.2, ordinary roots of unity, one-variable Laurent independence

Result family	Inputs actually used
Norm-one torus, Lie algebra, representations	Finite algebra over a real closed field, dual numbers, Laurent coactions; the torus language of [18]
Fine, standard-part and semialgebraic circle	The lower-bound cut, open infinitesimal subgroups, the chord comparison; $\pi_1^{\text{sa}} \cong \mathbb{Z}$ imported from [19, 20]
Period arithmetic: dictionary, defect, splitting	Theorem 17.4 , floors, profinite residues; classical $\mathbb{Z}$ -group facts [21]; a set-sized linear section
Minimal multiplier rings, reflection	Two-vector avoidance or fresh ordinal monomials; ZFC recursion on a continuum-sized field, or GBC recursion on No with conservativity; initial integer parts [8, Lemma 11.1]
Free action of translations	The fixed-shift flows of [23, Theorem 5.2], support control, standard parts

No result uses a proper-class sum, assumes that a positive valuation increment is cofinal, or identifies fine differentiation with a chosen derivation of the surreal field [16]. The several-variable Noetherianity, division and residue theorems of [2, 3] are not assumed as black boxes anywhere.

20.3 Reproducible checks and their limits

Sources 16–18, the three editions of the trigonometric core, carried accompanying Python scripts using exact SymPy algebra and formal power series, which between them check the Gram, rational-circle, Heron, circumradius, Euler and Ptolemy identities; the Cayley group law; Chebyshev formulas; the spread and triple-spread identities; flat-triangle, tangency and inverse-cosine coefficients; the coupled angular algebra after sine coordinates; the algebraization of trigonometric equations; line-circle formulas and the collision algebra; finite quadrature and Fourier convolution; a spectral-factor example; and the rational identities behind the spherical and hyperbolic laws. Those runs reported 67, 43 and 68 passing checks respectively. Source 19’s script checks, in 41 exact identities, the Cayley norm, inverse chart, affine and homogeneous group laws and associativity, the exceptional chart cases and [equation \(18.6\)](#), the invariant differential [equation \(18.20\)](#), rotation-matrix products, orthogonality and determinant, the displacement norm identity and the rational corrections of [example 18.9](#), the Clifford calculation [equation \(18.10\)](#), and truncated Taylor coefficients of the chart, of [equation \(18.7\)](#), of $\text{Exp}_m(ih) \text{Exp}_m(-ih) = 1$ and of the chord–angle ratio.

These checks validate the finite algebra and selected Taylor coefficients occurring in the proofs. They do not implement the full surreal class, verify transfinite support arguments or recursions, establish real closedness, or machine-check any mathematical proof. The support lemma, the branch arguments and the proper-class distinctions are justified in the text, not inferred from symbolic experiments, and no proof-assistant verification is claimed. The same holds for source 19’s checks: they cannot check the surreal normal-form theorem, a class-sized topology, Neumann summability for arbitrary supports, or the universal quantifiers of the structure theorem, and no Lean compilation or independent kernel verification of that manuscript was performed.

Sources 20 and 21 carried standard-library Python scripts with exact rational arithmetic; their scope and counts (47,939 checks, and 128 moduli, 645 compatibility pairs, 58,081 additivity pairs, 1,575 translation coefficients, 24 witnesses and 20 shadow checks, respectively) are stated in [section 19.12](#). They test CRT residues, root towers, shift coefficients and finite instances of the avoidance step, not the transfinite recursions, the reflection theorem or arbitrary supports; no Lean proof of either manuscript was built.

20.4 What is not proved

The following non-claims are made in the body of the article and are collected here. They are specific to individual results and are not pooled into a single blanket disclaimer.

1. (*proposition 2.5.*) No general transfer of arbitrary infinite families of inequalities is asserted; in particular, coefficients that are themselves infinitesimal are not automatically covered by an ordinary real-coefficient sign argument. The uniqueness assertion is deliberately about the Taylor rule, not about arbitrary functions satisfying the same differential equation in the fine topology.
2. (*proposition 3.2.*) The proof does not infer global monotonicity from the sign of a fine derivative; such an inference would require additional hypotheses on a general surreal function.
3. (*proposition 3.3, corollary 5.5.*) The asymptotic equivalences are statements about a single arbitrary infinitesimal. They are stronger than standard-part equalities and do not mean that a sequence of real arguments converges in the fine topology.
4. (*remark 5.2.*) The endpoint derivative of inverse sine does not exist as a surreal-valued derivative: its difference quotient grows beyond every fixed surreal bound as the increment becomes sufficiently small, and allowing infinitely large elements as derivative values does not repair this, because there is always a still smaller increment. Throughout, these statements are about differentiation of functions of a variable, not about a derivation on surreal numbers interpreted as transseries.
5. (*theorem 6.4.*) The usual side–side–angle ambiguity remains: a ray–circle intersection is a quadratic problem and may have zero, one or two admissible solutions; no additional infinite family of solutions appears merely because the coordinates are surreal.
6. (*equation (7.5).*) Quadrances and spreads forget orientation and distinguish neither an angle from its supplement nor a directed angle from its negative. That loss is useful for rational formulas but must be remembered.
7. (*theorem 7.2.*) A directed Menelaus or Ceva formula can be obtained in the same way after fixing signs, but is not needed for the present development.
8. (*theorem 7.4.*) Cyclic order is the order induced by finite representatives around the circle; it is not defined by fine-continuous motion of a real parameter. More generally, algebraic identities among finitely many vectors often survive unchanged, while any analytic interpretation of the associated angles must still be supplied.
9. (*theorem 5.4.*) Although cosine loses first-order information near zero, the full unit direction does not. At other angles one should use both coordinates, or a nonsingular tangent chart, rather than infer angular error solely from cosine error.
10. (*theorem 8.1.*) The conclusion about side valuations does not say that all three sides have equal valuation: a very short third side is possible.
11. (*theorem 8.4.*) The denominator $bc/(b+c)$ is essential. A defect that is small compared with the perimeter need not make the opposite angle close to π when one adjacent side is much shorter than the other.

12. (*theorem 9.2, theorem 9.9.*) The condition $e \prec \tau$ cannot be omitted from a uniform branch-matching theorem: $e = -\tau$ merges the roots and $e < -\tau$ destroys the two real intersections. Correspondingly the strict threshold $\sigma > 2\kappa$ cannot be weakened to equality uniformly near an extremum.
13. (*section 9.4.*) The dimension four of $\mathcal{B}_{s,u}$ is a *total* multiplicity of the entire finite zero set in the monad, not a claim that each individual simple root has a four-dimensional local algebra.
14. (*theorem 12.1.*) The support bound belongs to the factor coefficients, not to individual roots, which may require fractional scales; and no reality conclusion is asserted for an arbitrary multiple zero. The construction is a coefficient recursion, not a convergence claim for a Newton sequence.
15. (*corollary 11.2.*) The derivative statement bounds stationary angles; it does not invoke a general compactness or extreme-value theorem on a surreal interval.
16. (*theorem 13.2, equation (13.11).*) The finite Fourier identities establish no convergence or completeness for an unrestricted infinite Fourier series; infinitely many ordinary Fourier summands at one Hahn exponent do not form a strongly summable family merely because the corresponding real series converges. The Dirichlet and Fejér kernel identities are exact algebraic statements, not a fine-topological approximation theorem.
17. (*theorem 13.4.*) The theorem is a real-closed-field extension of a classical result, not a claim of a new spectral-factorization theorem. Multivariable trigonometric positivity is a different problem; no multivariable single-square theorem is asserted.
18. (*example 13.6.*) The example does not contradict [proposition 2.5](#): its coefficients depend on a non-real infinitesimal parameter, so the polynomial is not a single ordinary real-coefficient analytic function.
19. (*section 16.*) The coefficientwise arc-length convention is not an assertion that a nonconstant ordinary real-parameter path is continuous in the full fine topology, nor a definition of an unrestricted Riemann integral; other surreal integration frameworks have substantially different scope [\[15\]](#).
20. (*theorem 14.1, theorem 15.2.*) The spherical and hyperbolic sections give metric identities under explicit nondegeneracy hypotheses, not a complete global geometric or measure-theoretic foundation, and not a theory of hyperbolic area or of non-Archimedean geodesic completeness.
21. (*definition 17.1, theorem 17.2.*) The global normalization agrees with the established Ehrlich–Kaplan construction and is not claimed as a new definition; the published characterization of $\mathbf{Oz}$ is used to explain the word canonical, and its tree-theoretic proof is not reproduced. The local power-series argument establishes fine-local analyticity without asserting uniqueness from a differential equation.
22. (*theorem 17.4.*) The classification concerns group homomorphisms with the specified finite restriction; it is not a classification of all solutions of a differential equation, and by itself it does not answer the separate robustness questions formulated in [\[8, §11\]](#) for more general initial fields and their integer parts ([section 19.8](#) records what sources 20 and 21 add, and what stays open). It does not follow that defining a full-plane convention is impossible: the trivial-character convention is perfectly explicit. What fails is uniqueness from the stated analytic and algebraic axioms.

23. (*theorem 17.8, theorem 17.7.*) Both are restrictions on uniform analytic descriptions over the ring (R1) of [section 1.3](#), not failures of the algebraic identities or of arbitrary-radius circle geometry, and neither is asserted over the radius-free ring (R3) or the formal ring (R4). It is not claimed that a unique canonical value of circular sine at every infinite real argument follows from the finite-angle theory.
24. (*section 18, source 19.*) The following non-claims are source 19's, each also stated at its point of use.
- (a) Source 19 is a self-contained exposition organized around the rotation group, not a claim of priority or a machine-checked formalization; the finite-angle splitting is not presented as a new discovery, and it is compatible with, but logically prior to, the global construction of [\[8, §11\]](#).
 - (b) “Rotation” means the algebraic isometry; continuity is an extra property in a specified topology. Bounded matrix entries do not make the group set-sized or ordinarily compact.
 - (c) All statements about $\mathrm{SO}(2, \mathbf{No})$ are class-group statements: no proper-class Hahn sum, no quantification over a set of all class functions, no treatment of the group as a set-sized manifold, and topological statements use ball neighbourhoods with explicit quantifiers.
 - (d) $\mathbf{m}_{\mathbb{R}}$ is not an $\mathbf{No}$ -vector space, and the $\mathbb{R}$ -vector structure of $\mathbf{T}_{\mathbf{m}}$ does not extend to surreal scalars ([remark 18.1](#)).
 - (e) Preserving norms alone does not force linearity, and a field automorphism changing distances is not an isometry; the semidirect products [equation \(18.5\)](#) are algebraic, not statements about connected components.
 - (f) The rational chart parametrizes F -points; it is not an isomorphism of $\mathbb{P}_F^1$ with the affine torus ([remark 18.3](#)).
 - (g) Formal evaluation at infinitesimals is asserted for real or complex coefficients only; an arbitrary series with surreal coefficients need not be evaluable at every infinitesimal. Strong sums are not limits of partial sums in the fine topology.
 - (h) A branch of Arg is not an additive homomorphism; the group law belongs to angle classes. $\mathbf{No}/2\pi\mathbb{Z}$ is not the finite-angle model compatible with the finite phase; source 19 does not assert that some unrelated abstract class-group isomorphism $\mathbf{No}/2\pi\mathbb{Z} \cong \mathbf{T}(\mathbf{No})$ is impossible, the compatibility requirement being essential.
 - (i) Torsion means ordinary finite order; infinitesimal rotations give elements of infinite order, not new regular polygons. The square-root formula [equation \(18.9\)](#) is a local branch, not a multiplicative section; the $\mathrm{Spin}(2)$ calculation is finite algebra, not path lifting; finite orthogonal groups are conjugate to standard ones, which does not make their matrices real.
 - (j) [Theorem 18.8](#) is a filtration, not a direct-sum decomposition; $\sim$ is an algebraic relation, not a sequence limit; [equation \(18.14\)](#) gives no value to infinitely many identical turns, and its partial products do not converge finely.
 - (k) Cyclic order and the absence of fixed points are algebraic, not defined by ordinary real paths; [equation \(18.16\)](#) concerns polynomial functions, not all class functions on orbits.
 - (l) The algebraic group is applied at surreal points, without treating $\mathbf{No}$ as a set-sized coefficient field of scheme theory; the exponential series is not summable at infinite T and the canonical exponential lives on $\mathcal{O}_{\mathbb{R}}\mathbf{J}$; the invariant form defines no integration

over arbitrary full-class contours; a connected-component argument cannot replace the tangent calculation.

- (m) The representation classification supplies no Fourier convergence theorem, no Haar measure on the full class, and no classification of arbitrary abstract representations or of all abstract automorphisms; the automorphisms Ξ_a use no basis choice.
 - (n) The surreal exponent [equation \(18.15\)](#) is an analytic scaling on a restricted domain, not a transfinite iteration; beyond it a phase extension supplies further choices.
 - (o) Small-set discreteness is special to the full class, not a theorem about set-sized non-Archimedean fields with their intrinsic topology; it does not forbid a properly class-indexed net approaching zero. There is no ordinary real Lie-group interpretation of the fine circle, and classical compactness or Haar integration cannot be imported.
 - (p) The fine and standard-part uses of st must not be conflated. The semialgebraic fundamental group [equation \(18.27\)](#) is imported from [19, 20], not proved; the sequence $2\pi\mathbb{Z} \rightarrow \mathcal{O}_{\mathbb{R}} \rightarrow \mathbb{T}(\mathbf{No})$ is not a universal-covering claim.
 - (q) Solving the local rotation equation with $\Psi(0) = 1$ does not choose a global normalization; the classification of phase homomorphisms is not a classification of all solutions of the fine differential equation; mean-value and initial-value uniqueness arguments need further hypotheses ([example 18.16](#)); no compatibility with a scalar derivation is assumed.
 - (r) No representation of arbitrary surreal class elements by finite strings is asserted, and equality of effective Hahn-series names is not claimed decidable.
 - (s) Source 19 claims no novel construction of global surreal sine, no resolution of an Ehrlich–Kaplan robustness question (sources 20 and 21 later answer one part of the first one negatively, [section 19.8](#)), no classification of all abstract automorphisms, no Haar or Fourier theory on the full class group, and no global mean-value or initial-value theorem for fine differentiable functions; it does not assume that an infinitesimal Taylor sum is a fine limit. Its repository search covered the root and documentation overviews and the opening of this article, and no exhaustive search of the repository or of the literature is claimed; its arguments do not presuppose verification of the rest of this report, whose component manuscripts are unrefereed AI-assisted drafts.
25. ([section 19](#), *sources 20 and 21*.) The following non-claims are those of sources 20 and 21, each also stated at its point of use; the source is named when only one makes it.
- (a) [Theorem 19.33](#) is a partial negative answer to one sufficiency possibility in the first question of [8, §11.1]; it does not classify additional axioms that would suffice, does not assert that Ehrlich and Kaplan conjectured initiality to be sufficient, and does not answer their second question on exponentials from different initial embeddings (both). The first-order separation is proved in the broader class of all exponentials with the strip and modulus properties (source 20).
 - (b) Finite-angle uniformization, the canonical phase, the character classification [theorem 17.4](#), unavoidable infinite periods [theorem 17.6](#) and the local law are prerequisites from this article, not new; the canonical integer-part exponential is published [8, Proposition 11.2, Theorem 11.1]; not every elementary lemma is claimed new (both).
 - (c) The residue map, the maximal divisible subgroup and the splitting and homomorphism facts are classical $\mathbb{Z}$ -group theory [21]; the Leibnizian phenomenon is not a new theory of Presburger arithmetic (source 21; source 20 does not cite this literature).

- (d) The shift automorphisms are a prior type of construction [23, Theorem 5.2], [22, §4], re-proved; they are not asserted to preserve $\exp_{\mathbf{No}}$, Conway’s omega map or simplicity; “value-preserving” refers to the fixed value-group copy; the naturality obstruction concerns only the weaker valued-field data, is compatible with the Ehrlich–Kaplan construction, and no failure of invariance under exponential-field automorphisms is asserted (both).
 - (e) The integer-part criterion is an exact reduction, not a classification of every multiplier ring and not a substitute for independent hypotheses implying multiplicative closure; intermediate multiplier rings are not classified; the free action does not classify orbits under the full valued automorphism group (both).
 - (f) No claim is made that distinct noncanonical exponentials of source 21 are isomorphic or nonisomorphic to one another (source 21); their non-isomorphism with the canonical one is the merge’s [corollary 19.40](#). The split and nonsplit minimal examples are nonisomorphic, with no claim that their first-order theories differ (source 20).
 - (g) Reflection gives unboundedly many suitable fragments $\mathbf{No}(\lambda)$, not every fragment for a fixed phase, and no computable least λ ; no large-cardinal strength is hidden, the set-sized consequences resting on conservativity of global choice (source 20).
 - (h) The failure of countable saturation concerns the exponential expansion; $Z_E = \mathbb{Z}$ does not axiomatize standardness across the elementary class (source 20). Nothing is claimed about tameness, decidability, o-minimality or elementary equivalence of the phase expansion with the complex exponential field; no finite names, computable oracles or effective recognition of periods (source 21).
 - (i) $E' = E$ is the fine derivative, not the Berarducci–Mantova derivation; the local law is inherited, not a new uniqueness theorem; strong sums, including those in the cofinal-period theorem, are not asserted to be fine limits of partial sums (both).
 - (j) $\mathbf{No}(\varepsilon_0)$ is not asserted to be a full Hahn field; $\widehat{\mathbb{Z}}/\mathbb{Z}$ is an abstract group, not a topological quotient; the counts on $\mathbb{R}((t^{\mathbb{Q}}))$ are counts of embedded period groups, not of abstract isomorphism types, and use no continuum hypothesis; the explicit profinite examples establish neither a minimal multiplier ring nor reducedness (source 21).
 - (k) Both are AI-assisted, unrefereed drafts, not Lean-verified; neither modified or built the repository; their proposed Lean routes are architectures, not existing declarations; their finite checks (47,939 and the counts of [section 19.12](#)) test examples and finite identities only, not transfinite recursions, arbitrary supports, reflection or novelty. Novelty is provisional and rests on targeted audits (source 20: the root and documentation READMEs, this report’s guide and its Section 17, and targeted searches; source 21: the repository tree, root and documentation READMEs, this report and the automorphism report’s guide); failure to find a prior statement is not proof of priority.
26. (*Global.*) No proper-class sum, no arbitrary infinite polygon, no full-class Riemann integral, no ordinary sequential Fourier convergence theorem, and no uniqueness theorem for all fine-analytic differential equations is asserted anywhere. The three supplied manuscripts are identified as unpublished supplied documents, not as published literature. The literature search behind this report was targeted rather than exhaustive: a proof of an analogue is not, by itself, evidence that its exact formulation is absent from all previous literature, and no priority claim and no claim to have resolved a named published conjecture is made; the partial answer of [section 19.8](#) refutes one sufficiency possibility and leaves the Ehrlich–Kaplan questions open.

A A compact formula sheet with domain reminders

For finite real θ , nonzero $z, w \in \mathbf{SC}$, $u, v \in \mathbb{T}(\mathbf{No})$ and an ordinary integer n :

$$\begin{aligned} z &= |z| \operatorname{cis} \operatorname{Arg} z && \text{with } \operatorname{Arg} z \in \mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z}, \\ \cos \angle(z, w) &= \frac{\operatorname{Re}(\bar{z}w)}{|z||w|}, && \sin \angle(z, w) = \frac{\operatorname{Im}(\bar{z}w)}{|z||w|}, \\ \operatorname{cis}(n\theta) &= (\operatorname{cis} \theta)^n, && \tan(\theta/2) = \frac{\sin \theta}{1 + \cos \theta}, \\ |u - v| &= 2 \sin(\angle(u, v)/2), && \arctan t = \operatorname{Arg}(1 + it) \in (-\pi/2, \pi/2) \quad (t \in \mathbf{No}). \end{aligned}$$

The first argument is an angle class, and the sine in the directed-angle formula may be negative. In a triangle of positive area the interior sines are positive and

$$\begin{aligned} \alpha + \beta + \gamma &= \pi, & a^2 &= b^2 + c^2 - 2bc \cos \alpha, & a &= 2R \sin \alpha, \\ \Delta^2 &= s(s-a)(s-b)(s-c), & r &= \Delta/s, & R &= abc/(4\Delta), \\ \tan(\alpha/2) &= r/(s-a), & \ell_A &= \frac{2bc}{b+c} \cos(\alpha/2), & |O - I|^2 &= R(R - 2r), \\ v(\Delta) &= v(b) + v(c) + v(d(\alpha)), & v(R) &= v(a) + v(b) + v(c) - v(\Delta). \end{aligned}$$

For $z = x + iy$ with finite x and arbitrary $y \in \mathbf{No}$,

$$\operatorname{Sin} z = \sin x \cosh y + i \cos x \sinh y, \quad E(iz) = \exp_{\mathbf{No}}(-y) \operatorname{cis} x.$$

For infinite real x , circular sine and cosine require an explicitly selected extension; the lowercase finite-angle notation alone does not specify their values, and every such extension has an infinite period — on $\mathbf{No}$, cofinally many rationally divisible ones ([theorem 19.17](#)).

B Worked formal series

All of the following are strongly summable for infinitesimal h or τ , with the indicated positive square root in the last line:

$$\begin{aligned} \arctan h &= h - \frac{h^3}{3} + \frac{h^5}{5} - \frac{h^7}{7} + O(h^9), \\ \arcsin h &= h + \frac{h^3}{6} + \frac{3h^5}{40} + \frac{5h^7}{112} + O(h^9), \\ \operatorname{Log}_m(1 + h) &= h - \frac{h^2}{2} + \frac{h^3}{3} - \frac{h^4}{4} + O(h^5), \\ \sqrt{1 + h} &= 1 + \frac{h}{2} - \frac{h^2}{8} + \frac{h^3}{16} - \frac{5h^4}{128} + O(h^5), \\ \arccos(1 - \tau) &= \sqrt{2\tau} \left(1 + \frac{\tau}{12} + \frac{3\tau^2}{160} + \frac{5\tau^3}{896} + \frac{35\tau^4}{18432} + O(\tau^5) \right). \end{aligned}$$

For example, if $h = t^\lambda + t^\mu$ with $0 < \lambda < \mu$, the first terms of $\sin h$ are obtained by substituting this finite sum into its power series, mixed terms such as $t^{2\lambda+\mu}$ being collected with every other term of the same exponent. *The displayed order in a polynomial truncation is not necessarily the order of all Hahn monomials*: if $\mu \gg \lambda$, arbitrarily many powers $t^{n\lambda}$ precede t^μ . The support convention handles this automatically and does not replace it by a single real small parameter.

C Notation and domain guide

Symbol	Meaning and domain
$\mathbf{No}, \mathbf{SC} = \mathbf{No}[i]$	Surreal ordered class field and its complexification.
$t^\gamma = \omega^{-\gamma}$	Conway normal-form monomial; v the least support exponent, lc the leading complex coefficient, $v(0) = +\infty$.
$\mathcal{O}_{\mathbb{R}}, \mathbf{m}_{\mathbb{R}}; \mathcal{O}_{\mathbb{C}}, \mathbf{m}_{\mathbb{C}}$	Finite elements and infinitesimals of $\mathbf{No}$ and of $\mathbf{SC}$.
st, fin	Standard part of a finite element; the additive projection retaining all nonnegative-exponent terms of any real surreal.
$a \prec b, a \sim b$	a/b infinitesimal; respectively $a/b - 1$ infinitesimal.
$\mathbb{T}(\mathbf{No}), \mathbb{T}(\mathbb{R})$	The algebraic unit circle $\{u : u\bar{u} = 1\}$ and its ordinary counterpart; every point has finite coordinates.
$\sin, \cos, \text{cis}$	Taylor-lifted functions at finite real arguments; $\text{cis } \theta = \cos \theta + i \sin \theta$.
$C(t)$	The rational Cayley parametrization of the circle, never the cosine.
$\angle(u, v), d(\alpha)$	Angular metric on $\mathbb{T}(\mathbf{No})$; defect $\min\{\alpha, \pi - \alpha\}$ of an interior angle.
$h, \varepsilon, \tau, \delta$	Generic infinitesimal increment; generic positive infinitesimal; ramification/tangency parameter; triangle side gap $b + c - a$.
$\arctan, \arcsin, \arccos$	Real inverse branches with ranges $(-\pi/2, \pi/2)$, $[-\pi/2, \pi/2]$, $[0, \pi]$; the domain of $\arctan$ is all of $\mathbf{No}$.
$a, b, c; \alpha, \beta, \gamma$	Triangle sides and opposite interior angles; the angles are finite even when the sides are not.
Δ, R, r, s	Area, circumradius, inradius, semiperimeter. In section 9.4 only, s and u denote the two coupled parameters instead.
$\exp_{\mathbf{No}}, \log_{\mathbf{No}}$	The Gonshor real exponential and logarithm.
$\text{Exp}_{\mathbf{m}}, \text{Log}_{\mathbf{m}}$	Infinitesimal exponential and logarithm between $\mathbf{m}_{\mathbb{C}}$ and $1 + \mathbf{m}_{\mathbb{C}}$.
$\mathcal{H}, \mathcal{V}, E$	The two class strips and the strip exponential $E(x + iy) = \exp_{\mathbf{No}}(x) \text{cis } y$.
$\text{Sin}, \text{Cos}, \text{Tan}$	Canonical strip functions; after definition 17.1 also the canonical global functions, with which they agree.
$\Pi, \mathbf{Oz} = \Pi \oplus \mathbb{Z}$	Purely infinite normal forms; the omnific integer part, whose finite elements are exactly $\mathbb{Z}$.
χ, Ψ_χ, E_χ	A phase character on Π , the resulting full-real-class phase, and the resulting full-plane exponential.
$\int$	Coefficientwise Hahn integral over the ring (R1), with analytic endpoint corrections; not a fine-topological Riemann integral.
(R1)–(R4)	Fixed-domain, common-domain germ, radius-free and formal coefficient rings of section 1.3 .
$\text{SO}(2, \mathbf{No}), \text{Rot}(a, b)$	The rotation group and its matrix $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, $a^2 + b^2 = 1$ (section 18).
$\ell_{\mathbf{m}}, \mathbb{T}_{\mathbf{m}}$	Infinitesimal angle $-i \text{Log}_{\mathbf{m}}(u/\text{st } u)$ of a unit; the identity monad $\{u : \text{st } u = 1\} \cong (\mathbf{m}_{\mathbb{R}}, +)$.
$\mathbf{J}, \mathbf{c}, \Xi_a, \mathcal{S}_\rho$	Generator of $\mathfrak{so}(2)$; complex conjugation as a reflection; the rescaling automorphism equation (18.23) ; the circle of radius ρ .
$\mathcal{I}, \mathcal{I}^{\text{div}}, \text{Mult}, Z_E$	Normalized period group $\{a : \Psi(2\pi a) = 1\}$ of a phase, its maximal divisible subgroup, the multiplier ring, and the kernel-multiplier ring of an exponential (section 19).
$(\psi, \mathcal{L}), \text{pd}_\Psi, \widehat{\mathbb{Z}}$	Ordinary and infinitesimal coordinates of a character; the profinite defect $\Pi \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$; the profinite integers.
$\text{Sh}_c, [g]_0, F_{\mathbb{Q}}$	Strong shift automorphism with $\text{Sh}_c(\omega) = \omega + c$; the real coefficient of ω^0 in an exponent; the Hahn field $\mathbb{R}((t^{\mathbb{Q}}))$.

Symbol	Meaning and domain
$\mathbb{Z}, n, N, M$	Ordinary integers and ordinary finite degrees or sample sizes, never an unspecified class of “surreal integers.”

References

- [1] *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory*. User-supplied unpublished manuscript, September 21, 2026, [surcomplex_analysis\(2\).tex](#). Used for normal-form conventions, strong summability, finite Taylor lifting, coherent integrals, the fine-local logarithm, and the distinction between fine-local and common-domain coherent analyticity. Its global fine-analytic exponentials are the explicit predecessors of the infinite-phase family of [example 17.5](#).
- [2] *Finite Geometry of Hahn-Coherent Surcomplex Zeros: Conservation of Multiplicity, Normal Forms, and Collision-Stable Residues*. User-supplied unpublished manuscript, September 21, 2026, in [surcomplex_finite_geometry.zip](#). Background for positive-support conservation of finite zero clusters and collision-stable residue pairings; its several-variable theorems are not prerequisites for the geometry here.
- [3] *Finite Intersections in Surcomplex Analysis: Support-Controlled Division, Residue Duality, and Sharp Root Stability*. User-supplied unpublished manuscript, September 21, 2026, in [surcomplex_intersections.zip](#). Its §9 treats conditioned root stability with the threshold $\sigma > 2\kappa$; the trigonometric specializations in [theorem 9.7](#) and [theorem 9.9](#) are proved independently here.
- [4] John H. Conway, *On Numbers and Games*, Academic Press, 1976; second edition, A K Peters, 2001. Normal forms, the ordered-field structure, and omnific integers.
- [5] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series **110**, Cambridge University Press, 1986. [doi:10.1017/CBO9780511629143](#). Normal forms, real closedness, and the real exponential.
- [6] Norman L. Alling, *Foundations of Analysis over Surreal Number Fields*, North-Holland Mathematics Studies **141**, North-Holland, 1987.
- [7] Lou van den Dries and Philip Ehrlich, “Fields of surreal numbers and exponentiation,” *Fundamenta Mathematicae* **167** (2001), 173–188. [doi:10.4064/fm167-2-3](#). Erratum, *Fundamenta Mathematicae* **168** (2001), 295–297. The uses here are restricted analytic evaluation, real closedness and the exponential group law, not the ordinal estimates addressed by the erratum.
- [8] Philip Ehrlich and Elliot Kaplan, “Surreal ordered exponential fields,” *The Journal of Symbolic Logic* **86**, no. 3 (2021), 1066–1115. [doi:10.1017/jsl.2021.59](#); [arXiv:2002.07739](#). In particular §11: canonical trigonometric functions from an initial integer part, and canonical surcomplex exponentiation; its two open questions on robustness are in §11.1, on printed p. 34 of arXiv v3 (June 22, 2021), whose page and section numbers are those cited in [section 19](#). The paper credits an earlier idea of using **Oz** for trigonometric functions to Martin Kruskal.
- [9] B. H. Neumann, “On ordered division rings,” *Transactions of the American Mathematical Society* **66** (1949), 202–252. The ordered-support summability lemma.
- [10] Bjorn Poonen, “Maximally complete fields,” *L’Enseignement Mathématique* (2) **39** (1993), 87–106. [Author’s full text](#). The algebraic-closure input is Corollary 4.
- [11] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, Chapter 4, especially §§4.14, 4.21, 4.23–4.24. [dlmf.nist.gov/4](#). Consulted September 21, 2026, for ordinary trigonometric identities and inverse-function expansions; the surreal domains and support arguments are supplied here.

- [12] Michael A. Dritschel and James Rovnyak, “The operator Fejér–Riesz theorem,” in *A Glimpse at Hilbert Space Operators*, Operator Theory: Advances and Applications **207**, Birkhäuser, 2010, 223–254. [arXiv:0903.3639](#). Cited for the established factorization problem and its classical context; the scalar real-closed-field proof used here is given in full.
- [13] Norman J. Wildberger, “Rotor coordinates and vector trigonometry,” *Journal for Geometry and Graphics* **21** (2017), no. 1, 89–106. [Publisher record](#).
- [14] Charles Gunn, “Rational trigonometry via projective geometric algebra,” 2014. [arXiv:1401.2371](#).
- [15] Ovidiu Costin and Philip Ehrlich, “Integration on the surreals,” *Advances in Mathematics* **452** (2024), article 109823. [arXiv:2208.14331](#). A separate integration framework, not identified with the elementary coefficientwise functional of [section 16](#).
- [16] Alessandro Berarducci and Vincenzo Mantova, “Surreal numbers, derivations and transseries,” *Journal of the European Mathematical Society* **20** (2018), 339–390. [doi:10.4171/JEMS/769](#). Cited to distinguish a derivation on the scalar field from the function derivatives used here.
- [17] Alessandro Berarducci and Vincenzo Mantova, “Transseries as germs of surreal functions,” *Transactions of the American Mathematical Society* **371** (2019), 3549–3592. [doi:10.1090/tran/7428](#); [arXiv:1703.01995](#). Cited by source 19, especially §3, for the surreal analytic use of Neumann’s lemma and local Taylor calculus.
- [18] J. S. Milne, *Algebraic Groups: The Theory of Group Schemes of Finite Type over a Field*, Cambridge Studies in Advanced Mathematics **170**, Cambridge University Press, 2017, especially Chapters 10 and 12. Background for the norm-one torus and its representations in [sections 18.7](#) and [18.8](#).
- [19] Hans Delfs and Manfred Knebusch, *Locally Semialgebraic Spaces*, Lecture Notes in Mathematics **1173**, Springer, 1985. See also their survey “An introduction to locally semialgebraic spaces,” *Rocky Mountain Journal of Mathematics* **14** (1984), 945–963, [doi:10.1216/RMJ-1984-14-4-945](#).
- [20] Elías Baro and Margarita Otero, “Locally definable homotopy,” [arXiv:0812.2112](#) (2008), especially Corollary 6.7 on comparison with classical homotopy. With [19], the source of the imported [equation \(18.27\)](#).
- [21] Emil Jeřábek, “Rigid models of Presburger arithmetic,” *Mathematical Logic Quarterly* **65** (2019), no. 1, 108–115. [doi:10.1002/malq.201800019](#); [arXiv:1803.05797](#). Residue maps into $\widehat{\mathbb{Z}}$, maximal divisible subgroups and Leibnizian $\mathbb{Z}$ -groups (§§2–4), cited by source 21.
- [22] Elliot Kaplan, Lothar Sebastian Krapp and Michele Serra, “Decomposing the automorphism group of the surreal numbers,” preprint, [arXiv:2509.22374v3](#) (April 23, 2026). Class conventions and strongly linear automorphisms (§4).
- [23] *Automorphisms of the Surcomplex Numbers: Real Forms, Hahn Symmetries, Valuation, and Rigidity*. Report of this collection, [docs/surcomplex/surcomplex-field-automorphisms/](#); Theorem 5.2 (label `saut:thm:shiftflow`) is the fixed-shift flow used in [section 19.10](#). Unrefereed.